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Draft NSP/004/045 – Specification for EHV Wood Pole Lines operating up to 132kV with span lengths between 140m and 220m

1. Purpose

The purpose of this document is to provide a specification detailing the requirements for new single circuit EHV overhead lines with span lengths up to 220m designed for operation up to and including 132kV. This specification has been prepared to satisfy the requirements of ESQCR 2002. In essence the regulations require distribution plant to be fit for purpose and in the context of overhead lines this has been interpreted as requiring lines to be designed, constructed, used and maintained to adequately withstand all likely weather conditions.

Additionally this specification confirms the reference OHL 10/08 as the reference to be quoted to the BERR (Department for Business Enterprise and Regulatory Reform) for all new lines.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
Not Applicable	1.0	Woodhouse Mast Lines Specification for Yorkshire Electricity
NSP/004/045	1.0	Specification for EHV Wood Pole Lines operating up to 132kV with span lengths between 140m and 220m

2. Scope

This specification covers the design and constructional requirements for 3-phase overhead lines designed to operate at voltages up to and including 132 kV. It's prime purpose being to provide a single support design specification suitable for the replacement of Woodhouse Mast structures. The specification is based on the use of wood supports but additionally provides details on a range of self supporting steel structures for special design scenarios.

Note – Where possible new lines shall be constructed using shorter spans than those offered within this specification. Where this can be achieved the lines shall be designed to a companion specification NSP/004/046 "OHL9 - Specification for EHV Single Circuit Wood Pole Lines operating up to 132kV with span lengths up to 140m.

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3. Design Criteria

The specification has been designed in accordance with the requirements of BS EN 50341-1:2001 Overhead Electrical Lines Exceeding AC 45 kV. The structures in Appendix A have been designed utilising 200 mm² bare aluminium alloy (AAAC) conductors and comply with the "Empirical" design approach as detailed in BS EN 50423-3-9:2005.

To enable users to utilise this generic design when using alternative conductors and/or climatic loadings, this specification is provided with a suite of computer spreadsheets (Microsoft Excel Workbook filename Woodhouse Mast Replacement.XLS). A User Guide to the spreadsheet suite is provided as Appendix B of this Specification.

Appendix C of this specification contains details on a range of self supporting steel structures that may be used to supplement this design, where stayed structures cannot be accommodated.

3.1 Climatic Loads

The climatic loads are to comply with the requirements of Clause 4.3, Actions, and Empirical Approach as given in EN 50423-3-9:2005.

3.1.1 Wind Loads

The wind loads utilized are to correspond to the geographic area defined for the existing Woodhouse Mast Lines, that is, *Normal altitude*, site altitudes not exceeding 300 m, load cases 2 and 6.

Load Case	Load Condition	Wind Pressure (N/m ²)		Aerodynamic drag factors	
		q _x	q _c	C _x (structures)	C _c (conductors)
1	High wind (no ice)	1740	1740	0.8	1.0
2	Combined Wind and Ice (Normal altitude)	380	380	1.0	1.0
3	Combined Wind and Ice (High Altitude)		570		1.0
6	Construction and maintenance	0	0	-	-

Note: A modified span factor ($GC = (0.75L + 50)/L$) has been used for the structures in Appendix A. Also high wind condition and combined wind and ice (high altitude) are included for completeness only. (*High Altitude* is defined as site altitudes greater than 300m but not exceeding 500m.)

3.1.2 Ice Loads

In accordance with BS EN 50423-3-9:2005, a combined wind and ice condition only shall be considered in the design. A heavy ice condition need not be considered.

3.1.3 Combined Wind and Ice

A radial ice thickness of 9.5 mm shall be considered for *normal altitudes* and 12.5 mm for *high altitudes*.

For the design of the structures in Appendix A, a radial ice thickness of 9.5mm has been considered in conjunction with the combined wind and ice pressure given above with an assumed weight of ice of 9 kN/m³

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3.1.4 Temperature Effects

A minimum temperature of -5.6°C shall be considered for wind only loading conditions and combined wind and ice loading conditions.

3.1.5 Construction and Maintenance Loads

The following construction and maintenance loads have been considered in the design of the structures in Appendix A:

These loads assume an ambient temperature of 5°C in still air conditions and partial factors of 1.5 on static loads and 2.0 on dynamic loads.

The structure designs in Appendix A take account of the additional point loadings of 1.5 kN created as the result of a linesman working. A partial factor of 2.0 is assumed.

3.1.6 Security Loads

A security loading condition (broken wire condition) need not be considered for suspension or light angle timber pole structures therefore security against broken wire conditions has not been considered in the design of these structures in Appendix A.

Failure containment measures designed to limit damage in the event of cascade failure shall be incorporated. These should include, but not be limited to, the provision of longitudinal stays on selected angle/section structures which shall be designed to arrest cascades. Crossarms/insulators would then fail first on longitudinally stayed structures.

No section of line shall exceed 2000m or 10 spans without a section or angle structure which meets the requirements of failure containment as detailed above, being inserted into the line.

3.1.7 Short Circuit Loadings

This specification assumes that the forces due to short circuit currents on long, flexible conductors are less onerous than the climatic loadings.

3.2 Span Criteria

The Woodhouse mast replacement structures in Appendix A have been designed for the following span criteria:

Span Criteria	
Standard or Basic Span	200 m
Maximum Span Length	220 m
Maximum Sum of Adjacent Spans	440 m
Maximum Weight Span	400 m

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3.3 Partial Factors

3.3.1 Partial Factors for Actions

The following partial factors shall be used:

Action (load)	Partial factor
Normal load cases – variable actions	
Climatic loads and conductor tension	
High wind (Load case 1)	1.1
Combined wind and ice (Load case 2)	2.5
Permanent actions	
Self weight	
High wind (Load case 1)	1.1
Combined wind and ice (Load case 2)	2.5
Static cantilever loads (All load cases)	1.0
Construction and maintenance (Load case 6)	1.5 on static loads 2.0 on dynamic loads

3.3.2 Partial Strength Factors

The following partial strength factors shall be used for overhead line components.

Component	Clause	Material Property	γ_m
Steel Members (Grade S275) used as ancillaries on wood poles	5.5	Resistance of cross sections and buckling of sections (based on yield strength)	0.64
		Resistance of bolted connections: (based on ultimate tensile strength)	
		- Shear	1.33
		- Tension	1.0
		- Bearing	2.0
		Resistance of welded connections (based on yield strength of parent steel)	0.64
Timber Poles	5.7.1	Resistance of body of pole, cross-section, elements and bolted connections (based on mean ultimate strength)	1.0
Guyed Structures	5.7.2	Resistance of Guys (based on nominal failing load)	1.0
Foundations	5.11		1.0
Conductor	5.1	Resistance of conductors (based on nominal breaking load)	
		Combined wind and ice	2.0
		Wind only	2.0
Tension and Post Insulator Sets (b)	5.1	All string components (based on nominal failing load)	1.0

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3.4 Electrical Design

Unless specified otherwise, the overhead line shall be designed to operate at 132 kV with a nominal system frequency of 50 Hz.

3.4.1 Current Ratings

Unless specified otherwise, the line shall be designed to operate with 200 mm² AAAC Poplar conductor with a maximum operating temperature of 75°C

This will provide the following thermal ratings (3% excursion values):

132 kV @ 75 deg C						
Conductor		Summer		Spring/Autumn		Winter
Poplar		Amps	MVA	Amps	MVA	Amps
200	AAAC	624	143	678	155	708

Further information on alternative line ratings can be found by reference to IMP/001/011" Code of practice for Overhead Line ratings"

3.4.2 Internal Electrical Clearances

Wire clearances at the support shall be in accordance with ENA practice, that is the minimum clearance from live metal to structure steelwork and earthed fittings when tension insulators are used shall not be less than:

1.2 m in still air

0.9 m for jumper loops with a maximum swing angle of 20°

Note: The internal clearances shall also comply with the requirements of BS EN 50341-1:2001 where the conductor to earth clearances shall exceed the following:

1.2 m in still air

0.76 m under 3-year return period wind speeds (jumpers swung to 30°)

0.28 m under 50-year return period wind speeds (jumpers swung to 50°)

1.4m phase to phase in still air

To facilitate increased phase to earth clearances at section structures, all pilot insulators shall incorporate the support stool assembly detailed on drawing 1091010487 sht34

Minimum downlead phase-to-phase clearances shall be 2.5 m in still air and 1.5 m when conductors are subjected to a 3-year return period wind speed. (One phase subjected to 3-year return gust wind speed with adjacent phase subjected to 3-year mean hourly wind speed in the same direction.)

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3.4.3 External Electrical Clearances

External overhead line clearances shall fully meet the requirements of CE Electric UK code of practice document NSP/004/011, titled "Overhead Line Clearances", being designed for operation at 132 kV at the maximum operating temperature of 75°C.

3.5 Components

3.5.1 Conductors

Conductors shall be manufactured and supplied in accordance with CE Electric UK technical specification NPS/001/007. All conductors shall be greased in accordance with Engineering Recommendation L38/1 and the weight of greased conductors shall be used in the design. Aluminium Alloy conductors shall be provided in an inert condition as detailed in BS EN 50182, clause 5.5.7 and have a maximum resistivity of 32.53 nΩm at 20°C.

Unless specified otherwise, conductors shall be 200 mm² (AAAC), (37/2.87mm) code-named Poplar. The minimum factor of safety shall be 2.0 on the conductor nominal breaking loads. The conductor Maximum Working Tension (MWT) shall be calculated at a minimum conductor temperature of -5.6°C.

3.5.1.1 Conductor Spacing to Avoid Clash

Phase separation shall comply with the requirements of EN 50423-3-9: 2005, Clause 5.4.3.1.

For the structures in Appendix A, the assumed smoothed gust and lull wind pressures of 1.832 times the mean and 0.546 times the mean wind pressures respectively for a gust factor of 1.5 have been considered for a "2C" weather zone. That is a wind pressure of 380 N/m² and diametric ice thicknesses from 0 mm - 30 mm in increments of 10 mm. The worst combination of wind and ice loading is used for the calculation, with the actual wind and ice loads employed enhanced by a withstand factor of 1.1.

Maps of the UK showing the appropriate weather zones at varying altitudes from sea level up to 500 m can be found in EN50423-3-9:2005 and ENA TS 43-40

3.5.1.2 Conductor Dampers

Vibration dampers shall be fitted to the phase conductors and they shall be designed to:

- introduce an additional damping effect to that of the phase conductor and to control Aeolian vibration to ensure that the strain level in the conductors at the clamps, both suspension and tension, is below the fatigue limit of the conductor strands,
- Guarantee that the conductor bending strain at suspension and tension clamps shall not exceed the limit of 150 microstrain.

Vibration dampers shall be of the Stockbridge type and shall satisfy the performance and test requirements specified in IEC 61897. The damper shall have an adequate clamping surface with a radius at the clamp mouth. Where installed over Armour Rods the clamp of the damper shall be correctly sized for this application. Clamp bolts shall be locked in an approved manner.

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For structures in Appendix A, the damper shall be designed for 'Poplar' Conductor and shall have an adequate clamping surface with a radius at the clamp mouth. Stockbridge dampers shall have a nominal mass of 5 kg and shall be installed on all spans, positioned 800 mm from the mouth of the suspension or tension clamp. One damper shall be installed at each end (2 per span) on all span lengths up to and including 220 metres. See drawing 1091010188 sht1 for details on vibration dampers.

3.5.1.3 Conductor Terminations, Clamps and Joints

Conductor terminations and joints shall be designed for 'Poplar' conductor and supplied in accordance with CE Electric UK technical specification NPS/001/005 and the requirements of IEC 61284. Tensile tests shall be carried out to the requirements of Clause 11.5.1, alternative A of IEC 61284.

3.5.1.4 Conductor Terminations

Conductor terminations shall be full tension compression dead-end types supplied complete with jumper terminal flags. See drawing 1091010653 sht3 for details.

Jumper palms for connecting jumper loops to dead ends, shall be of the straight compression type. All jumper palms shall be supplied complete with the required bolt assemblies.

Bolt assemblies shall consist of bolts and nuts to grade 8.8/8.0 in accordance with BS 4190 complete with load spreading washers. Load spreading washers shall be 44mm diameter and 9.5mm thick for M16 bolts. To prevent loosening in service the minimum installation torque shall be 90Nm.

3.5.1.4 Conductor - line post trunnion clamps

Conductors shall be secured at the line post insulator positions using pivot type trunnion clamps which are supplied with the insulators and have been designed to accommodate a conductor diameter of 20.1mm. The trunnion (pivot) shall be on the centre-line of the conductor.

Where alternative conductor types are used, the project engineer may be required to specify alternative clamp types.

Conductors shall be secured at intermediate pole positions by direct clamping onto the conductors, however all pilot insulator positions have been designed to accommodate armour grip suspension rods to improve the stability of long jumpers.

3.5.1.5 Conductor Tension Joints

The fullest possible use shall be made of the maximum conductor lengths in order to reduce to a minimum the number of mid-span tension joints. Where this cannot be avoided, no more than one mid-span tension joint per conductor will be allowed in any one span.

Mid span Joints shall not be used in the following locations:-

- Within 3 m of the line post insulator attachment or dead-end tension joint.
- In spans over railways, navigable rivers, motorways, buildings or spans covered by special wayleave conditions.

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3.5.1.6 Conductor Non-tension Joints

Non-tension joints shall comply with the performance and test requirements of partial tension fittings as detailed in IEC 61284. The maximum resistance of all joints shall not exceed 75% of the resistance of an equivalent length of conductor. A record of all joint resistances shall be made and retained as part of the overhead line records.

3.5.2 Supports & Steelwork

3.5.2.1 Crossarm Assemblies

Mild steel crossarm assemblies shall be designed in accordance with BS5950 Part 1 except that a partial material factor γ_m of 0.64 shall be applied to the yield strength of grade S275 steel to EN 10025. For other grades of steel a partial factor equal to the ratio of yield strength to ultimate tensile strength shall be used. Crossarm assemblies shall be supplied in accordance with CE Electric UK technical specification NPS/001/005.

Following the completion of all fabrication processes on the support steelwork, including nuts, bolts and washers, they shall be hot dip galvanised and tested in accordance with the requirements of BS EN ISO 1461.

3.5.2.2 Fasteners and Washers

All bolts shall be grade 4.6 in accordance with BS EN 20898

All nuts, bolts, fasteners and washers shall be supplied in accordance with CE Electric UK technical specification NPS/001/010 and ENA TS 43-96.

3.5.2.3 Galvanising

Galvanizing of all ferrous material except messenger cable wires used in Stockbridge type vibration dampers shall be in accordance with ISO 1461 and shall be applied by the hot dip process to provide a mass of zinc coating of not less than 610 gm of zinc per square metre of surface on steel bars, plates, sections and fittings. Threaded work shall have a coating mass of 305 gm of zinc per square metre. Messenger wires shall be galvanized to BS EN 10244 Part 2 with a mass of zinc coating in accordance with Table 1. Sheradizing, zinc or cadmium plating or other similar processes shall not be used.

3.5.2.4 Supports

Supports shall be designed to resist the specified loads due to the climatic conditions, accidental-security conditions and construction and maintenance conditions taking into consideration the specified partial load factors and partial material strength factors.

The supports in Appendix A may be used for different conductors, spans and loading conditions subject to satisfactory analysis using the spreadsheet "Woodhouse Mast Replacement.XLS". A user guide can be found in Appendix B.

The structures capabilities detailed below have been matched to Poplar conductor as defined in clause 4.1 of this specification.

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3.5.2.5 Timber Poles

The strength of timber supports shall be derived from formulae contained in BS 1990 part 1 and in respect of *Pinus Sylvestris* wooden products, the pole producers in the UK have confirmed the following parameters:

Mean ultimate extreme fibre stress (bending stress)	53.3 N/mm ²
Mean modulus of elasticity	10054 N/mm ²

These parameters are based on a *Pinus Sylvestris* population whose southern border occurs at 60° latitude. Where some other species and / or pole population is considered, other strength parameters may be applicable and in this respect, amended support capabilities may be used.

Poles shall be fabricated and supplied in accordance with CE Electric UK technical specification NPS/001/001 and the fabrication drawings detailed within this specification. All Rutter type poles will be supplied as composite structures with the re-enforcing steel or hardwood keys pre-installed.

Windspan Capability Tables – Intermediate Structures

Available Pole Heights	Available Windspan (Medium Grade Rutter)	Available Windspan (Stout Grade Rutter)	Sinking Depth
12m	91m	221m	3.2m
13m	99m	230m	3.2m
14m	100m	226m	3.2m
15m	107m	225m	3.2m
16m	115m	236m	3.2m
17m	129m	224m	3.2m

Windspan Capability Tables – Section Angle Structures

Pole Height	Max Allowable Windspan on Angle Structures									
		Angle of Deviation								
	(m)	5	10	15	20	25	30	35	40	60 (H Pole)
	12	220 (220)	220 (220)	220 (220)	220 (220)	220 (220)	220 (200)	220 (173)	220 (146)	220 *
	13	220 (220)	220 (220)	220 (220)	220 (220)	220 (220)	220 (203)	220 (170)	220 (142)	220 *
	14	220 (220)	220 (220)	220 (220)	220 (220)	220 (197)	220 (162)	220 (133)	220 (108)	220 *
	15	220 (220)	220 (220)	220 (220)	220 (207)	220 (167)	220 (134)	220 (108)	217 (na)	220 *
	16	220 (220)	220 (220)	220 (220)	220 (195)	220 (157)	220 (125)	220 (na)	206 (na)	220 *
17	220 (220)	220 (220)	220 (218)	220 (168)	220 (133)	220 (na)	200 (na)	175 (na)	220 *	

Notes

* The above table assumes structures are stayed with spreads at 45 deg to the pole. Alternative spans shown in brackets are the maximum allowed windspan with stay spreads applied at a minimum spread of 30 Deg. Where these show n/a it confirms that reduced stay spreads cannot be installed without discussions with the overhead standards section as the calculated figures are less than 100m at the minimum angle (larger spans can be used with larger stay spreads)

** Stay spreads on H pole structures are limited to 40 deg to obtain maximum windspan.

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3.5.2.6 Type and Range of Timber Supports in Appendix A

The range of the supports in Appendix A has been selected such that the existing Woodhouse Mast lines can be replaced and uprated without the need to insert additional support structures. The new pole designs are based on overall pole lengths of 12 m – 17 m in 1 m intervals. This range of pole lengths will allow the existing poles to be replaced without exceeding the existing structure heights. If any ground clearance infringements are encountered then these can be overcome by using the next height of pole.

Note (Only a 10% increase in structure height of the highest structure in the route is allowed if a line is required to stay with it's original Section 37 planning consent) .

The structures detailed provide for the following angles of deviation.

Structure Type	Deviation
Intermediate (Rutter)	0 – 2°
Section (unstayed) (Rutter)	0 – 2°
S20	2° - 20°
S40	20° – 40°
S60	40° – 60° *
Large Angle H-Pole	40° - 60°
Tee Off	N/A
Terminal (H-Pole)	0° entry.

* The default pole assembly for 40° - 60° deviations shall be the Large Angled H pole. The single structure pole to drawing no 1091390015 shall only be used in special circumstances and only with the prior approval of the companies Overhead Line Standards Manager.

3.5.2.7 Steel Poles

This specification includes the option for a range of self-supporting steel pole structures removing the requirement for stays. The steel poles have been designed to comply with EN 50423-3-9:2005. Details can be found in Appendix C.

3.5.3 Guyed Structures

Where stays are used, the ability of the poles to resist the crippling loads due to the vertical component of the forces in the stays shall be calculated in accordance with BS 1990: Part 1: 1984.

Stay spreads shall normally be such that an angle of 45° is provided between stay and pole. In circumstances when tighter stay spreads are unavoidable due to physical obstructions, reduced angles can be used, subject to the approval of the company's Overhead Line Standards Manager. However the minimum allowable stay angle shall be 30°.

At single pole angle supports where a single stay is used the stay shall be set so as to bisect the angle of deviation to counteract the transverse component of the conductor tension. When two or more stays

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are fitted, they shall be arranged in a splayed formation symmetrical about the angle of deviation bisector.

Triple stays shall comprise two splayed stays as above, set at the approved angle to the pole, the third stay being placed to bisect the two and entering the ground at least 2 metres behind them.

Where structures with multiple stays are required to provide resistance for failure containment purposes, the location of stays shall be chosen to provide adequate longitudinal resistance. The ideal location for the outer stays in such cases is in line with the conductors in the opposite span, but if this does not provide adequate transverse resistance, a compromise position may need to be selected.

Multiple stay baulks shall be installed so that the length of undisturbed ground between stay pits or anchors is not less than 3.0m.

3.5.3.1 Loading limitations on Guyed Single Structures

To maximise the allowable strut load capability on single Extra Stout supports, this specification has amended the minimum acceptable pole diameters as detailed in BS 1990: Part 1 :1984. The tabulated "min" value has been increased to reflect a calculated mean value. The new effective min values are shown in the pole fabrication drawing 1091390003.

Additionally to ensure that no loading situation exceeds the maximum allowable design values, an artificial maximum windspan limit has been set at 170m.

Variations in allowable pole diameters outside of these limits, stay spreads of less than 45° or maximum windspans in excess of these tabulated values may only be allowed with the approval of the CE Electric UK Overhead Standards Manager.

Stay requirements and pole crippling loads due to the vertical component of the forces in the stays for all angle structures shall be determined using the accompanying spreadsheet "Woodhouse Mast Replacement Design .XLS". A user's guide for this is provided as Appendix B.

An indication of stay requirements for the structures in Appendix A is provided below.

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Structure	Pole Type	Length	Stays	Structure Drawing ref
Section 0° - 2°	Rutter 2 x Medium/Stout	16m	None (**)	1091390017 sht1
S20 0° - 10°	E/Stout	13m	Bisector x 1	1091390013 sht1
S20 0° - 10°	E/Stout	17m	Bisector x 1	
S20 10°-20°	E/Stout	13m	Bisector x 2	
S20 10°-20°	E/Stout	17m	Bisector x 2	
S40 20°-30°	E/Stout	13m	Bisector x 2	1091390014 sht1
S40 20°-30°	E/Stout*	17m	Bisector x 3	
S40 30°-40°	E/Stout	13m	Bisector x 3	
S40 30°-40°	E/Stout*	17m	Bisector x 3	
S60 40°-50°**	E/Stout*	13m	Bisector x 3	1091390015 sht1
S60 40°-50°**	E/Stout*	17m	Bisector x 3	
Large Angle H Pole 40° - 60°	Stout	13m	Bisector x 4	1091390018 sht1
Large Angle H Pole 40° - 60°	Stout	17m	Bisector x 4	
Terminal	E/Stout H - Pole	13m	Longitudinal x 4	1091390019 sht1
Terminal	E/Stout H - Pole	17m	Longitudinal x 4	

*Only suitable where max allowable pole diameter has been specified on pole fabrication drawing.

Allowable pole size may be confirmed using the spreadsheet detailed in Appendix B. Alternatively use a Large Angle H pole.

** The default pole assembly for 40° - 60° deviations shall be the Large Angled H pole. The single structure pole to drawing no 1091390015 shall only be used in special circumstances and only with the prior approval of the company's Overhead Line Standards Manager.

3.5.4 Insulators

Unless specified otherwise, all insulators shall be manufactured from composite materials and be fully compliant with the CE Electric UK technical specification NPS/001/006 and ENA TS 43-93: issue 4: 2004, line Insulators.

Insulator Types	Insulator Drawing No.	Structure type used
Vertical Post	1091010487 sht 18	Intermediate
Horizontal Post *	1091010487 sht 16	
Tension Insulator	1091010487 sht 30 or 1091010487 sht 31	Section / Terminal
Vertical post/pilot + Support Stool	1091010487 sht 18	Section
Stay insulators	1091010372 sht 2	All Unearthed stayed supports
Stay insulators	1000439107 sht 2	All Earthed stayed supports

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* By default this insulator may ship complete with a removable base to allow its use on OHL9 construction. This must be removed to allow the insulator to integrate into the mounting steelwork.

All lines shall be assumed to be in areas designated as “heavy pollution”. Creepage distances for 132kV insulators shall therefore be **3625mm** and **2538mm** for vertical and horizontal composite insulators respectively.

The insulation co-ordination requirements for the replacement lines shall be as follows:

Specific Requirements	Nominal System Voltages		
	33 kV	66 kV	132 kV
Wet power frequency withstand voltage (60s)	90	140	275
Dry lightning impulse withstand voltage (1.2/50µs wave)	200	325	650

3.5.4.1 Post Insulators

Post insulators shall comply with the requirements of ENA TS 43-93, in particular Section 6, Quality Assurance and Section 8, Mechanical Performance. Design, type, sample and routine tests shall be carried out as required. Tracking and Erosion Tests, RIV Requirements and Corona Extinction Voltage Tests shall be conducted as specified.

Unless agreed otherwise, the specified cantilever load (SCL) shall comply with Clause 8.2 of ENA TS 43-93. The SCL shall therefore be at least 21kN.

It should be noted that suppliers generally use **Routine Cantilever Load (RCL)** or **Maximum Design Cantilever Load (MDCL)** when specifying the mechanical strength of post insulators. This is defined as the maximum recommended working load and is usually specified by suppliers as being 50% of the ultimate cantilever strength rating of the insulator defined as the **Specified Cantilever Load (SCL)** or the **Specified Mechanical Load (SML)**

If the SCL is not specified by the supplier, then to comply with the specification, the RCL or MDCL of the insulator shall not be less than 50% of specified cantilever load as calculated in the spreadsheet “Woodhouse Mast Replacement. XLS”. If the SCL is less than 21kN, suppliers are required to demonstrate the 50 year lifetime SCL in accordance with IEC 61109 Annex A.

3.5.4.2 Tension Insulators

Unless otherwise specified composite tension insulators shall be ball and socket and have a mechanical rating of 120 kN with a standard coupling size of 20 mm. End fittings shall be in accordance with BS EN 61466-1. Sockets shall have “W” type phosphor bronze security clips in accordance with IEC 60372.

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3.5.4.3 Arc horns

Arc horns shall only be fitted to intermediate type structures fitted with post insulators in the approach zone (up to 1.6km from the substation)

Arc gap requirements on tension sets shall be as tabulated below:

Arc gaps	Nominal system Voltage		
	33kV	66kV	132kV
Normal (beyond initial 1.6km from substation on unearthed wood poles)	1120mm	1120mm	1120mm
Approach – (Wood poles assumed to be earthed structures within 1.6km of substation)	453mm*	453mm	995mm

*normal 268mm arc gap increased to 453mm to reduce bird strike risks,

Where replacement lines are installed with 132kV insulator arrangements but are required to be initially operated at a reduced nominal system voltage, adjustable arcing horns shall be provided on tension sets such that the arc gaps for the appropriate system voltage can be provided whilst providing a simple upgrade option.

3.5.4.4 Insulator Fittings

All ball, socket, tongue, clevis and eye couplings shall be supplied in accordance with the CE Electric UK technical specification NPS/001/005, have a mechanical rating of 120 kN and be in accordance with the requirements of BS 3288. All ball and socket couplings shall also comply with the requirements of IEC 60372.

3.5.5 Stays

All stays shall be adjustable, utilising 7/4.00 mm grade 1150 galvanised steel stay strand as detailed in ENA TS 43-91. Stay strand shall be supplied in accordance with the CE Electric UK technical specification NPS/001/005.

The tensile strength of all components of the stay assembly shall be not less than the stay wire which has a MFL = 101kN.

3.5.5.1 Stay Insulators

All stays shall be fitted with stay insulator(s) supplied in accordance with the CE Electric UK technical specification NPS/001/006 and shall be positioned such that under a broken jumper or a broken stay condition the insulator shall prevent the stay becoming 'live'. The insulator shall therefore be placed below any likely point of contact with live metal, but must remain a minimum of 3.0 m vertically above the ground.

All stay insulators (except approach) shall be provided with insulator protective devices with an arc gap of 1120mm and comply with the electrical and mechanical requirements of ENA TS 43-91 as given below:

Nominal system voltage	132kV
Minimum wet power frequency withstand voltage	190kV
Minimum dry lightning impulse withstand voltage	200kV

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Within the approach zone (up to 1.6km from the substation) all stays shall be fitted with insulators that comply with drawing 439107 Type 2 as detailed in ENA TS 43-91

3.5.6 Foundations

3.5.6.1 Stay Foundations

Stay blocks shall be to ENA TS 43-91 drawing no. 439103 type 2 and installed at a depth of 1.8 m below ground level. Alternatively screw anchors supplied in accordance with the CE Electric UK technical specification NPS/001/020 may be used.

3.5.6.2 Pole Foundations

Structure foundations shall be designed for three soil conditions with ultimate lateral rupture capacities of 628 kN/m², 471 kN/m² and 314 kN/m² per metre depth for good, good / average and average / poor soils respectively. Unless specific soil strength information is available, all calculations shall be based on 471 kN/m² or "average" soil on the premise that soil additives will be used.

All un-stayed structures shall be fitted with four wood foundation blocks (1300 x 250 x 125mm) as detailed in drawing 1091010675 sht 30. The blocks shall be placed as follows:

Ground line to 2 x Top Block Centres = 700 mm

Ground line to 2 x Bottom Block Centres = 2800 mm.

Note.

Where Rutter type structures are installed in poor ground the default top block types shall be replaced by (2500 x 250 x 125mm) as detailed in drawing 191010675 sht 30.

For stayed structures an ultimate soil bearing capacity of 429 kN/m² (4 tons/ft²) shall be used. Concrete pads or wooden blocks shall be used under the pole to reduce the bearing loads when necessary.

All pole foundations shall incorporate the use of the backfill "Perma-Soil" backfill additive as detailed below.

25kg of Perma-Soil shall be added per cubic metre of excavated spoil. The mixture shall be replaced in the excavation and fully compacted in layers not exceeding 300mm thick. Initially the bottom baulks only shall be fitted to unstayed structures to facilitate this compaction. The top baulks shall only be fitted when the compacted fill reaches the underside of these baulks.

Research work has shown that where average to poor foundations are identified, the Perma-Soil stabiliser will normally provide sufficient improvements in the foundation capability to improve the soil quality to that of a ' Good ' backfill.

All un-stayed structures shall be installed to a depth of 3.2m. All stayed structures shall be installed to a depth of 2.4m

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3.5.7 Lightning Protection and Earthing

With the exception of structures located in the substation approach zone, all structures have been designed to be of unearthed construction.

To avoid a capacitive induced potential on unearthed supports, all insulator support steelwork shall be bonded together using flexible insulated earthwire bonds. Similarly, all stays shall be bonded at the top of the pole steelwork.

3.5.7.1 Earthing Structures – Substation Approach Zone

Over the last 1.6km of route into any substation, pole crossarms shall be earthed by means of a 70mm PVC (Green) insulated copper cable down the pole, fixed to the pole at intervals of not more than 600 mm and connected to earth rods. Rods shall be driven into the ground as necessary to obtain a maximum earth resistance of 10 ohms. Earth rods and their connectors shall comply with the requirements of ENA TS 43-94 and CE Electric UK technical specification NPS/002/001

In order to enable earth electrodes to be bolted to steel poles, earthing lugs shall be attached to the pole shaft 300mm above the top of the concrete foundation.

3.5.8 Anti Climbing Devices

Anti-climbing devices (ACD) shall be designed in accordance with the CE Electric UK technical specification NPS/001/005 and be installed in accordance with NSP/004/109.

They shall consist of pre-wrapped barbed wire Anti-climbing guards (Enhanced Design) and shall as a default be installed on all rutter type poles. See drawing 1091010675 sht 33 for further details. The installation of standard types of ACD on all other structure types will be dependant upon their location and other equipment secured to the pole.

3.5.9 Safety Signs, Labels and Notices

All supports shall be fitted with at least one Safety Sign. The sign shall be mounted approximately 3 m above the ground level, above any anti-climbing guard and clearly visible to an observer on the ground. Safety Signs, labels & notices shall be designed in accordance with the CE Electric UK material specification NPS/001/010 and shall be installed in accordance with CE Electric UK guidance note NSP/004/109.

3.5.10 Climbing systems and work positioning attachment points

In accordance with the requirements of the Working at Height Regulations 2005, these structures have been designed to accommodate the installation of an assisted temporary climbing system. This system involves the permanent installation of fixing brackets at specific points up the structures during the initial construction process which allow the later installation of a lightweight temporary ladder system. (See Appendix D, item D4 for details). The ladder system provides a combined access and fall arrest system via its internal track system. The internal track system allows a travelling device (See Appendix D, Item D6 for Details) located in the track to follow the direction of the climber and in the event of a fall, the travelling device locks onto the track.

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The ladder system is constructed from individual small ladder sections approx 2m in length (See Appendix D, item D5 and D5.1 for details) allowing the climber to provide an access system from a point above the structure ACD to a point above the pole top. A selection of ladders will be provided at a central point within CE Electric UK to provide access to structures when required.

Additionally all poles have been designed to incorporate fall arrest / work positioning attachment points. These are provided via the installation of M20 eyebolts at strategic points around the pole top working area.

Although this climbing system has been designed into the structures, it is envisaged that in most cases pole top access is likely to be achieved through independent means ie mobile access platforms.

Ladders are available in 5 standard sizes. To accommodate the varying pole heights and arrangements the following matrix has been produced to indicate the combinations required in each situation.

Type	Description	Cat Number	No Rungs	Weight in Kgs
A	Common base ladder section 2240mm in length	313080	8	9.2
B	Intermediate ladder section 1960mm in length	313081	7	8.3
C	Intermediate ladder section 1680mm in length	313082	6	7.4
D	Intermediate ladder section 1400mm in length	313083	5	6.5
E	Shaft Entry Top Section 1120mm in length	313084	1	4.5
-	Permanent Female Mounting Brackets	313086	-	-
-	Glidelock Traveller	313085	-	-

Ladder Selection table

Structure Type	(Rutter Pole)	Bkts Req	Single section poles & H section / terminal poles	Bkts Req
12m	(1 x A) + (1 x B) + (2 x E)	8	(1 x A) + (3 x D)	8
13m	(1 x A) + (1 x C) + (1 x D) + (2 x E)	10	(1 x A) + (1 x B) + (1 x C) + (1 x D)	8
14m	(1 x A) + (3 x D) + (2 x E)	12	(1 x A) + (1 x C) + (3 x D)	10
15m	(1 x A) + (3 x C) + (2 x E)	12	(1 x A) + (5 x D)	12
16m	(1 x A) + (2 x B) + (2 x D) + (2 x E)	14	(1 x A) + (3 x C) + (2 x D)	14
17m	(1 x A) + (5 x D) + (2 x E)	16	(1 x A) + (2 x C) + (4 x D)	16

Note : H pole require the number of brackets and ladders to be doubled (ladders fitted to each leg)

3.5.11 Specialist Pole top working Equipment

3.5.11.1 Combined Platform / Crane Assembly

To allow insulators and conductors to be manipulated whilst working at the pole top, a combined pole top platform / crane assembly has been designed. The assembly has been constructed in a modular

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fashion to allow it to be carried up the pole and assembled. Once in place the crane allows conductors to be manipulated in or out of the insulator conductor clamps and or the complete insulator assemblies to be replaced in a controlled manner. The crane assembly has been designed and to incorporate fall arrest attachment points for work required above the pole top. (See Appendix D, item D1 for details)

3.5.11.2 Pole Platforms

The pole top has been designed to allow installation of the standard 2m long pole top temporary access platform. Once in position the platform is capable of providing full access to the outer insulators.

3.5.11.3 Insulator Mounted Running blocks

Specialist insulator mounted running blocks have been designed to allow the conductors to be installed and manipulated into the insulator conductor clamps in a safe controlled manner without the need for any other additional equipment. (See Appendix D, Item D2 and D3 for details)

3.5.12 Conductor Erection Guidance

3.5.12.1 Conductor Sagging Bases

Conductors shall be installed to comply with the sagging bases detailed below:

Condition	Temp (°C)	Radial ice thickness (mm)	Ice Density (k/m ³)	Wind Pressure (N/m ²)	Tension Limit	Comment
High wind	-5.6	N/A	N/A	1740		
Combined wind and ice (normal altitude)	-5.6	9.5	913	380	RBL/ γ_m	Maximum working tension limit
Combined wind and ice (high altitude)	-5.6	12.5	913	570		Maximum working tension limit
Still air	15	N/A	N/A		BL/5.0	Aeolian vibration limit
Max. operating	75	N/A	N/A	N/A		

Where RBL is the rated breaking load of the conductor.

3.5.12.2 Conductor Creep Compensation

Suspended conductors are subject to longitudinal stresses that cause permanent long-term elongation or conductor creep; this results in an increase in sag in the conductors. In order to ensure satisfactory ground clearance exists throughout the life of the line, a combination of compensation techniques have been employed within this Specification.

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(i) Over-Tensioning

The reduction of sag due to the increased tension compensates for the increase in sag due to conductor creep during the life of the line. The over tension can be applied either as a temperature shift or as a percentage increase over the design tension. The following over-tension values should be used.

Conductor Type	Spans less than 100m	Spans greater than 100m
AAAC	10% increase in tension	-20°C temperature shift
ACSR	10% increase in tension	-20°C temperature shift

The spreadsheet "Woodhouse Mast Replacement. XLS" utilises this approach.

(ii) Ordinate Shift

Longitudinal profiles for overhead lines commonly use the "Optimal, PoleCad Software" to design overhead lines and to ensure that all statutory clearance requirements have been met. The calculations used in the production of ground clearance curves apply a technique of modifying the clearance ordinates by reducing the conductor tension by a specific percentage. This process builds in a small safety factor into the clearances. All percentage reduction values are based on an application temperature of 15°C. Lines to this specification shall utilise a 10 % reduction

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3.5.12.3 Conductor Details**CONDUCTOR SAG/TENSION PARAMETERS for DESIGN**

Conductor Code Name (if any)	Poplar
Greased Conductor Weight (N/m)	6.768
Cross Sectional Area of Conductor (mm ²)	239.4
Conductor Overall Diameter (mm)	20.09
Coefficient of Linear Expansion (/Degree C)	0.000023
Modulus of Elasticity (kN/mm ²)	56
Rated Breaking Strength of Conductor (kN)	70.6
Basic / Recommended Span (m)	200
Wind Pressure on Conductor (N/m ²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kN/m ³)	8.95
Maximum Working Tension (MWT) (kN)	35.3
Resultant load/m at MWT (kN/m)	0.020409
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) (kN)	14.13
Temperature at EDT Limit (Degrees C)	15
Resultant load/m at EDT (kN/m)	0.006768
Tension at 0 deg,C (kN)	16.80

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	Poplar	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C	10%

	Sag (m) for	Sag (m) for Range of Hot Curve Temperatures					
Span (m)	Cold Curve -5.6	50	55	60	65	70	75
40	0.08	0.14	0.15	0.16	0.16	0.17	0.18
80	0.30	0.58	0.60	0.63	0.66	0.68	0.71
120	0.68	1.30	1.36	1.42	1.48	1.54	1.60
160	1.20	2.31	2.42	2.53	2.63	2.74	2.84
200	1.88	3.61	3.78	3.95	4.11	4.28	4.44
240	2.70	5.20	5.44	5.68	5.92	6.16	6.39
280	3.68	7.07	7.41	7.74	8.06	8.38	8.70
320	4.80	9.24	9.67	10.10	10.53	10.95	11.36
360	6.08	11.69	12.24	12.79	13.33	13.86	14.38
400	7.50	14.44	15.12	15.79	16.45	17.11	17.76
440	9.08	17.47	18.29	19.10	19.91	20.70	21.48
480	10.80	20.79	21.77	22.74	23.69	24.64	25.57
520	12.68	24.40	25.54	26.68	27.81	28.92	30.01

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DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name Poplar
 Basic / Recommended Span (m) 200
 Temperature Shift for Creep (Deg. C) 0 Insert minus sign as necessary
 Equivalent Percentage Increase in Tension (%) 0.00% at 15°C
Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kN)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	120	200	210	230
-5.6	18.04	0.17	0.23	0.30	0.38	0.47	0.68	1.88	2.07	2.48
0	16.80	0.18	0.25	0.32	0.41	0.50	0.73	2.01	2.22	2.66
5	15.75	0.19	0.26	0.34	0.44	0.54	0.77	2.15	2.37	2.84
10	14.77	0.21	0.28	0.37	0.46	0.57	0.82	2.29	2.53	3.03
15	13.87	0.22	0.30	0.39	0.49	0.61	0.88	2.44	2.69	3.23
20	13.03	0.23	0.32	0.42	0.53	0.65	0.94	2.60	2.86	3.44
25	12.26	0.25	0.34	0.44	0.56	0.69	0.99	2.76	3.04	3.65
30	11.56	0.26	0.36	0.47	0.59	0.73	1.05	2.93	3.23	3.87
35	10.93	0.28	0.38	0.50	0.63	0.77	1.11	3.10	3.41	4.09
40	10.36	0.29	0.40	0.52	0.66	0.82	1.18	3.27	3.60	4.32
45	9.84	0.31	0.42	0.55	0.70	0.86	1.24	3.44	3.79	4.55
50	9.38	0.32	0.44	0.58	0.73	0.90	1.30	3.61	3.98	4.77
55	8.96	0.34	0.46	0.60	0.77	0.94	1.36	3.78	4.17	5.00
60	8.57	0.36	0.48	0.63	0.80	0.99	1.42	3.95	4.35	5.22
65	8.23	0.37	0.50	0.66	0.83	1.03	1.48	4.11	4.54	5.44
70	7.91	0.38	0.52	0.68	0.87	1.07	1.54	4.28	4.72	5.66
75	7.62	0.40	0.54	0.71	0.90	1.11	1.60	4.44	4.89	5.87
80	7.36	0.41	0.56	0.74	0.93	1.15	1.66	4.60	5.07	6.08

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name Poplar
 Basic / Recommended Span (m) 200
 Temperature Shift for Creep (Deg. C) -20 Insert minus sign as necessary
 Equivalent Percentage Increase in Tension (%) 22.56% at 15°C
Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kN)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	120	200	210	230
-5.6	23.00	0.13	0.18	0.24	0.30	0.37	0.53	1.47	1.62	1.95
0	21.54	0.14	0.19	0.25	0.32	0.39	0.57	1.57	1.73	2.08
5	20.28	0.15	0.20	0.27	0.34	0.42	0.60	1.67	1.84	2.21
10	19.07	0.16	0.22	0.28	0.36	0.44	0.64	1.77	1.96	2.35
15	17.90	0.17	0.23	0.30	0.38	0.47	0.68	1.89	2.08	2.50
20	16.80	0.18	0.25	0.32	0.41	0.50	0.73	2.01	2.22	2.66
25	15.75	0.19	0.26	0.34	0.44	0.54	0.77	2.15	2.37	2.84
30	14.77	0.21	0.28	0.37	0.46	0.57	0.82	2.29	2.53	3.03
35	13.87	0.22	0.30	0.39	0.49	0.61	0.88	2.44	2.69	3.23
40	13.03	0.23	0.32	0.42	0.53	0.65	0.94	2.60	2.86	3.44
45	12.26	0.25	0.34	0.44	0.56	0.69	0.99	2.76	3.04	3.65
50	11.56	0.26	0.36	0.47	0.59	0.73	1.05	2.93	3.23	3.87
55	10.93	0.28	0.38	0.50	0.63	0.77	1.11	3.10	3.41	4.09
60	10.36	0.29	0.40	0.52	0.66	0.82	1.18	3.27	3.60	4.32
65	9.84	0.31	0.42	0.55	0.70	0.86	1.24	3.44	3.79	4.55
70	9.38	0.32	0.44	0.58	0.73	0.90	1.30	3.61	3.98	4.77
75	8.96	0.34	0.46	0.60	0.77	0.94	1.36	3.78	4.17	5.00
80	8.57	0.36	0.48	0.63	0.80	0.99	1.42	3.95	4.35	5.22

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3.5.13 Survey and profiling

Survey and profiling shall be carried out in accordance with the company's guidance document NSP/004/024. All proposed routes shall be surveyed to ensure conformance with company's design and clearance criteria.

The survey data together with the proposed pole type and position shall be presented back to the Company in an electronic design file format for inspection and approval.

The company uses the Optimal Software PoleCad package and will provide any necessary Cell or Conductor Libraries for use with the system.

In addition way-leaves shall be obtained and forwarded together with plans to a 1 : 10,000 scale to enable Form B, Section 37 DTI consents to be applied for. Upon completion of a satisfactory design and receipt of way-leaves and consents, line schedules shall be produced for construction and line record purposes. All CAD designs shall be given a unique file reference, related to the line feeder route number and archived.

When the line is ready for connection to the Companies distribution system, all network diagrams and asset databases shall be updated.

3.5.13.1 Conductor Design Checks

- a) Proposed lines shall be checked for possible conductor uplift conditions through the use of an unloaded, -6.0°C conductor catenary curve. The check shall ensure that 0.5m exists between the top of the insulator and the cold curve position.
- b) Where uplift conditions are designed out of the line, this process shall not result in excessive pole heights (ie under normal circumstances due to ground clearance requirements only it is envisaged that poles will not exceed 16m, poles in excess of this value shall be sectioned).
- c) Lines shall be checked for conformance with the clearance requirements of CE Electric UK guidance document NSP/004/011.

3.6 Inspection

All lines shall be inspected before commissioning.

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4 References

4.1 External Documentation

Reference	Title
ENA TS 43-8	Overhead Line Clearances
ESQCR	The Electricity Safety, Quality and Continuity Regulations 2002, SI 2665
BS EN 50341-1	Overhead electrical lines exceeding AC 45kV Part 1: General Requirements - Common Specifications
BS EN 50423-3-9	Overhead electrical lines exceeding 1 kV, Part 3: National Normative Aspects for Great Britain and Northern Ireland
BS 1990-1	Wood Poles for Overhead Power and Telecommunication Lines
ENA TS 43-88	Selection and treatment of wood poles and associated timber for overhead lines.
ENA TS 43-90	Anti-Climbing devices and Safety signs for HV lines up to and including 400kV.
ENA TS 43-91	Stay strands and stay fittings for overhead lines
ENA TS 43-93	Line insulators
ENA TS 43-94	Specification for Earth Rods and Connectors
ENA TS 43-95	Steelwork for overhead lines
ENA TS 43-96	Fasteners and washers for wood pole overhead lines
IEC 61109	Composite Insulators for a.c. Overhead lines with a nominal voltage greater than 1000V – definitions, test methods and acceptance criteria
IEC 61284	Overhead lines. Requirements and tests for fittings

4.2 Internal Documentation

Reference	Title
NPS/001/001	Technical Specification for Wood Poles and Associated Products for Overhead Lines
NPS/001/002	Technical Specification for Helical Products
NPS/001/005	Technical Specification for Steelwork, Conductor/Insulator Fittings and Stay Fittings for Overhead Lines and General Construction Works
NPS/001/006	Technical Specification for Insulators for Overhead Lines Up to And Including 132kV
NPS/001/007	Technical Specification for Conductors for Overhead Lines
NPS/001/010	Technical Specification for Notice Plates and Signs for Use on NEDL and YEDL Distribution Systems
NPS/001/011	Technical Specification for Fasteners And Fixings For Wood Pole Overhead Lines and General Construction Works
NPS/001/013	Technical Specification for Stay wire for Overhead Lines
NPS/001/015	Technical Specification for Barbed Wire
NPS/001/020	Technical Specification for Ground Anchor Systems used in Overhead Lines
NPS/002/001	Technical Specification for Earthing Materials

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Reference	Title
NSP/004/011	Overhead Line Clearances
NSP/004/101	(OHI 1) Guidance on Pole and Stay Holes
NSP/004/102	(OHI 2) Guidance on Erecting Single or 'H' poles
NSP/004/103	(OHI 3) Guidance on the type and fabrication of overhead line supports
NSP/004/104	(OHI 4) Guidance on the types and installation requirements for stays
NSP/004/105	(OHI 5) Guidance on the selection, erection and sagging of O/H line conductors
NSP/004/106	(OHI 6) Guidance on the selection and application of conductor joints, termination & binders
NSP/004/107	(OHI 7) Guidance on the selection of conductor jumpers and non-tension joints
NSP/004/108	(OHI 8) Guidance on the installation of compression joints
NSP/004/109	(OHI 9) Guidance on anti-climbing devices, safety signs and labels required on O/H lines
NSP/004/127	(OHI 27) Guidance on the selection and application of insulators and insulator assemblies.

5 Definitions

Term	Definition
Aeolian Vibration	Conductor Oscillation caused by low winds (1 m/s to 7 m/s) blowing steadily across the conductor.
Average Span	The arithmetic average of a number of spans in a line or section of line
Basic Span	The span length adopted for sag/tension calculations
Clashing	The electrical contact between two dissimilar line conductors of an overhead line displaced from their normal position by environmental forces such that the conductors are likely to touch.
Conductor Creep	Permanent long term elongation of the conductor
Conductor Downpull	The vertical loading imposed by conductors corresponding to a gradient measured between adjacent pole tops
Everyday Tension (EDT)	The design stress in an unloaded conductor at 5°C. Used in sag / tension calculations to limit harmful conductor vibrations
Freezing Point Tension (FPT)	The design tension of an unloaded conductor at 0°C in still air.
Ice Co-ordinate	The intercept on the ice axis of the weather incidence load line (See ENA TR 111) whose value is given for differing height and UK locations in the associated weather maps
Intermediate Support	A support in a straight run of line on which the conductors are supported on post insulators.
Maximum Conductor Pressure (MCP)	The maximum transverse component of applied conductor load when subjected to a wind load.
Maximum Conductor Weight (MCW)	The maximum vertical component of applied conductor load, including the weight of accreted ice, if present.

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Term	Definition
Maximum Span	The maximum permitted length of any span
Maximum Working Tension (MWT)	The absolute maximum conductor tension assessed at -5.6°C with wind and ice loading.
Minimum Failing Load	The minimum applied mechanical load that will cause failure by breakage of a component.
Over-tensioning	Excess tension applied above normal tension at time of erection to compensate for conductor creep.
Pre- tensioning	The tension treatment applied to a conductor for a short duration before final erection tension is established to remove a proportion of conductor creep.
Sag	The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured at mid-span
Section Angle Support	A support at which a line deviates and the conductors are made off on either side of the crossarm on tension insulator sets
Section Support	A support in a straight line on which the conductors are made off on either side of the crossarm on tension insulator sets
Span	The horizontal distance between adjacent supports. Individual spans will normally be within 20% of the chosen basic span
Wind Co-ordinate	The intercept on the wind axis of the weather incidence load line (See ENA TR 111) whose value is given for differing height and UK locations in the associated weather maps
Weather Zone	A geographical area in which the likely mean wind pressure and absolute maximum ice accretion thickness may be described by a numeral and letter respectively. The wind co-ordinate is described in 190N/m ² increments, whilst the ice co-ordinate is measured in 10mm diametric thickness increments.
Wind Loading Span	Half the sum of the spans adjacent to the support.

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6 Authority for Issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G.Hammel	Overhead Standards Manager	<<Signature>>	<<Date>>

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M.Nicholson	Head Of System Strategy	<<Signature>>	<<Date>>

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C.Holdsworth	Standards & Assessment Manager	<<Signature>>	<<Date>>

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	<<Signature>>	<<Date>>

6.5 Approval

Approval is given for the content of this document

		Sign	Date
Sean Caskey	Network Construction Manager	<<Signature>>	<<Date>>

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Neil Denley	Primary Engineering Projects Manager	<<Signature>>	<<Date>>

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Appendix A – Drawing List

200m Basic Span – Poplar Conductor

Drawing Number	Sheet No	Title
Pole Arrangement Drawings		
1 09 139 0016	1	132KV OVERHEAD LINES RUTTER TYPE POLE ARRANGEMENT. TYPICAL 17m POLE
1 09 139 0013	1	132KV OVER HEAD LINES ASSEMBLY OF SINGLE POLE 2°-20° DEVIATION
1 09 139 0014	1	132KV OVER HEAD LINE SINGLE POLE 20-40 DEGREE DEVIATION
1 09 139 0015	1	132KV OVERHEAD LINE ASSEMBLY OF SINGLE POLE 40-60° DEVIATION
1 09 139 0017	1	132KV OVERHEAD LINE RUTTER POLE TENSION ARRANGEMENT TYPICAL 17m MEDIUM POLE SHOW 0-2° DEVIATION
1 09 139 0018	1	132KV SINGLE CIRCUIT OVERHEAD LINES LARGE ANGLE STRUCTURE 2.9m CENTRES
1 09 139 0019	1	132KV SINGLE CIRCUIT OVERHEAD LINES TERMINAL STRUCTURE 2.9m CENTRES
Pole Fabrication Drawings		
1 09 139 0003	2	132KV SINGLE CIRCUIT OVER HEAD LINES DRILLING, SCARFING & MARKINGS FOR SINGLE ANGLE POLE ARRANGEMENT
1 09 139 0003	3	ADDITIONAL FABRICATION DETAILS FOR THE ATTACHEMENT OF THE HCL CLIMBING ACCESS SYSTEM TO DRAWING 1091390003 SHT 2, 1091390018 sht 2 & 1091390019 sht2
1 09 139 0008	2	132KV OVERHEAD LINES DRILLING, SCARFING & MARKINGS FOR RUTTER TYPE MEDIUM AND STOUT POLES FOR USE WITH TIMBER AND HEAVY DUTY STEEL SHEAR BLOCKS
1 09 139 0008	3	ADDITIONAL FABRICATION DETAILS FOR THE ATTACHEMENT OF THE HCL CLIMBING ACCESS SYSTEM TO DRAWING 1091390008 SHT 2
1 09 139 0018	2	132KV SINGLE CIRCUIT OVERHEAD LINES DRILL, SCARFING & MARKING FOR LARGE ANGLE H-POLE
1 09 139 0019	2	132KV SINGLE CIRCUIT OVERHEAD LINES DRILL, SCARFING & MARKING FOR TERMINAL H-POLE
Insulators & Insulator fittings		
1 09 101 0487	16	132KV COMPOSITE HORIZONTAL POST INSULATOR – TECHNICAL PARAMETERS
1 09 101 0487	18	132KV COMPOSITE VERTICAL LINE/PILOT POST POLYMER INSULATOR – TECHNICAL PARAMETERS
1 09 101 0487	29	132KV COMPOSITE TENSION / STAY INSULATOR (120KN WITH 20MM BALL / SOCKET COUPLING) – TECHNICAL PARAMETERS
1 09 101 0487	30	132KV COMPOSITE TENSION SET INSULATOR ASSEMBLY DRAWING – (NORMAL UNEARTHED SITUATIONS) FOR USE WITH DRAWING 1091010487 SHT 29
1 09 101 0487	31	132KV TENSION SET INSULATOR ASSEMBLY DRAWING – (APPROACH / EARTHED SITUATIONS) FOR USE WITH DRAWING 1091010487 SHT 29
1 09 101 0487	32	132KV COMPOSITE VERTICAL LINE/PILOT POST INSULATOR - ARCING HORN ASSEMBLY FOR (APPROACH / EARTHED SITUATIONS) FOR USE ON DRAWING 1091010487 SHT18

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Drawing Number	Sheet No	Title
1 09 101 0487	33	132KV COMPOSITE HORIZONTAL POST INSULATOR - ARCING HORN ASSEMBLY FOR (APPROACH/EARTHED SITUATIONS) FOR USE ON DRAWING 1091010487 SHT16
1 09 101 0487	34	SUPPORT STOOL FOR USE WITH COMPOSITE LINE POST INSULATORS ON DRAWING 1091010487 SHT 18 AT PILOT POSITIONS
Stays and Stay Fittings		
1 09 101 0372	2	132KV COMPOSITE STAY INSULATOR ASSEMBLY DRAWING (UNEARTHED ARRANGEMENTS C/W ARCING HORN ASSEMBLIES)
1 09 101 0429	8	132kv SINGLE CIRCUIT METHOD OF FIXING STAYS
1 09 101 0675	16	132kv OVERHEAD LINES STAY YOKE LEG DETAIL (1000 LONG x 65 x 8)
1 09 101 0675	24	132kv OVERHEAD LINES STAY YOKE ARRAY FOR 1, 2 & 3 STAY OPTIONS
1 09 101 0675	46	132kv OVERHEAD LINES 10° CRANKED STAY STRAP
1 09 101 0675	47	132kv OVERHEAD LINES 45° CRANKED STAY STRAP
1 09 101 0675	48	132kv OVERHEAD LINES 40° CRANKED STAY STAP FOR TERMINAL H POLE STAYS
Pole top steelwork		
1 09 101 0675	1	132kv OVERHEAD LINES POST INSULATOR SUPPORT / TENSION PLATE (565 LONG X 220 X 15)
1 09 101 0675	2	132kv OVERHEAD LINES CROSSARM 40-60 DEGREES (5260 LONG x 150 x 90 x 15)
1 09 101 0675	4	132kv OVERHEAD LINES CROSSARM BRACE (518 LONG x 50 x 50)
1 09 101 0675	5	132kv OVERHEAD LINES CROSSARM SUPPORT CHANNEL (1118 LONG x 230 x 75 CHANNEL)
1 09 101 0675	6	132kv OVERHEAD LINES CROSSARM 0-40 DEGREES (4860 LONG x 150 x 90 x 15)
1 09 101 0675	7	132kv OVERHEAD LINES SHORT CROSSARM FOR IN-LINE POLES (650 LONG x 150 x 90 x 15)
1 09 101 0675	8	132kv OVERHEAD LINES SIDE INSULATOR SUPPORT PLATE FOR IN-LINE POLES (409 LONG x 250 x 12)
1 09 101 0675	9	132kv OVERHEAD LINES INSULATOR SUPPORT PLATE BRACKET FOR IN-LINE POLES (250 LONG x 150 x 90 x 15)
1 09 101 0675	10	132kv OVERHEAD LINES SHORT XARM SUPPORT CHANNEL FOR IN-LINE POLES (1263 LONG x 230 x 74 CHANNEL)
1 09 101 0675	29	132KV OVERHEAD LINES M20 TIE ROD SCHEDULE
1 09 101 0675	30	132KV OVERHEAD LINES "RUTTER" TYPE POLE SHEAR BLOCK SCHEDULE
1 09 101 0675	31	132KV OVERHEAD LINES "RUTTER" POLE CROSSARM 0° - 2° DEVIATION (4860 X 150 X 90 X 15mm UNEQUAL ANGLE)
1 09 101 0675	32	132KV OVERHEAD LINES HORIZONTAL INSULATOR SUPPORT ANGLE (250 LONG x 100 x 75 x 12)
1 09 101 0675	34	ENHANCED ANTI-CLIMBING DEVICE FOR EXTRA STOUT POLE ASSEMBLY
1 09 101 0675	39	132kv OVERHEAD LINES RUTTER TYPE POLE HEAVY DUTY WELDED SHEAR BLOCK

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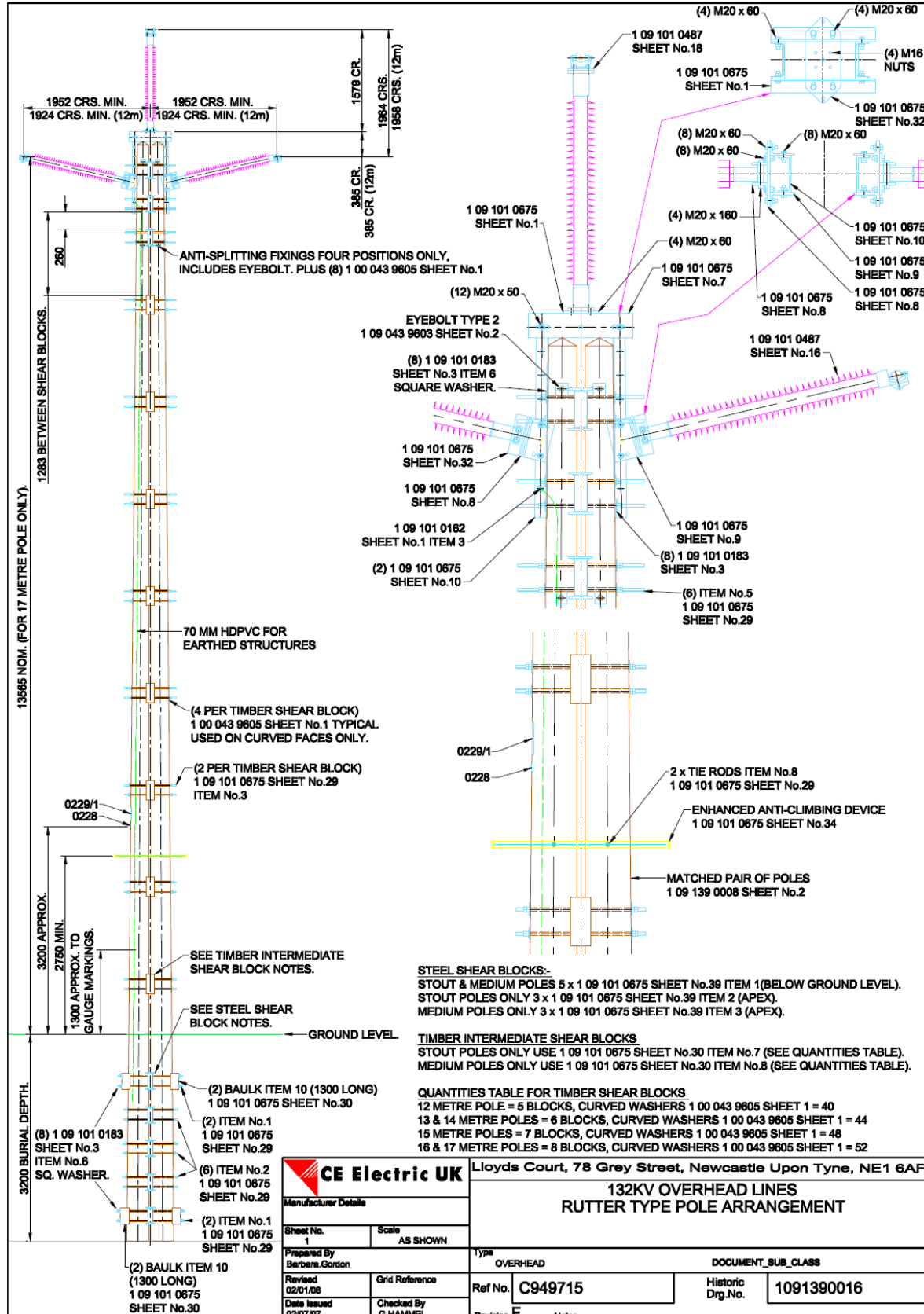
Drawing Number	Sheet No	Title
1 09 101 0675	40	132kV OVERHEAD LINES CROSSARM STRUT (1460 LONG x 50 x 50 x 5)
1 09 101 0675	50	132kV OVERHEAD LINES H POLE CROSSARM COMPONENTS
Miscellaneous Items		
1 09 101 0653	3	COMPRESSION ANCHOR CLAMP – POPLAR CONDUCTOR
1 09 101 0188	1	Vibration Dampers
1 09 101 0461	1	125kN Shackle

Appendix C – Self supporting steel masts - Drawing list

Drawing Number	Sheet No	Title
1 09 139 0034	1	132kV OVERHEAD LINES FABRICATED ASSEMBLY OF OCTAGONAL SECTION STEEL SUSPENSION POLES
1 09 139 0035	1	132kV OVERHEAD LINES FABRICATED ASSEMBLY OF OCTAGONAL SECTION STEEL POLES LINE DEVIATION 0° – 10°
1 09 139 0036	1	132kV OVERHEAD LINES FABRICATED ASSEMBLY OF OCTAGONAL SECTION STEEL POLES LINE DEVIATION 10° - 30°
1 09 139 0037	1	132kV OVERHEAD LINES FABRICATED ASSEMBLY OF OCTAGONAL SECTION STEEL POLES LINE DEVIATION 30° - 45°

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Arrangement Drawings and Materials Lists



CAUTION! - This document may be out of date if printed

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MATERIAL LIST FOR 1 09 139 0016 SHEET No.1 (Intermediate)

THE PAIRS OF TIMBER POLES MUST BE SELECTED 'MEDIUM' OR STOUT' & ITS LENGTH FROM THE LIST BELOW

<u>CAT. No.</u>	<u>Pole Height</u>	<u>Fabrication Drawing</u>
233986	12m Medium Pole	1 09 139 0008 Sht2 Item 'A Medium'
233987	13m Medium Pole	1 09 101 0008 Sht2 Item 'B Medium'
233988	14m Medium Pole	1 09 101 0008 Sht2 Item 'C Medium'
233989	15m Medium Pole	1 09 101 0008 Sht2 Item 'D Medium'
233990	16m Medium Pole	1 09 101 0008 Sht2 Item 'E Medium'
233991	17m Medium Pole	1 09 101 0008 Sht2 Item 'F Medium'
233992	12m Stout Pole	1 09 139 0008 Sht2 Item 'A Stout'
233993	13m Stout Pole	1 09 101 0008 Sht2 Item 'B Stout'
233994	14m Stout Pole	1 09 101 0008 Sht2 Item 'C Stout'
233995	15m Stout Pole	1 09 101 0008 Sht2 Item 'D Stout'
233996	16m Stout Pole	1 09 101 0008 Sht2 Item 'E Stout'
233997	17m Stout Pole	1 09 101 0008 Sht2 Item 'F Stout'

Note

Poles are supplied fabricated into structures using the materials listed below

<u>VARIABLE QUANTITIES</u>	<u>17M</u>	<u>16M</u>	<u>15M</u>	<u>14M</u>	<u>13M</u>	<u>12M</u>
Curved Washer 1 00 043 9605 Sht1	56	56	52	48	48	44
Wood Shear Blocks 1 09 101 0675 Sht 30 Item 7 (Stout Poles)	8	8	7	6	6	5
Wood Shear Block 1 09 101 0675 Sht 30 Item 8 (Medium Poles)	8	8	7	6	6	5

<u>QTY</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
5	Steel Shear Block (Below GL)	1 09 101 0675 Sht39 Item 1 (Med & Stout)
3	Steel Shear Block (Below GI)	1 09 101 0675 Sht39 Item 1 (Stout Only)
3	Steel Shear Block (Below GI)	1 09 101 0675 Sht39 Item 1 (Med Only)
4 * 234476	Wood Baulks (1300mm)	1 09 101 0675 Sht 30 Item 10
4	M20 Tie Rod (1200 Lg)	1 09 101 0675 Sht29 Item 1
6	M20 Tie Rod (980 Lg)	1 09 101 0675 Sht29 Item 2
16	M20 Tie Rod (940 Lg)	1 09 101 0675 Sht29 Item 3
6	M20 Tie Rod (700 Lg)	1 09 101 0675 Sht29 Item 5
* Where structures are installed in poor ground the top block type shall be changed as shown below		
2 234477	Wood Baulks (2500mm)	1 09 101 0675 Sht 30 Item 9

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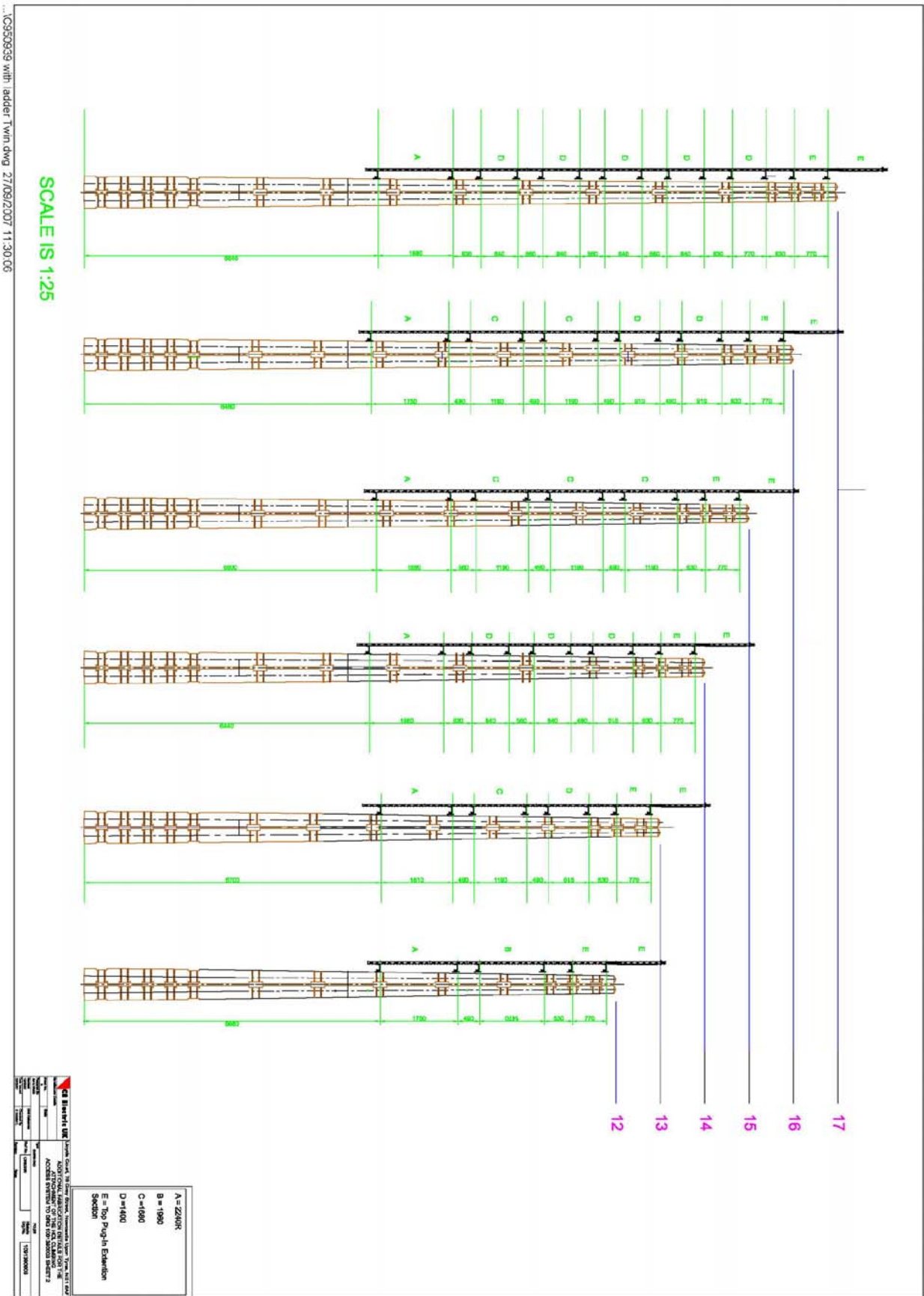
Additional materials to complete pole arrangement not associated with the pole fabrication

Materials Not supplied with Poles

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>	
1	346416	Vertical Post Insulator	1 09 101 0487 Sht 18	
1	346397	Arcing horn kit for vertical post insulator	1 09 101 0487 sht 32 (approach Ins only)	
2	346401	Horizontal Post Insulators	1 09 101 0487 Sht 16	
2	346398	Arcing horn kit for Horiz. post insulator	1 09 101 0487 sht 33 (approach Ins only)	
1	346474	Insulator/ Tension Plate	1 09 101 0675 Sht1	
2	346479	Short Crossarm	1 09 101 0675 Sht7	
2	346482	Crossarm Support Channel	1 09 101 0675 Sht10	
4	346481	Insulator Support Plate Bkt	1 09 101 0675 Sht9	
2	346480	Insulator Support Plate	1 09 101 0675 Sht8	
4	346501	Horiz. Ins. Support Angle	1 09 101 0675 Sht32	
1	To Suit	Pole Number Plate	1 09 101 0228 sht 3	
1	363318	Pole Safety Sign	1 09 101 0229 Sht1	
1	244981	Enhanced Anti-Climbing Device	1 09 101 0675 Sht34	(If Required)
2	346496	M16 Tie Rod (474 Lg)	1 09 101 0675 Sht29 Item 8	
3	346487	M20 Tie Rod (360 Lg)	1 09 101 0675 Sht29 Item 4	(Anti-Split)
1	36 8059	Eye Bolt Type 2	1 09 043 9603 Sht1	
8	368078	Square Curved Washer	1 00 043 9605 Sht1	
1	370504	Earthing Fastenings	1 09 101 0162 Sht1	
8	372816	Bolt Ms Hex Galv M20 X 160 Lg		
40	378703	Bolt Ms Hex Galv M20 X 60 Lg		
72	375616	Plain Washer Ms Galv M20		
36	378915	Nut Ms Hex Galv M20		
4	337630	Nut Ms Hex Galv M16		
4	378987	Plain Washer Ms Galv M16		



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MATERIAL LIST FOR 1 09 139 0017 SHEET No.1 (0° - 2° Deviation)

<u>CAT. No.</u>	<u>Pole Height</u>	<u>Fabrication Drawing</u>
233992	12m Stout Pole	1 09 139 0008 Sht2 Item 'A Stout'
233993	13m Stout Pole	1 09 101 0008 Sht2 Item 'B Stout'
233994	14m Stout Pole	1 09 101 0008 Sht2 Item 'C Stout'
233995	15m Stout Pole	1 09 101 0008 Sht2 Item 'D Stout'
233996	16m Stout Pole	1 09 101 0008 Sht2 Item 'E Stout'
233997	17m Stout Pole	1 09 101 0008 Sht2 Item 'F Stout'

Note

Poles are supplied fabricated into structures using the materials listed below

<u>VARIABLE QUANTITIES</u>	<u>17M</u>	<u>16M</u>	<u>15M</u>	<u>14M</u>	<u>13M</u>	<u>12M</u>
Curved Washer 1 00 043 9605 Sht1	56	56	52	48	48	44
Wood Shear Blocks 1 09 101 0675 Sht 30 Item 7 (Stout Poles)	8	8	7	6	6	5

<u>QTY</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
3	Steel Shear Block (Below GI)	1 09 101 0675 Sht39 Item 1 (Stout Only)
4 234476	Wood Baulks (1300mm)	1 09 101 0675 Sht 30 Item 10
4	M20 Tie Rod (1200 Lg)	1 09 101 0675 Sht29 Item 1
6	M20 Tie Rod (980 Lg)	1 09 101 0675 Sht29 Item 2
16	M20 Tie Rod (940 Lg)	1 09 101 0675 Sht29 Item 3
6	M20 Tie Rod (700 Lg)	1 09 101 0675 Sht29 Item 5

* Where structures are installed in poor ground the top block type shall be changed as shown below

2 234477	Wood Baulks (2500mm)	1 09 101 0675 Sht 30 Item 9
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Additional materials to complete pole arrangement not associated with the pole fabrication

Materials Not supplied with Poles

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
1	251550	Pilot Post Insulator + (support stool)	1 09 101 0487 sht 18
1	inc above	Pilot Insulator Support Stool (See above)	1 09 101 0487 sht 34
6	251553	120kn Ten. Ins. Assembly (132kv Approach)	1 09 101 0487 sht 31 or
6	251554	120kn Ten. Ins. Assembly (33 & 66kv Approach)	1 09 101 0487 sht 31 or
6	251552	120kn Ten. Ins. Assembly (33-132kv Normal)	1 09 101 0487 sht 30
6	243240	Poplar Comp. dead end c/w straight lug	1 09 101 0653 sht 3
3	346474	Insulator/ Tension Plate	1 09 101 0675 Sht1
4	346502	Crossarm Strut	1 09 101 0675 Sht40 Item 40
4	346476	Crossarm Brace	1 09 101 0675 Sht4
2	346500	Rutter Pole Crossarm	1 09 101 0675 Sht31
2	346482	Crossarm Support Channel	1 09 101 0675 Sht10
1	To Suit	Pole Number Plate	1 09 101 0228 Sht 3
1	363318	Pole Safety Sign	1 09 101 0229 Sht1

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1	244981	Enhanced Anti-Climbing Device	1 09 101 0675 Sht34	(If Required)
2	346496	M16 Tie Rod (474 Lg)	1 09 101 0675 Sht29 Item 8	
3	346487	M20 Tie Rod (360 Lg)	1 09 101 0675 Sht29 Item 4 (Anti-Split)	
1	36 8059	Eye Bolt Type 2	1 09 043 9603 Sht1	
8	368078	Square Curved Washer	1 00 043 9605 Sht1	
3	370504	Earthing Fastenings	1 09 101 0162 Sht1	
8	372816	Bolt Ms Hex Galv M20 X 160 Lg		
36	378703	Bolt Ms Hex Galv M20 X 60 Lg		
72	375616	Plain Washer Ms Galv M20		
36	378915	Nut Ms Hex Galv M20		
4	337630	Nut Ms Hex Galv M16		

CASCADE CONTAINMENT

The Following Items Are Required Per Pole To Prevent Cascade Effect On Unstayed Sections

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
2	346503	45° Cranked Stay Strap	1 09 101 0675 Sht46
2	226365	Shackle 28/29a	
2	337631	Bolt Ms Hex Galv M24 X 60 Lg c/w M24 Nut	
4	229195	Plain Washer Ms Galv M24	

Stays

See separate material list for drawing 1091010429 sht 8

- 2 Stays assemblies (using the double stay solution shown on the drawing above, but replacing the 10° cranked strap with the 45° cranked strap listed above) are to be installed in both directions against the direction of the overhead conductors.



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MATERIAL LIST FOR 1 09 139 0013 SHEET No.1 (2° - 20° Deviation)

Pole fabrication Details

233999	12m Extra Stout Pole	1 09 139 0003 Sht2 Item 'A'
234000	13m Extra Stout Pole	1 09 101 0003 Sht2 Item 'B'
234001	14m Extra Stout Pole	1 09 101 0003 Sht2 Item 'C'
234002	15m Extra Stout Pole	1 09 101 0003 Sht2 Item 'D'
234003	16m Extra Stout Pole	1 09 101 0003 Sht2 Item 'E'
234004	17m Extra Stout Pole	1 09 101 0003 Sht2 Item 'F'

Poles supplied complete with foundation blocks and fixings

2	235124	439103 Type 2 Blocks Or 1 X 1000 Dia. X 150 Thk. C25 Concrete Pad.
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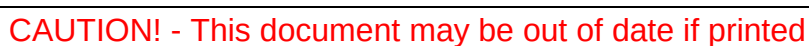
Materials Not supplied with Poles

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>	
2	251550	Pilot Post Insulator + (support stool)	1 09 101 0487 sht 18	
2	inc above	Pilot Insulator Support Stool (See above)	1 09 101 0487 sht 34	
6	251553	120kn Ten. Ins. Assembly 132kV Approach)	1 09 101 0487 sht 31	or
6	251554	120kn Ten. Ins. Assembly 33 & 66kV Approach)	1 09 101 0487 sht 31	or
6	251552	120kn Ten. Ins. Assembly (33-132kV Normal)	1 09 101 0487 sht 30	
6	243240	Poplar Comp. dead end c/w straight lug	1 09 101 0653 sht 3	
5	346474	Insulator/ Tension Plate	1 09 101 0675 Sht1	
2	346478	0-40° Crossarm	1 09 101 0675 Sht6	
4	346502	Crossarm Strut	1 09 101 0675 Sht40 Item 40	
4	346476	Crossarm Brace	1 09 101 0675 Sht4	
2	346477	Crossarm Support	1 09 101 0675 Sht5	
1	To Suit	Pole Number Plate	1 09 101 0228 Sht 3	
1	363318	Pole Safety Sign	1 09 101 0229 Sht1	
1	251362	Enhanced Anti-Climbing Device	1 09 101 0408 Sht 8 (If Required)	
1	346487	M20 Tie Rod (360 Lg)	1 09 101 0675 Sht29 Item 4 (Anti-Split)	
1	36 8059	Eye Bolt Type 2	1 09 043 9603 Sht1	
4	368078	Square Curved Washer	1 00 043 9605 Sht1	
1	370504	Earthing Fastenings	1 09 101 0162 Sht1	
36	378703	Bolt Ms Hex Galv M20 X 60 Lg		
60	375616	Plain Washer Ms Galv M20		
2	346493	M20 Tie Rod (700 Lg)	1 09 101 0675 Sht29 item 5	
8	337630	Nut Ms Hex Galv M16		
8	378987	Plain Washer Ms Galv M16		
20m	220146	70mm HD Green PVC Insulated earthing Conductor –		Earthed Poles
1	075234	70mm compression Lug (m20 clearance hole	1091010237	Earthed Poles

For details on stay requirements see the following detailed stay drawings:

* For 1 To 3 Stays (1 Per Stay) Refer To Drawing 1 09 101 0429 Sht8

* For 1 To 3 Stays Refer To Stay Yoke Array Drawing 1 09 101 0675 Sht24



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MATERIAL LIST FOR 1 09 139 0014 SHEET No.1 (20° - 40° Deviation)

Pole fabrication Details

233999	12m Extra Stout Pole	1 09 139 0003 Sht2 Item 'A'
234000	13m Extra Stout Pole	1 09 101 0003 Sht2 Item 'B'
234001	14m Extra Stout Pole	1 09 101 0003 Sht2 Item 'C'
234002	15m Extra Stout Pole	1 09 101 0003 Sht2 Item 'D'
234003	16m Extra Stout Pole	1 09 101 0003 Sht2 Item 'E'
234004	17m Extra Stout Pole	1 09 101 0003 Sht2 Item 'F'

Poles supplied complete with foundation blocks and fixings

2	235124	439103 Type 2 Blocks Or 1 X 1000 Dia. X 150 Thk. C25 Concrete Pad.
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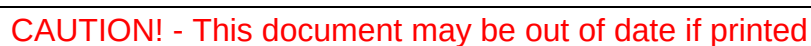
Materials Not supplied with Poles

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
2	251550	Pilot Post Insulator + (support stool)	1 09 101 0487 sht 18
2	inc above	Pilot Insulator Support Stool (See above)	1 09 101 0487 sht 34
6	251553	120kn Ten. Ins. Assembly (132kV Approachl)	1 09 101 0487 sht 31 or
6	251554	120kn Ten. Ins. Assembly (33 & 66kV Approachl)	1 09 101 0487 sht 31 or
6	251552	120kn Ten. Ins. Assembly (33 - 132kV Normal)	1 09 101 0487 sht 30
6	243240	Poplar Comp. dead end c/w straight lug	1 09 101 0653 sht 3
5	346474	Insulator/ Tension Plate	1 09 101 0675 Sht1
2	346478	0-40° Crossarm	1 09 101 0675 Sht6
4	346502	Crossarm Strut	1 09 101 0675 Sht40 Item 40
4	346476	Crossarm Brace	1 09 101 0675 Sht4
2	346477	Crossarm Support	1 09 101 0675 Sht5
1	To Suit	Pole Number Plate	1 09 101 0228 sht3
1	363318	Pole Safety Sign	1 09 101 0229 Sht1
1	251362	Enhanced Anti-Climbing Device	1 09 101 0408 Sht 8 (If Required)
1	346487	M20 Tie Rod (360 Lg)	1 09 101 0675 Sht29 Item 4 (Anti-Split)
1	36 8059	Eye Bolt Type 2	1 09 043 9603 Sht1
4	368078	Square Curved Washer	1 00 043 9605 Sht1
1	370504	Earthing Fastenings	1 09 101 0162 Sht1
36	378703	Bolt Ms Hex Galv M20 X 60 Lg	
60	375616	Plain Washer Ms Galv M20	
2	346493	M20 Tie Rod (700 Lg)	1 09 101 0675 Sht29 item 5
8	337630	Nut Ms Hex Galv M16	
8	378987	Plain Washer Ms Galv M16	
20m	220146	70mm HD Green PVC Insulated earthing Conductor –	Earthed Poles
1	075234	70mm compression Lug (m20 clearance hole	1091010237 Earthed Poles

For details on stay requirements see the following detailed stay drawings:

* For 1 To 3 Stays (1 Per Stay) Refer To Drawing 1 09 101 0429 Sht8

* For 1 To 3 Stays Refer To Stay Yoke Array Drawing 1 09 101 0675 Sht24



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MATERIAL LIST FOR 1 09 139 0015 SHEET No.1 (40° - 60° Deviation)

Pole fabrication Details

233999	12m Extra Stout Pole	1 09 139 0003 Sht2 Item 'A'
234000	13m Extra Stout Pole	1 09 101 0003 Sht2 Item 'B'
234001	14m Extra Stout Pole	1 09 101 0003 Sht2 Item 'C'
234002	15m Extra Stout Pole	1 09 101 0003 Sht2 Item 'D'
234003	16m Extra Stout Pole	1 09 101 0003 Sht2 Item 'E'
234004	17m Extra Stout Pole	1 09 101 0003 Sht2 Item 'F'

Poles supplied complete with foundation blocks and fixings

2	235124	439103 Type 2 Blocks Or 1 X 1000 Dia. X 150 Thk. C25 Concrete Pad.
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Materials Not supplied with Poles

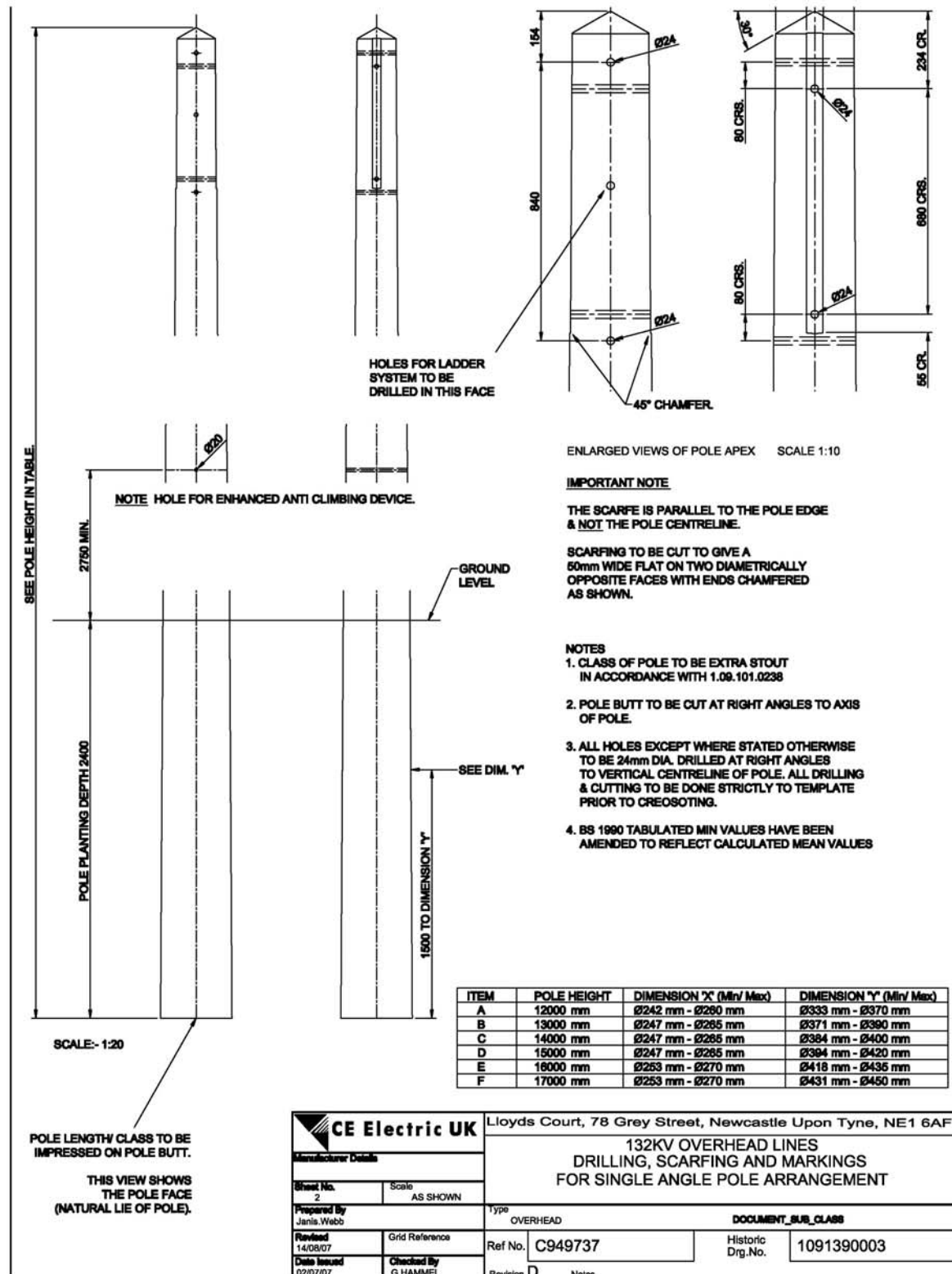
<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>	
2	251550	Pilot Post Insulator + (support stool)	1 09 101 0487 sht 18	
2	inc above	Pilot Insulator Support Stool (See above)	1 09 101 0487 sht 34	
6	251553	120kn Ten. Ins. Assembly (132kV Approach)	1 09 101 0487 sht 31	or
6	251554	120kn Ten. Ins. Assembly (33 & 66kV Approach)	1 09 101 0487 sht 31	or
6	251552	120kn Ten. Ins. Assembly (33-132kV Normal)	1 09 101 0487 sht 30	
6	243240	Poplar Comp. dead end c/w straight lug	1 09 101 0653 sht 3	
5	346474	Insulator/ Tension Plate	1 09 101 0675 Sht1	
2	346475	40-60° Crossarm	1 09 101 0675 Sht2	
4	346502	Crossarm Strut	1 09 101 0675 Sht40 Item 40	
4	346476	Crossarm Brace	1 09 101 0675 Sht4	
2	346477	Crossarm Support	1 09 101 0675 Sht5	
1	To Suit	Pole Number Plate	1 09 101 0228 sht3	
1	363318	Pole Safety Sign	1 09 101 0229 Sht1	
1	251362	Enhanced Anti-Climbing Device	1 09 101 0408 Sht8 (If Required)	
1	346487	M20 Tie Rod (360 Lg)	1 09 101 0675 Sht29 Item 4	(Anti-Split)
1	36 8059	Eye Bolt Type 2	1 09 043 9603 Sht1	
4	368078	Square Curved Washer	1 00 043 9605 Sht1	
1	370504	Earthing Fastenings	1 09 101 0162 Sht1	
36	378703	Bolt Ms Hex Galv M20 X 60 Lg		
60	375616	Plain Washer Ms Galv M20		
2	346493	M20 Tie Rod (700 Lg)	1 09 101 0675 Sht29 item 5	
8	337630	Nut Ms Hex Galv M16		
8	378987	Plain Washer Ms Galv M16		
20m	220146	70mm HD Green PVC Insulated earthing Conductor –		Earthed Poles
1	075234	70mm compression Lug (m20 clearance hole	1091010237	Earthed Poles

For details on stay requirements see the following detailed stay drawings:

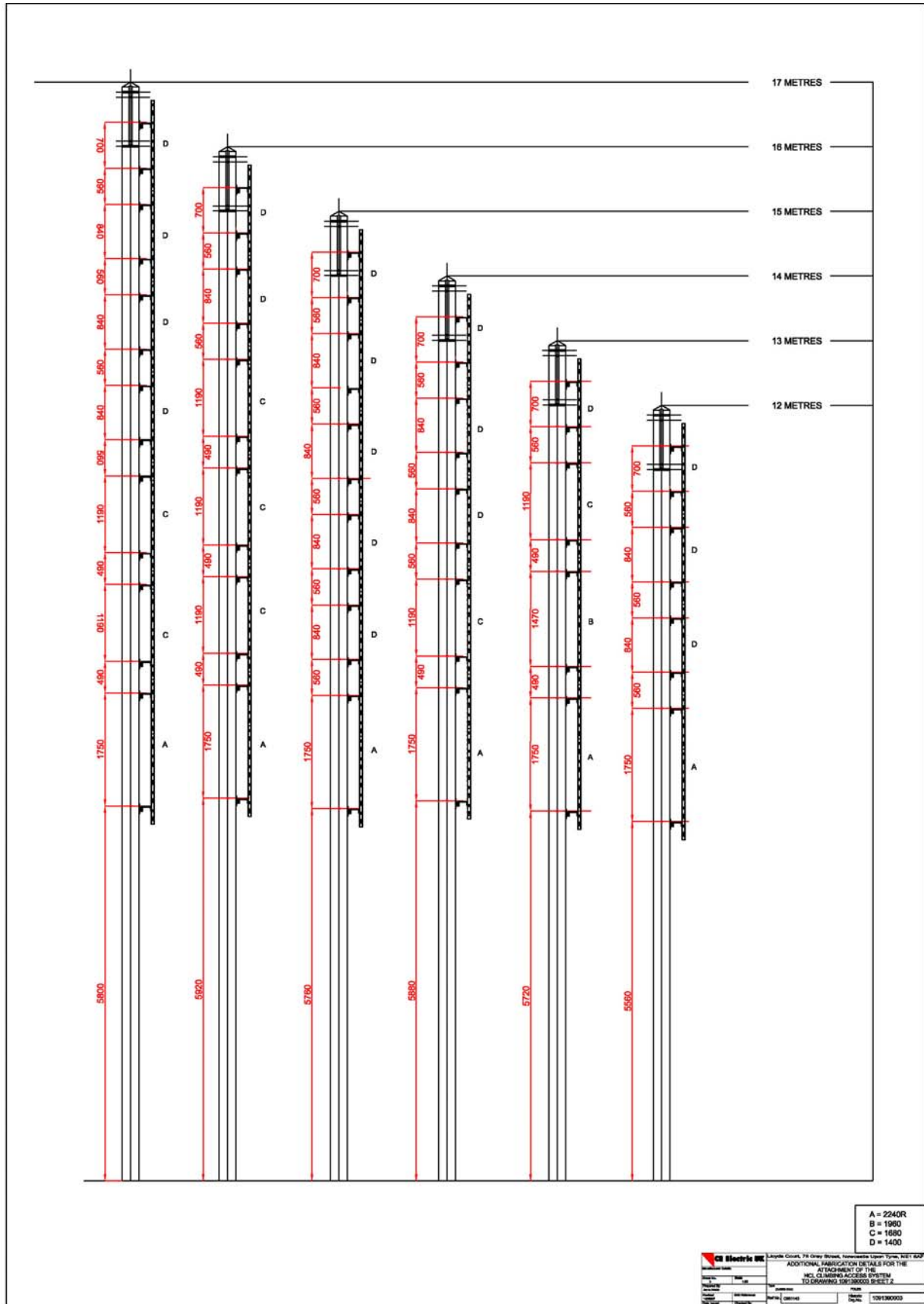
* For 1 To 3 Stays (1 Per Stay) Refer To Drawing 1 09 101 0429 Sht8

* For 1 To 3 Stays Refer To Stay Yoke Array Drawing 1 09 101 0675 Sht24

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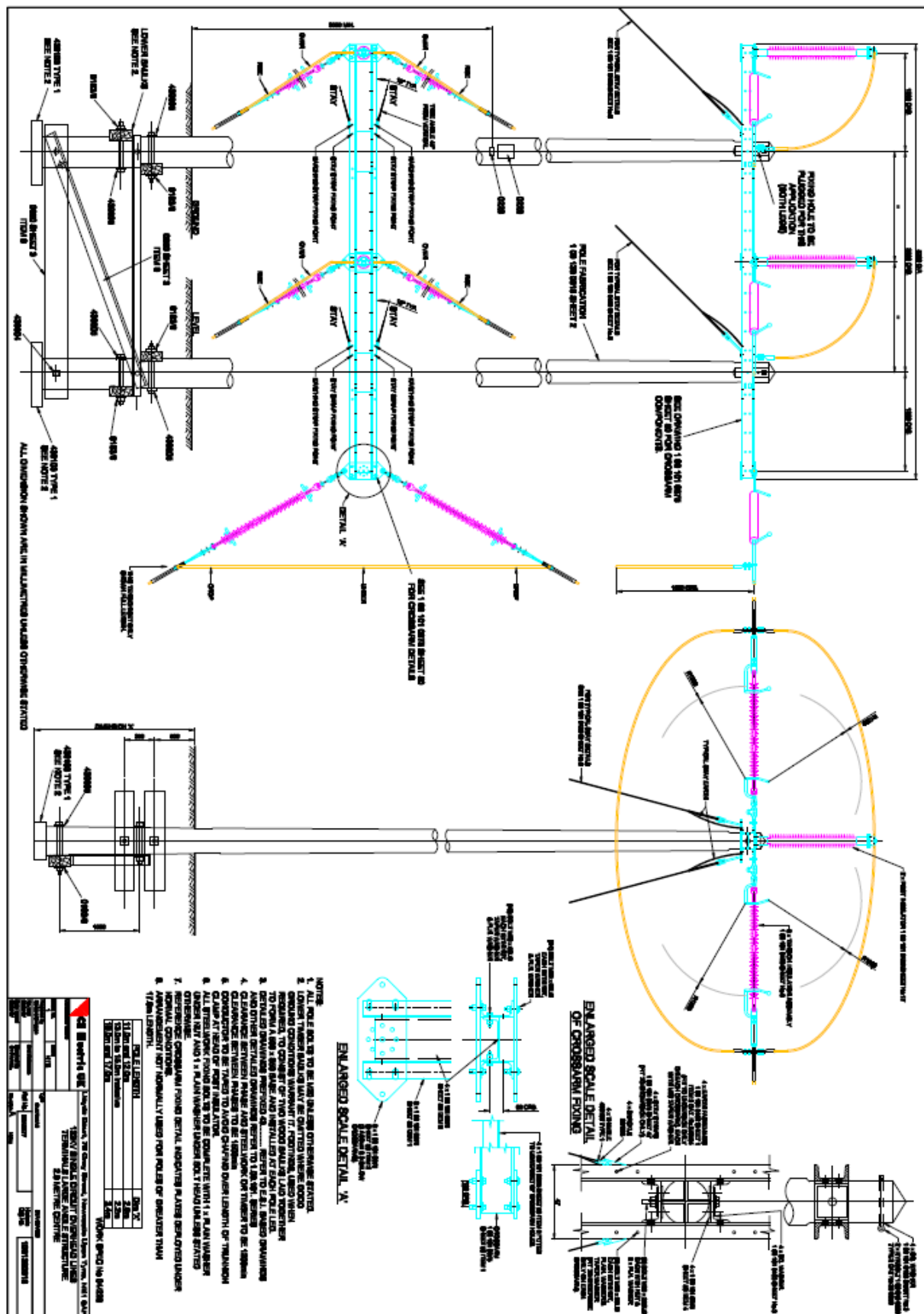
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MATERIAL LIST FOR 1 09 139 0018 SHEET No.1(40° - 60° Deviation)

Pole fabrication Details (H Pole Assembly select pole)

234005	12m E/Stout Pole	1 09 139 0018 Sht2
234006	13m E/Stout Pole	1 09 101 0018 sht2'
234007	14m E/Stout Pole	1 09 101 0018 sht2'
234008	15m E/Stout Pole	1 09 101 0018 sht2'
234009	16m E/Stout Pole	1 09 101 0018 Sht2
234010	17m E/Stout Pole	1 09 101 0018 Sht2 '

Note

Poles are supplied fabricated into structures using the materials listed below

<u>QTY</u>		<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
8	372815	Bolt Ms Hex Galv M20 X 530 Long	
2	375616	Plain Washer Ms Galv M20	
5	368078	Square Curved Washer	1 00 043 9605 Sht1
5	368063	Square Flat washers	1091010183 sht 3 item 6
2	240315	Foundation Braces	1091010650 sht 3 item 9
1	234507	Wood Block (3600 x 250 x 125mm)	1091010650 sht3 item 8
4	235124	Wood Blocks 439103 type 2	1000439103 type2

Materials Not supplied with Poles

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
2	251550	Pilot Post Insulator (+ support stool)	1 09 101 0487 Sht 18
2	inc above	Pilot Insulator Support Stool (See above)	1 09 101 0487 sht 34
6	251553	120kn Ten. Ins. Assembly (132kV Approach)	1 09 101 0487 Sht 31
6	251554	120kn Ten. Ins. Assembly (33 & 66kV Approach)	1 09 101 0487 Sht 31
6	251552	120kn Ten. Ins. Assembly (33-132kV Normal)	1 09 101 0487 Sht 30
6	243240	Poplar Comp. Dead End C/W Straight Lug	1 09 101 0653 sht 3
2	346507	Crossarm	1 09 101 0675 Sht 50 item 1
6	346508	Post Insulator & Tension Plate	1 09 101 0675 Sht 50 item 2
4	252670	Pole Cheek Plate	1 09 101 0675 Sht 50 item 4
2	372904	Bolt Ms Hex Galv M20 X 400 Long	
4	368078	Square Flat Washer	1 09 101 0183 Sht 3 item 6
4	240739	Square Plate For Item 4	1 09 101 0675 Sht 50 item 5
4	346510	Crossarm Spacer Channel	1 09 101 0675 Sht 50 item 6
4	368078	Square Curved Washer	1 00 043 9605 Sht1
2	368059	Eyebolt Type 2	1 00 043 9303 Sht 1
84	378703	Bolt Ms Hex Galv M20 X 60 Long	
88	374488	Taper Washer Galv M20	1 09 101 0183 item4
86	378915	Nut Ms Hex Galv M20	
88	375616	Plain Washer Galv M20	
4	346504	45 Degree Stay Strap	1 09 101 0675 Sht 47

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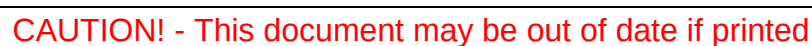
Stay Note:

4 STAYS GENERALLY TO DRAWING 1 09 101 0675 SHT8

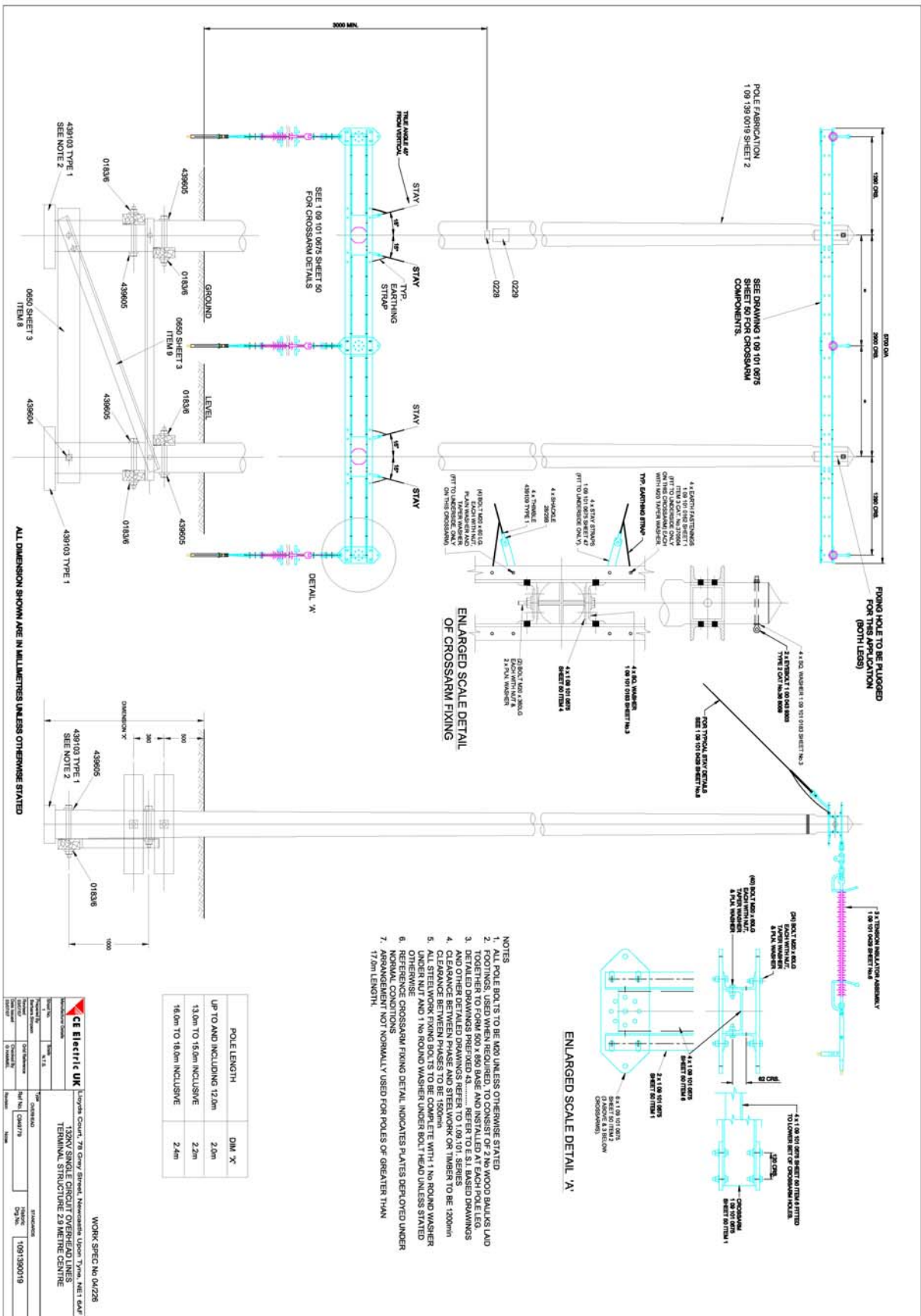
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4	370504	Earthing Fastenings	1 09 101 0162 Sht1	
1	243258	Pole Number Plate	1 09 101 0228 sht 3 (number to suit)	
1	(number to suit)	Pole Numbers	1 09 101 0228 sht 3	
2	363318	Pole Safety Sign	1 09 101 0229 Sht1	
1	251362	Enhanced Anti-Climbing Device	1 09 101 0408 Sht 8	(If Required)
4	375777	Brass Slotted Head screws 1" No8		
4	374581	Fibre washers (Thin)		
4	374577	Fibre washers (Thick)		
20m	220146	70mm HD Green PVC Insulated earthing Conductor –		Earthed Poles
1	075234	70mm compression Lug (m20 clearance hole	1091010237	Earthed Poles

Manufacturer Details		DRILLING, SCARFING AND MARKING FOR LARGE ANGLE 'H' POLE STRUCTURE	
Sheet No.	Scale		
	1:20		
Prepared By B. Barlow, Gordon	Grid Reference	Type OVERHEAD	STANDARDS
Revised 05/10/07	Checked By G. JAMES	Ref No.	Historic Dwg No.
		C949778	1091390018
Revision	Notes		
0			



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MATERIAL LIST FOR 1 09 139 0019 SHEET No.1(Terminal H)

Pole fabrication Details (H Pole Assembly select pole)

234005	12m Stout Pole	1 09 139 0019 Sht2
234006	13m Stout Pole	1 09 101 0019 sht2'
234007	14m Stout Pole	1 09 101 0019 sht2'
234008	15m Stout Pole	1 09 101 0019 sht2'
234009	16m Stout Pole	1 09 101 0019 Sht2
234010	17m Stout Pole	1 09 101 0019 Sht2 '

Note

Poles are supplied fabricated into structures using the materials listed below

<u>QTY</u>		<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
8	372815	Bolt Ms Hex Galv M20 X 530 Long	
2	375616	Plain Washer Ms Galv M20	
5	368078	Square Curved Washer	1 00 043 9605 Sht1
5	368063	Square Flat washers	1091010183 sht 3 item 6
2	240315	Foundation Braces	1091010650 sht 3 item 9
1	234507	Wood Block (3600 x 250 x 125mm)	1091010650 sht3 item 8
4	235124	Wood Blocks 439103 type 2	1000439103 type2

Materials Not supplied with Poles

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>	
6	251553	120kn Ten. Ins. Assembly (132kV Approach)	1 09 101 0487 Sht 31	Or
6	251554	120kn Ten. Ins. Assembly (33 & 66kV Approach)	1 09 101 0487 Sht 31	Or
6	251552	120kn Ten. Ins. Assembly (33 – 132kV Normal)	1 09 101 0487 Sht 30	
6	243240	Poplar Comp. Dead End C/W Straight Lug	1 09 101 0653 sht 3	
2	346507	Crossarm	1 09 101 0675 Sht 50 item 1	
6	346508	Post Insulator & Tension Plate	1 09 101 0675 Sht 50 item 2	
4	252670	Pole Cheek Plate	1 09 101 0675 Sht 50 item 4	
2	372904	Bolt Ms Hex Galv M20 X 400 Long		
4	368078	Square Flat Washer	1 09 101 0183 Sht 3 item 6	
4	240739	Square Plate For Item 4	1 09 101 0675 Sht 50 item 5	
4	346510	Crossarm Spacer Channel	1 09 101 0675 Sht 50 item 6	
4	368078	Square Curved Washer	1 00 043 9605 Sht1	
2	368059	Eyebolt Type 2	1 00 043 9303 Sht 1	
84	378703	Bolt Ms Hex Galv M20 X 60 Long		
88	374488	Taper Washer Galv M20	1 09 101 0183 item4	
86	378915	Nut Ms Hex Galv M20		
88	375616	Plain Washer Galv M20		
4	346504	40 Degree Stay Strap	1 09 101 0675 Sht 47	

Stay Note:

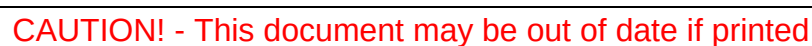
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4 STAYS GENERALLY TO DRAWING 1 09 101 0675 SHT8

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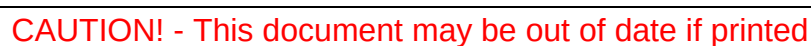
4	370504	Earthing Fastenings	1 09 101 0162 Sht1	
1	243258	Pole Number Plate	1 09 101 0228 sht 3 (number to suit)	
1	(number to suit)	Pole Numbers	1 09 101 0228 sht 3	
2	363318	Pole Safety Sign	1 09 101 0229 Sht1	
1	251362	Enhanced Anti-Climbing Device	1 09 101 0408 Sht 8	(If Required)
4	375777	Brass Slotted Head screws 1" No8		
4	374581	Fibre washers (Thin)		
4	374577	Fibre washers (Thick)		
20m	220146	70mm HD Green PVC Insulated earthing Conductor –		Earthed Poles
1	075234	70mm compression Lug (m20 clearance hole	1091010237	Earthed Poles



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Cable Termination Structures

- 33 or 66kV Cable Terminations shall be provided in accordance with drawing 1091231173 sht 4.
- 132kV Cable Terminations shall be provided in accordance with the drawing and material lists shown in ENA TS 43-50 fig 12 and the poles fabricated in accordance with figs 4 &



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33 - 66 KV 'H' TERMINAL

Based on Drg Nos 1.09.123.1173 Sheet 4

	DRAWING NO	QUANTITY	CAT NO
Poles			
12m Stout H Pole	1.09.138.0019 Sheet 2	1	234005
13m Stout H Pole	1.09.101.0019 Sheet 2	1	234006
14m Stout H Pole	1.09.101.0019 Sheet 2	1	234007
15m Stout H Pole	1.09.101.0019 Sheet 2	1	234008
16m Stout H Pole	1.09.101.0019 Sheet 2	1	234009
17m Stout H Pole	1.09.101.0019 Sheet 2	1	234010
Foundation Components			
Bolt Ms Hex Galv M20 X 530 Long	-	8	372815
Plain Washer Ms Galv M20	-	2	375616
Square Curved Washer	1 00 043 9605 Sht1	5	368078
Square Flat washers	1091010183 sht 3 item 6	5	368063
Foundation Braces	1091010650 sht 3 item 9	2	240315
Wood Block (3600 x 250 x 125mm)	1091010650 sht3 item 8	1	234507
Wood Blocks 439103 type 2	1000439103 type2	4	235124
Pole, Insulator and Conductor Fittings			
120kn Ten. Ins. Assembly (132kV Approach)	1 09 101 0487 Sht 31 <u>or</u>	6	251553
120kn Ten. Ins. Assembly (33 & 66kV Approach)	1 09 101 0487 Sht 31	6	251554
120kn Ten. Ins. Assembly (33 – 132kV Normal)	1 09 101 0487 Sht 30	6	251552
Poplar Comp. Dead End C/W Straight Lug	1 09 101 0653 sht 3	6	243240
Crossarm	1 09 101 0675 Sht 50 item 1	2	346507
Post Insulator & Tension Plate	1 09 101 0675 Sht 50 item 2	6	346508
Pole Cheek Plate	1 09 101 0675 Sht 50 item 4	4	252670
Bolt Ms Hex Galv M20 X 400 Long		2	372904
Square Flat Washer	1 09 101 0183 Sht 3 item 6		368078
Square Plate For Item 4	1 09 101 0675 Sht 50 item 5	4	240739
Crossarm Spacer Channel	1 09 101 0675 Sht 50 item 6	4	346510
Square Curved Washer	1 00 043 9605 Sht1	4	368078
Eyebolt Type 2	1 00 043 9303 Sht 1		368059
Bolt Ms Hex Galv M20 X 60 Long		84	378703
Taper Washer Galv M20	1 09 101 0183 item4	88	374488
Nut Ms Hex Galv M20		86	378915
Plain Washer Galv M20		88	375616
40 Degree Stay Strap	1 09 101 0675 Sht 47	4	346504
Earthing Fastenings	1 09 101 0162 Sht1	4	370504
Pole Number Plate	1 09 101 0228 sht 3	1	243258
Pole Numbers	1 09 101 0228 sht 3	1	(number to suit)
Pole Safety Sign	1 09 101 0229 Sht1	2	363318
Enhanced Anti-Climbing Device	1 09 101 0408 Sht 8	1	251362

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Brass Slotted Head screws 1" No8	-	4	375777
Fibre washers (Thin)	-	4	374581
Fibre washers (Thick)	-	4	374577
70mm HD Green PVC Insulated earthing Conductor –	-	20m	220146
70mm compression Lug (m20 clearance hole)	1091010237	1	075234
Stays (2 per leg)			
Pole Top Stay Dead End	0541/2	4	250209
7/4.00mm Grade 1150 Staywire	-	10	231481
Stay Dead End	0541/1	12	255217
Stay thimble	9109	12	244922
Polymeric Stay Insulator	0372/2	4	346596
66/132kv Stay Insulator Earthed	9107/2	2	253743
Type 2 Stay Rod	9101	4	231477
Wood Stay Block	9103/2	4	235124
Barbed wire	-	2	366829
66/132kv Stay Insulator Arcing Horn	0490/9	8	238014
Cable and Diverter Support			
Channel	0650 item 11	2	240508
Channel	0650 item 12	1	240512
Channel	0650 item 13	2	240289
Channel	0551/10	1	243421
Channel	0551/11	1	243493
Angle	0551/3	2	243578
Angle	0551/4	2	243582
SB Support Plates	0308 sheet 7 or 8	3	
Arrester Plates	0308/45	3	243987
Cleat Support	0309/7	3	245268
Bolt, M20 x 360		8	374609
Bolt, M16 x 50		46	378544
Bolt, M12 x 55 Earthing	0162/1	1	370487
Bolt, M12 x 40 Earthing	0162/2	3	370491
Washer Sq Curved	9605	8	368078
Washer, M16 Round		46	378987
Washer, M20 Round		8	375616
Washer, M16 Galvanised Tap		22	379852
Surge Arrester to suit Voltage		3	
Cable Support Cleat	0650 item 14	3	244640
Cable Sealing Bell Support Bars	0208 item 47	6	232732
Cable Sealing Bell Support Bars	0308 item 49	6	232747
Bolt, M12 x 40		24	378421
Cable Ladder			
Ladder Assy. To suit Pole Height	0550 sheet 2	1	To suit

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Bolts, M16 x 50		2	378544
Coach Screws ½ x 3"		2	372891
Baulk Type 2	9103	1	235124
Bolts, M20 x 530		1	372815
Washer, Flat Sq	0183/3	2	368063
Washer, M20 Round		2	375616
Flat Cleat Support	0308/22	5	242895
Flat Cleat Support	0309/37	1	245272
Bolts, M12 x 40		38	378421
Washer, M12 Round		38	375601

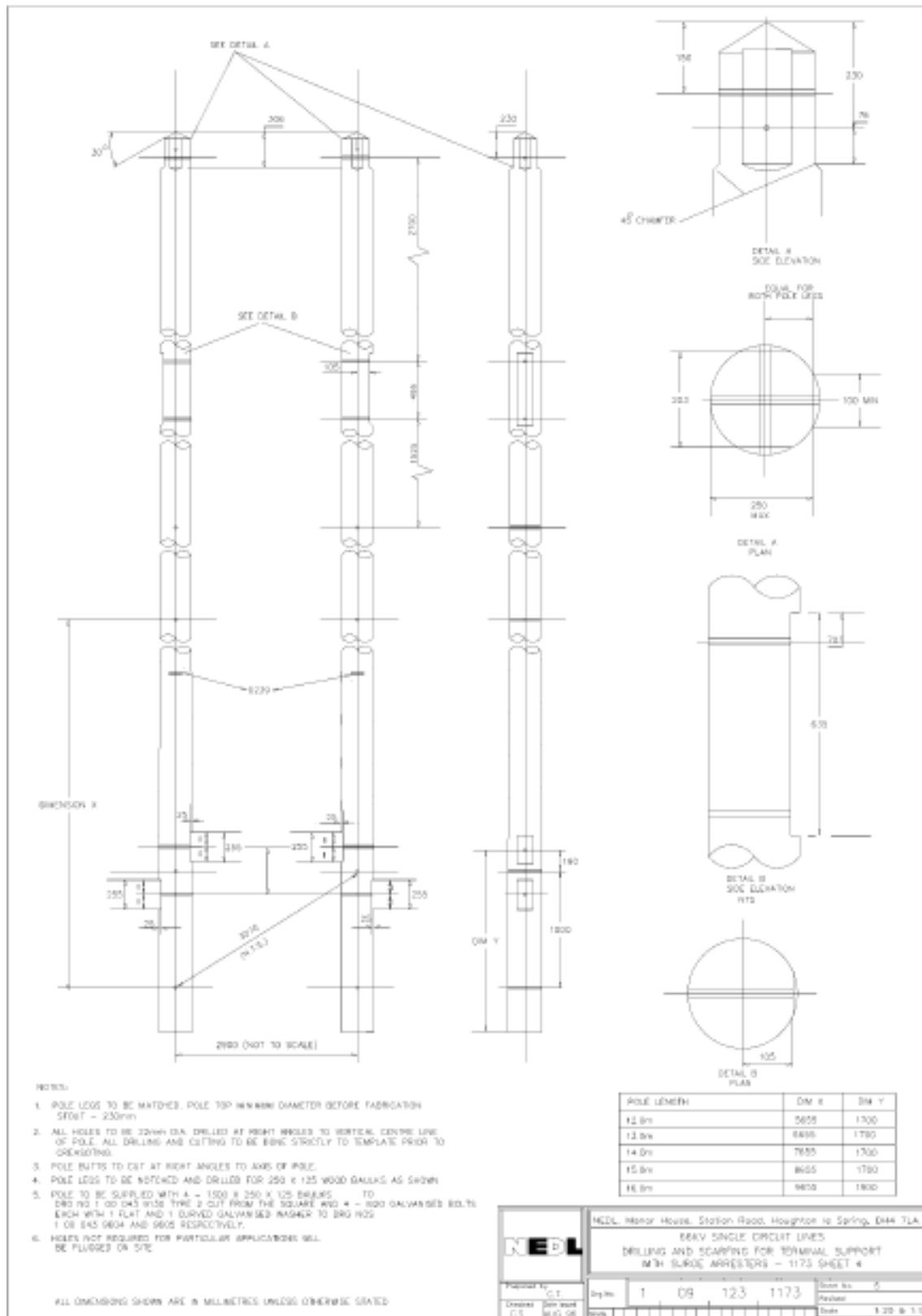
Protection

Cable Casing	0308/40	1	243084
Bolts, M16 x 40		18	378525
Bolts, M12 x 40		6	378421
Washer, M12 Round		6	375601
ACG	0408/12	1 pair	251362
ACG	0408/7	1 pair	243192
Bolt, M16 x 400		2	378671
Washer, M16 Round		20	378987
Steel Wire Galvanised		2m	330013
Barbed Wire		50m	366829
Staples, ¼ x 9G Galvanised		20	375599

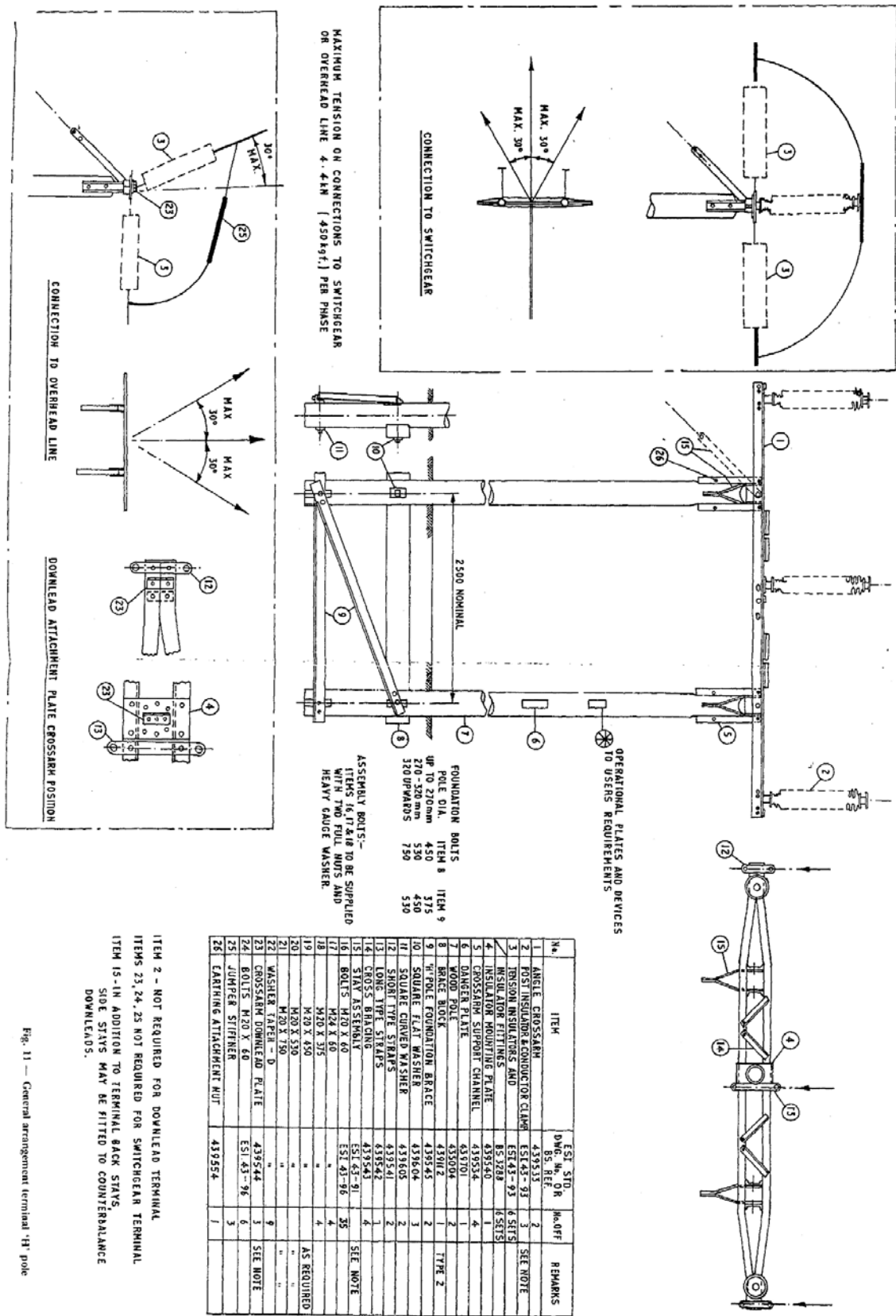
Earth Materials

Comp Lug – M22 Hole (70mm PVC Cu)	0237	1	268819
70mm (7/3.55)HD CU Cond. PVC Covered type 8	-	20m	224774
		Approx.	
Staples Copper 1½ x 8G	-	As Req.	374399
Earthing Lead casing	0149/3	1	235548
Earth Rod	3143	As Req.	354249
Rod Coupling	3143	As Req.	354361
Driving Bolt	3143	As Req.	354408
Saddle Clamp	3143/2	As Req.	354338

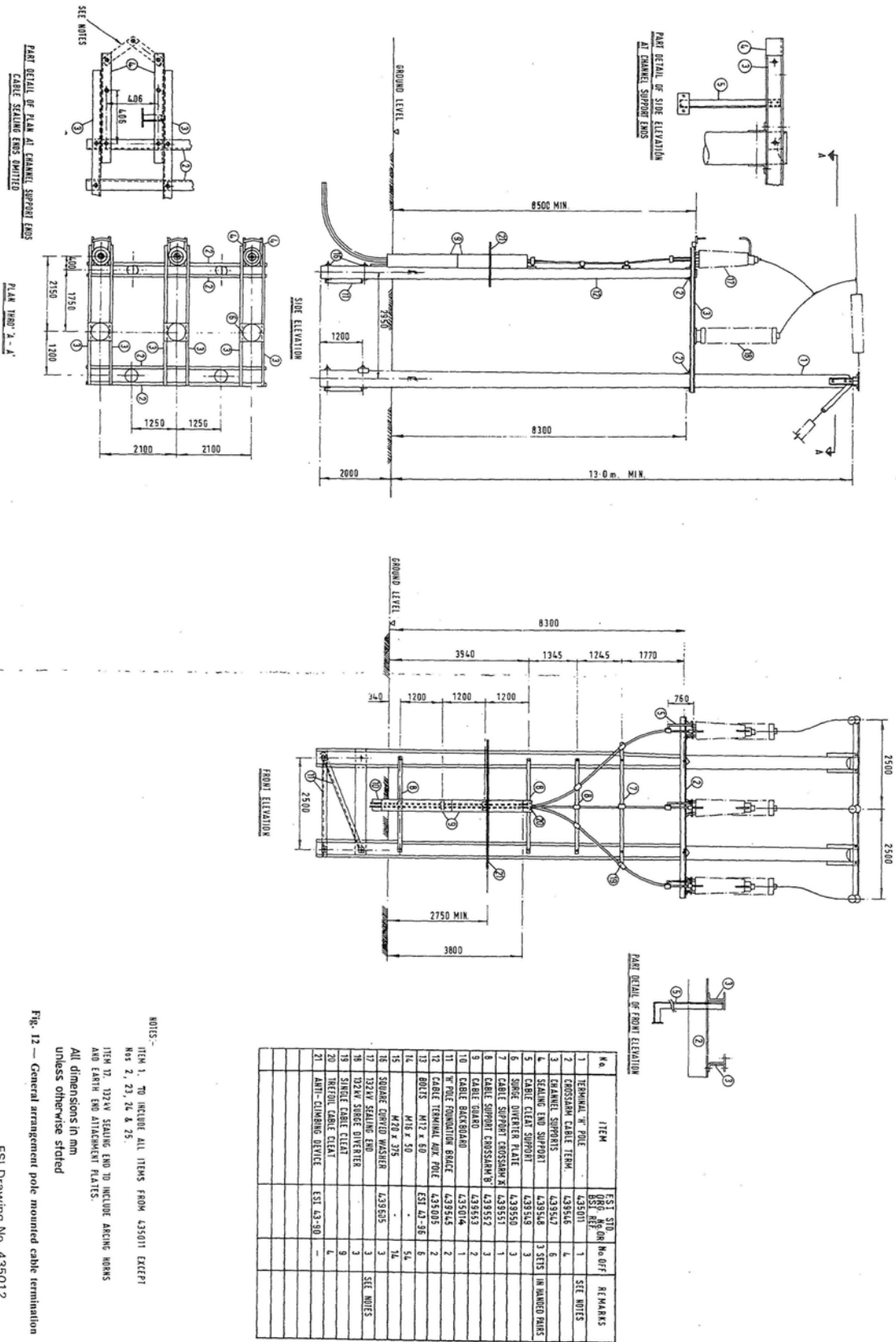
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132 KV 'H' Cable Terminal

Based on Drg Nos ENA TS 43-50 sht 12

Item Description	Item Ref	Drawing No	Quantity	Cat No
Materials for Terminal Support leg				
Wood pole	435011/7	ENA TS 435004		To Suit
Foundation Components				
Bolt Ms Hex Galv M20 X 530 Long	435011/18 – 435011/20	ENA TS 43-96	8	372815
Square Curved Washer	435011/11	1 00 043 9605 Sht1	5	368078
Square Flat washers	435011/10	1091010183 sht 3 item 6	5	368063
Foundation Braces	435011/9	ENA TS 439545	2	TBA
Wood Block (3600 x 250 x 125mm)	435011/8	ENA TS 439112 (Type 2)	1	TBA
Pole, Insulator and Conductor Fittings				
Angle Crossarm	435011/1	ENA TS 439533	2	TBA
Short Type Straps	435011/12	ENA TS 439541	2	TBA
Long Type Straps	435011/13	ENA TS 439542	1	TBA
Cross bracing	435011/14	ENA TS 439543	4	TBA
Insulator Mounting Plate	435011/4	ENA TS 439540	1	TBA
Crossarm Support Channel	435011/5	ENA TS 439534	4	TBA
Crossarm Downlead Plate	435011/23	ENA TS 439544	3	TBA
120kn Ten. Ins. Assembly (132kV Approach)	435011/3	1 09 101 0487 Sht 31	6	251553
Poplar Comp. Dead End C/W Straight Lug	435011/3	1 09 101 0653 sht 3	6	243240
Bolt Ms Hex Galv M20 X 60 Long	435011/16	ENA TS 43-96	36	378703
Bolt Ms Hex Galv M24 X 60 Long	435011/17	ENA TS 43-96	4	TBA
Washer Taper - D	435011/22	ENA TS 43-96	9	TBA
Bolt Ms Hex Galv M20 X 60 Long	435011/24	ENA TS 43-96	6	378703
Earthing & Bonding				
Earthing Fastenings	435011/26	1 09 101 0162 Sht1	4	370504
70mm HD Green PVC Insulated earthing Conductor –	-	-	20m	220146
70mm compression Lug (m20 clearance hole)	-	1091010237	1	075234
Signs and Notices				
Pole Safety Sign	435011/6	1 09 101 0229 Sht1	2	363318
Pole Number Plate	435011/6	1 09 101 0228 sht 3	1	243258
Pole Numbers	435011/6	1 09 101 0228 sht 3	1	to suit)
Brass Slotted Head screws 1" No8	435011/6	-	4	375777
Fibre washers (Thin)	435011/6	-	4	374581
Fibre washers (Thick)	435011/6	-	4	374577
Enhanced Anti-Climbing Device		1 09 101 0408 Sht 8	1	251362

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Item Description	Item Ref	Drawing No	Quantity	Cat No
Stays (2 per leg)	435011/15			
190kN Heavy Duty Stay Straps (pair)	435011/15	1000439113	2	252647
Cranked Links (190kN) - 1091010469 sht 2	435011/15	BS 3288 ref 42/87	3	
Bolt Ms Hex Galv M24 X 60 Long Thread length 36mm with two round washers	435011/15	ENA TS 43-96	4	TBA
Bolt Ms Hex Galv M20 X 45 Long Thread length 36mm with two round washers	435011/15	ENA TS 43-96	2	378703
190KN Shackle	435011/15	BS 3288 ref 42/103	4	250938
190kN Yoke Plate	435011/15	BS 3288 ref 42/42	2	TBA
120kN Insulator Stay Assembly	435011/15	1 09 123 0372 Sht2	4	251547
Stay thimble	435011/15	1000439109 type 1	8	244922
7/4.00mm Grade 1150 Staywire	435011/15	-	20	231481
Stay Dead End	435011/15	0541/1	8	255217
Type 2 Stay Rod	435011/15	9101	4	231477
Wood Stay Block	435011/15	9103/2	4	235124
notes items 16, 17 & 18 to be supplied with two full nuts and a washer				
item 2 not required for download terminal				
items 23,24 and 25 not req for switchgear terminals				
item 15 - in addition to terminal back stays, side stays may be fitted to counterbalance downloads				
Cable Support Pole				
Cable Terminal Aux Pole Fabrication Drawing	435012/12	ENA TS 435005	2	TBA
Materials for Cable Support leg				
Crossarm - cable termination	435012/2	ENA TS 439546	4	TBA
Channel Supports	435012/3	ENA TS 439547	6	TBA
Sealing End Support	435012/4	ENA TS 439548	3 sets	TBA
Cable Cleat Support	435012/5	ENA TS 439549	3	TBA
Surge Diverter plate	435012/6	ENA TS 439550	3	TBA
Cable suport Crossarm (a)	435012/7	ENA TS 439551	1	TBA
Cable support crossarm (b)	435012/8	ENA TS 439552	3	TBA
Cable Guard	435012/9	ENA TS 439553	1	TBA
Cable backboard	435012/10	ENA TS 435014	1	TBA
H Pole Foundation Brace	435012/11	ENA TS 439545	2	TBA
Bolt Ms Hex Galv M12 X 60 Long	435012/13	ENA TS 43-96	6	TBA
Bolt Ms Hex Galv M16 X 50 Long	435012/14	ENA TS 43-96	54	TBA
Bolt Ms Hex Galv M20 X 375 Long	435012/15	ENA TS 43-96	14	TBA
Square Curved Washer	435012/16	1 00 043 9605 Sht1	3	TBA

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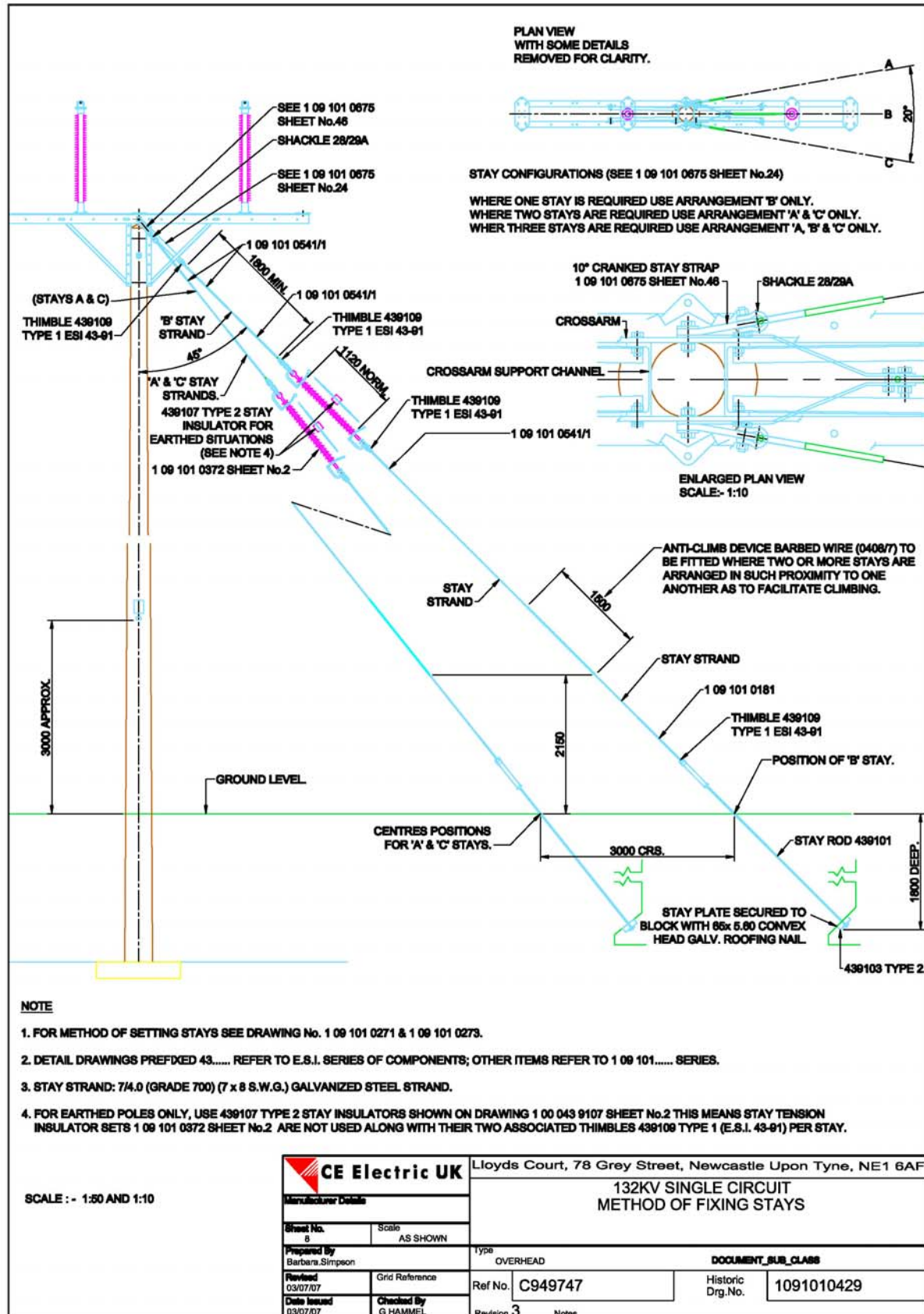
Item Description	Item Ref	Drawing No	Quantity	Cat No
132kV sealing End	435012/17		3	TBA
132kV Surge Diverter	435012/18		3	368078
Single cable Cleat	435012/19	1091010309 item 1	9	TBA
Trfoil cable cleat	435012/20	1091010309 item 5	4	TBA
Earth Materials				
Comp Lug – M22 Hole (70mm PVC Cu)		1091010237	1	075234
70mm (7/3.55)HD CU Cond. PVC Covered type 8		-	20m Approx.	220146
Polyprop cleats		-	As Req.	248815
Earthing Lead casing		1091190149/3	1	235548
Earth Rod		3143	As Req.	354249
Rod Coupling		3143	As Req.	354361
Driving Bolt		3143	As Req.	354408
Saddle Clamp		1091193143/2	As Req.	354338
Item 17 . 132kV sealing end to include arcing horns and earth end attachment plates				
Protection				
Anti Climbing Device - cable trunking	435012/21	1091010408 sht 11 item 7	2	243192
Anti Climbing Device - Cable Pole leg	435012/21	1090100408 sht 8	2	251362
Barbed Wire			50m	366829
Signs and Notices				
Pole Safety Sign		1 09 101 0229 Sht1	2	363318
Brass Slotted Head screws 1" No8		-	4	375777
Fibre washers (Thin)		-	4	374581
Fibre washers (Thick)		-	4	374577

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MATERIAL LIST FOR 1 09 101 0429 SHEET No.8 REV 1

IF ONE STAY IS USED THE FOLLOWING ITEMS ARE REQUIRED: -

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
1	253743	Single Stay Insulator Type 2 Earthed Poles	1 00 043 9107 Sht2 Or
1	346596	Stay Insulator Assembly Unearthed Poles	1 09 123 0372 Sht2
4	244922	Stay Thimble	1000439109 Type 1
4	255217	Helical Stay Dead End	1091010541
1	231477	Stay Rod	1000439101 Type 2 Item 3
3	337632	Bolt Ms Hex Galv M24 X 80 Long c/w full nut	
3	229195	Plain Washer Galv M24	
1	378703	Bolts MS Hex Galv M20 x 60 long	
1	375616	Washers, M20, round gald.	
Notes for more details See		STAY YOKE ARRAY (SEE DRAWING)	1 09 101 0675 SHT24
2	346483	Stay Yoke leg	1091010675 sht 16
1	235124	Stay Block 439103 type 2	1000439103 type 2

IF TWO STAYS ARE USED THE FOLLOWING ITEMS ARE REQUIRED: -

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
2	346503	10° Cranked Stay Strap	1091010675 Sht46
2	226365	Shackle BS3288 28/29a	1091010461 sht1
2	253743	Single Stay Insulator Type 2 Earthed poles	1 00 043 9107 Sht2 Or
1	346596	Stay Insulator Assembly Unearthed Poles	1 09 123 0372 Sht2
8	244922	Thimble	1000439109 Type 1
8	255217	Helical Stay Dead End	1091010541
2	231477	Stay Rod	1000439101 Type 2 Item 3
2	337632	Bolt Ms Hex Galv M24 X 80 Long c/w full nut	
2	229195	Plain Washer Galv M24	
1	235124	Stay Block 439103 type 2	1000439103 type 2

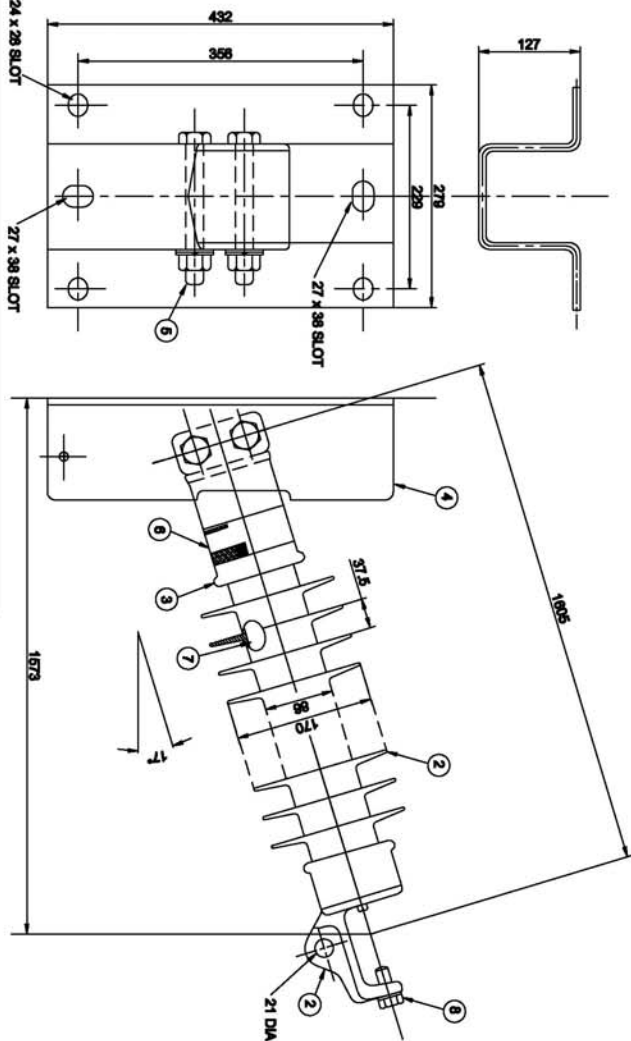
IF THREE STAYS ARE USED THE FOLLOWING ITEMS ARE REQUIRED: -

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
3	253743	Single Stay Insulator Type 2 Earthed Poles	1 00 043 9107 Sht2 Or
1	346596	Stay Insulator Assembly Unearthed Poles	1 09 123 0372 Sht2
2	346503	10° Cranked Stay Strap	1091010675 Sht46
2	226363	Shackle BS3288 28/29a	1091010461 sht1
2	346483	Stay Yoke leg	1091010675 sht 16
12	244922	Stay Thimble	1000439109 Type 1
12	255217	Helical Stay Dead End	1091010541
3	231477	Stay Rod	1000439101 Type 2 Item 3
3	337632	Bolt Ms Hex Galv M24 X 80 Long c/w full nut	
3	229195	Plain Washer Galv M24	
1	378703	Bolts MS Hex Galv M20 x 60 long	
1	375616	Washers, M20, round gald.	
1	235124	Stay Block 439103 type 2	1000439103 type 2

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ITEM	COMPONENT	MATERIAL
1	HORIZONTAL CLAMP TOP	DUCTILE IRON, HDG
2	WEATHERSHEDS	SILICONE RUBBER
3	BOX END	DUCTILE IRON, HDG
4	BENDABLE FLAT BASE	STEEL, HDG
5	BOLT SET	STEEL, HDG
6	LABEL	POLYESTER
7	CORE	HIGH QUALITY PULTRUDED FRP ROD
8	TRUNNION BOLT SET	STEEL, HDG

1. DIMENSIONS ARE IN METRIC (mm)
2. MAXIMUM DESIGN CANTILEVER LOAD IS 50% OF THE MINIMUM AVERAGE CANTILEVER BREAKING LOAD
3. TOLERANCES ARE IN ACCORDANCE WITH IEC 61862
4. POSITIVE TOLERANCE TO LEADAGE (CREEPAGE) DISTANCE IS NOT LIMITED
5. CLAMP TOP DIMENSIONS ARE IN ACCORDANCE WITH ANSI C29.7
6. INSULATORS ARE TO BE SUPPLIED WITH CONDUCTOR CLAMPS SUITABLE FOR 200mm MMC
7. REMOVABLE BASE NOT BE SUPPLIED WHEN ORDERED FOR USE ON WOOD-ORIENTED MAST JOBS



CAT. No	251548	WOODHOUSE MAST LINES NO RENDABLE BASE
CAT. No	251549	CHL9 LINES TO INCLUDE RENDABLE BASE

TECHNICAL DATA
Specification Applied: IEC 61852
1 tested herein

CHARACTERISTICS

DIMENSIONS

Adding Distance, mm
Leakage (Creepage) Distance, mm
Number of Shields, "N"
Core Diameter, mm

RATING

1318
3825
34
762

MECHANICAL VALUES

Average Cantilever Failing Load, kN

APPROX. NET WEIGHT, kg

44.4

CHARACTERISTICS

ELECTRICAL VALUES

Wet Power-Frequency Withstand Voltage, kV
Dry Lightning Impulse Withstand Voltage, kV

RATING

360
700

		Manufacturer's Details: F.A.S.1352 ALLED INSULATORS LTD	
Sheet No.	Scale 1:5		
Prepared By J. Allen, Wexho			
Revised 03/06/07	Grid Reference		
Date Issued By 22/03/07	Checked By G. HAMMILL		
Type OVERHEAD		STANDARDS	
Ref No. CS46738	Historic Dwg No.	1091010487	
Revision A	Notes		

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ITEM	COMPONENT	MATERIAL
1	FLANGE END	DUCTILE IRON, HDG
2	WEATHERSHEDS	SILICONE RUBBER
3	FLANGE END	DUCTILE IRON, HDG
4	LABEL	POLYESTER
5	CORE	HIGH QUALITY PULTRUDED FRP ROD
6	VERTICAL CLAMP ADAPTOR (LINDSEY#2121)	DUCTILE IRON, HDG
7	TRUNNION BOLT SET	STEEL, HDG
8	BOLT SET (LE326-P-1,2,3)	STAINLESS STEEL OR STEEL, HDG

CHARACTERISTICS

DIMENSIONS

Arching Distance, mm
Leakage (Creepage) Distance, mm
Number of Sheds, N°
Core Diameter, mm

MECHANICAL VALUES

Maximum Design Cantilever Load, kN (Without Vertical Clamp Adaptor)
Routine Tension-Load, kN (Without Vertical Clamp Adaptor)
Average Cantilever Failing Load, kN (Without Vertical Clamp Adaptor)

APPROX. NET WEIGHT, kg

CHARACTERISTICS

ELECTRICAL VALUES

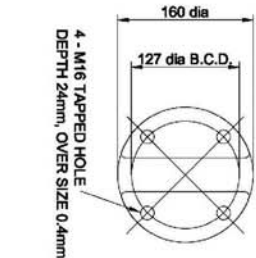
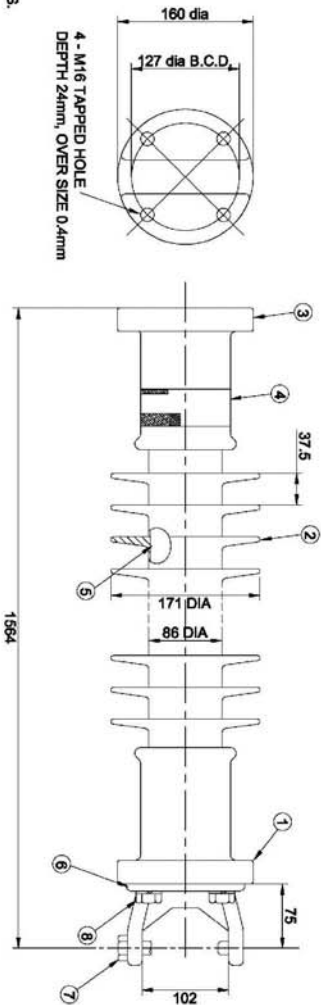
Wet Power Frequency Withstand Voltage, kV
Dry Lightning Impulse Withstand Voltage, kV

RATING

1168
3376
30
76.2

RATING

36.74
350
650



NOTES.

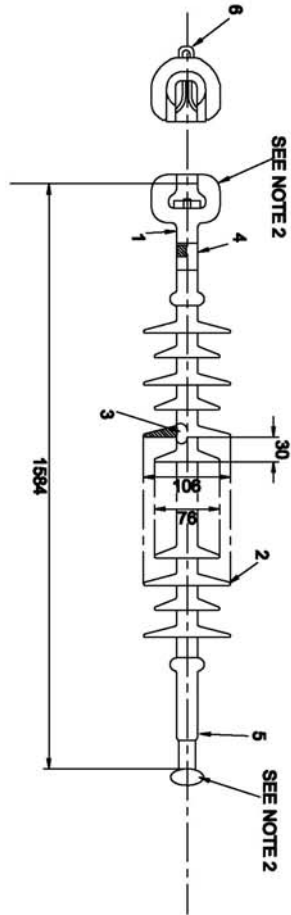
1. DIMENSIONS ARE IN METRIC (mm)
2. MAXIMUM DESIGN CANTILEVER LOAD IS 50% OF THE MINIMUM AVERAGE CANTILEVER BREAKING LOAD
3. TOLERANCES ARE IN ACCORDANCE WITH IEC 61952
POSITIVE TOLERANCE TO LEAKAGE (CREEPAGE)DISTANCE IS NOT LIMITED
4. ELECTRICAL TEST : IEC 61952 - 2002
5. BOTTOM FLANGE SHOULD BE INSTALLED BY 4 BOLTS OF THE FOLLOWING MATERIAL IN ORDER TO WITHSTAND AVERAGE CANTILEVER FAILING LOAD
6. WHERE USED AS A PILOT INSULATOR THIS UNIT MUST BE USED WITH A SUPPORT STOOL SEE DRAWING 1091010487 SHEET 34
7. TO BE SUPPLIED COMPLETE WITH CONDUCTOR TRUNNION CLAMPS SUITABLE FOR 200mm AAC
- MATERIAL : SAE GRADE 9 (ISO 898 - 1:1999, CLASS 12.9) OR EQUIVALENT, GALVANISED

CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details F.S.S.4134	ALLIED INSULATORS LTD	132KV COMPOSITE VERTICAL POST / PILOT (127MM PCD PEDASTEL BASE)	
Sheet No. 18	Scale 1:5	Type OVERHEAD	
Prepared By Jervis Webb	Grid Reference	Ref No. C946788	Historic Drg No. 1091010487
Revised 03/06/07	Checked By G HAMMILL	Revision B	Notes
Date Issued 23/03/07			

TECHNICAL DATA
Specification Applied : IEC 61952
Latest Issue

CAT. No 251551 SUPPLIED WITHOUT STOOL
CAT. No 251550 SUPPLIED WITH SUPPORT STOOL
AND ARMOUR GRIP CLAMPS

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ITEM	COMPONENTS	MATERIAL	REQ.
1	SOCKET FITTING	HIGH GRADE F.S. OR D.I. HDG	1
2	WEATHERSHIELDS	SILICONE RUBBER	1
3	CORE	HIGH QUALITY PULTRUDED FRP ROD	1
4	LABEL	POLYESTER	1
5	BALL FITTING	HIGH GRADE FORGED STEEL, HDG	1
6	SPLIT PIN	STAINLESS STEEL	1

TECHNICAL DATA
SPECIFICATION APPLIED : IEC81108, LATEST ISSUE
CHARACTERISTICS

DIMENSIONS

ARCING DISTANCE mm
LEAKAGE (CREEPAGE) DISTANCE mm
NUMBER OF SHEDS "N"
CORE DIAMETER mm

MECHANICAL VALUES

**SPECIFIED MECHANICAL LOAD IN
ROUTINE TENSION LOAD IN**

APPROX. NET WEIGHT. kg

ELECTRICAL VALUES

**WET POWER FREQUENCY WITHSTAND VOLTAGE (KV)
DRY LIGHTNING IMPULSE WITHSTAND VOLTAGE (KV)**

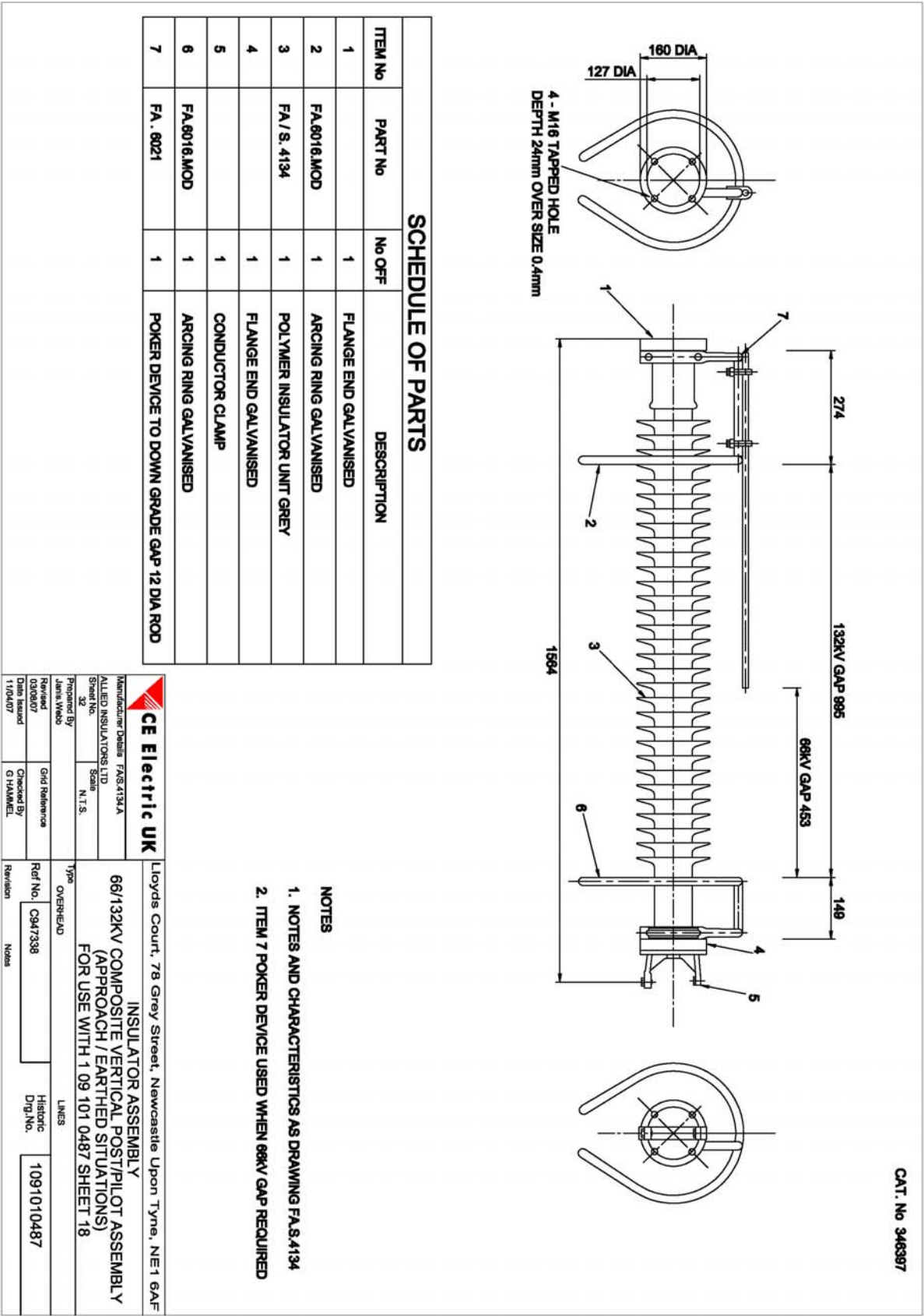
NOTES

1. DIMENSIONS ARE IN MILLIMETRES
2. SOCKET AND BALL COUPLINGS CONFORM TO GAUGES FOR IEC PUB90120 / 20
3. TOLERANCES ARE IN ACCORDANCE WITH IEC 61109, LATEST ISSUE.

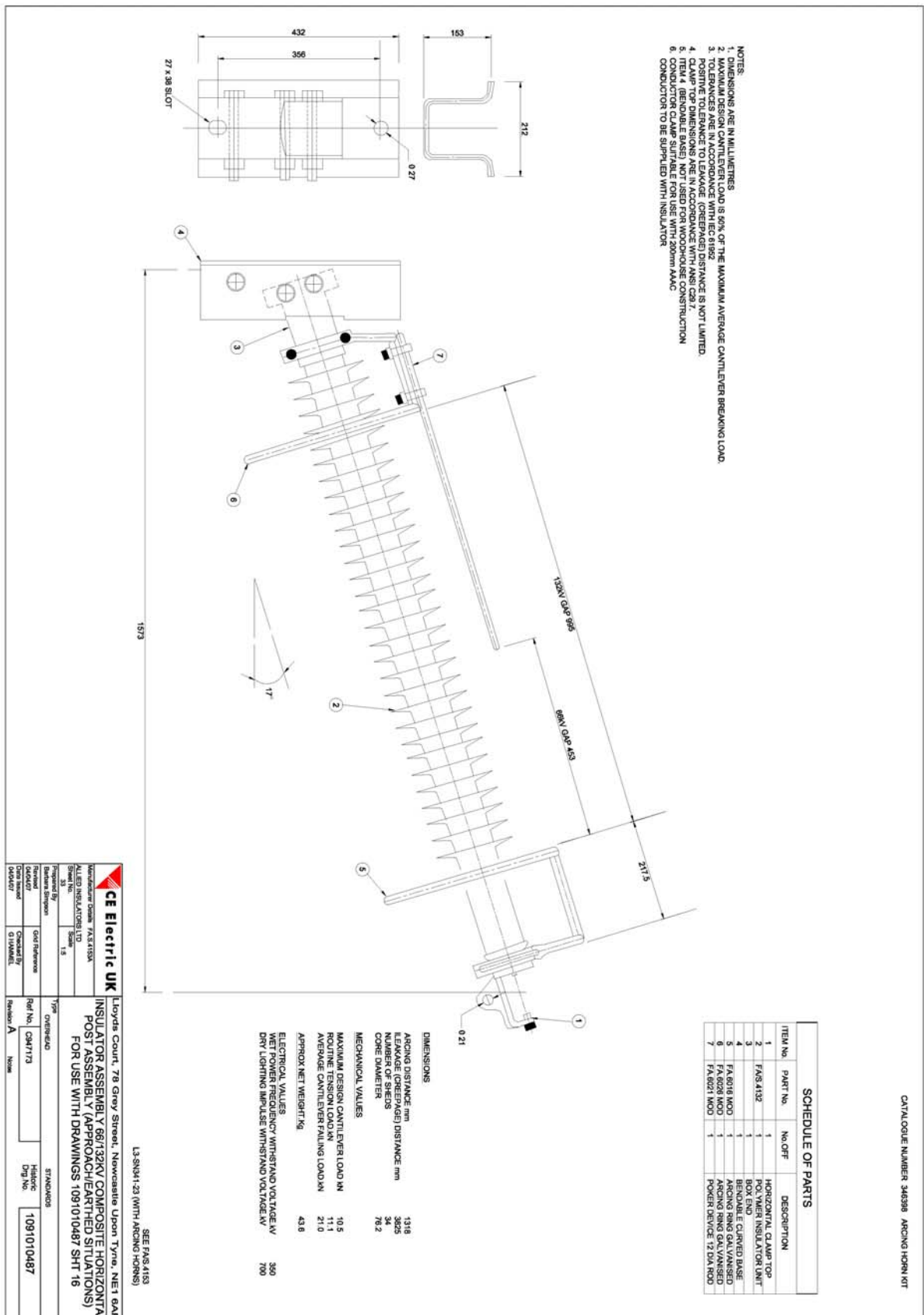
POSITIVE TOLERANCE TO LEAKAGE (CREEPAGE) DISTANCE IS NOT LIMITED

	Manufacturer Details FAS-4117 ALLIED INSULATORS LTD Street, 15 Stretton, 15		Revision 15	
	Reviewed G3036017 Date Issued G304007		Checked By G304007	
	Prepared By J. W. Webb		Grid Reference G3036017	
INSULATOR 132KV COMPOSITE STAY (120KN WITH 20MM BALL / SOCKET COUPLING)				
Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AN				
OVERHEAD				
Ref No. C94/326		Historic Dwg. No. 1091010487		
Revision		Notes		

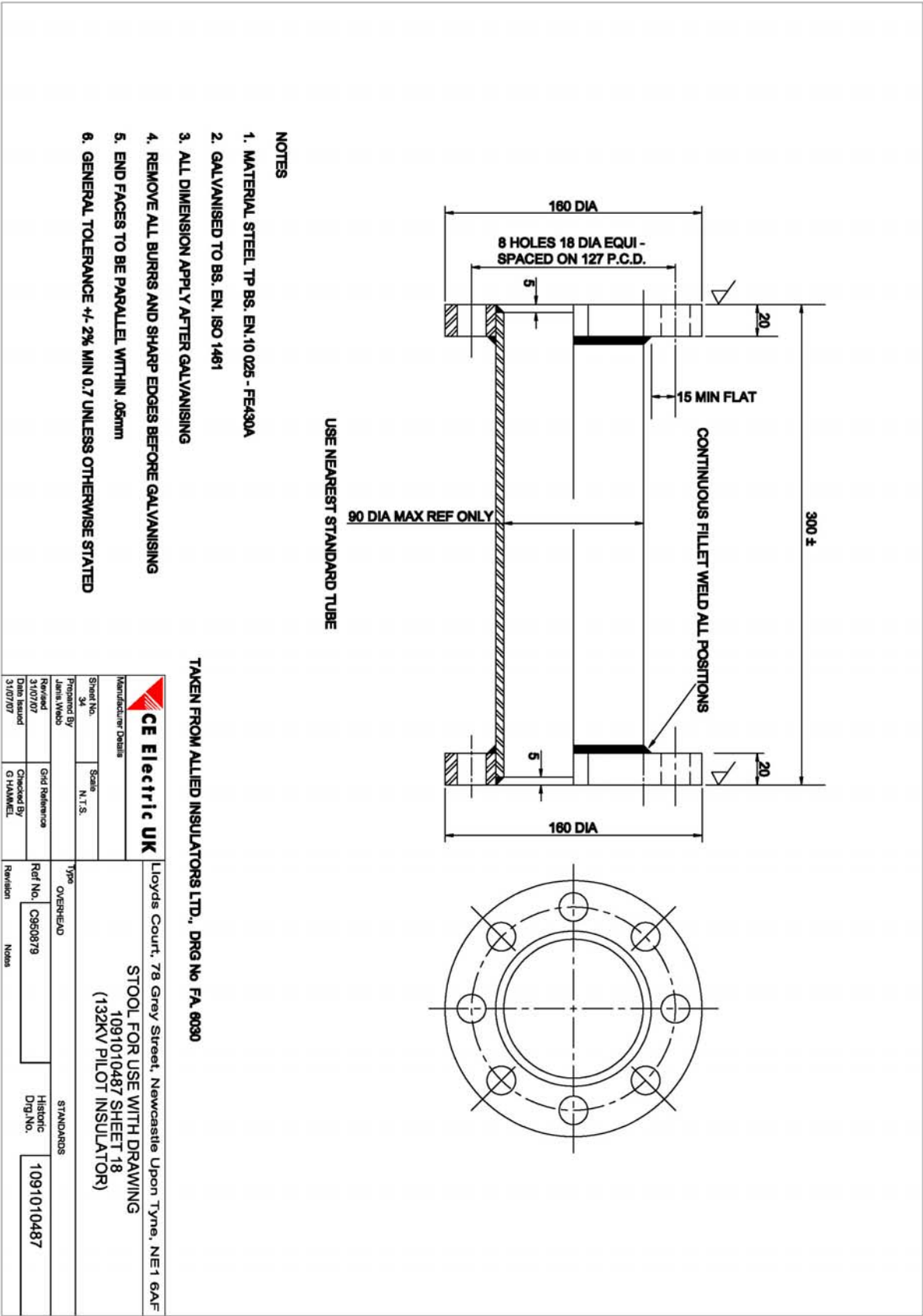
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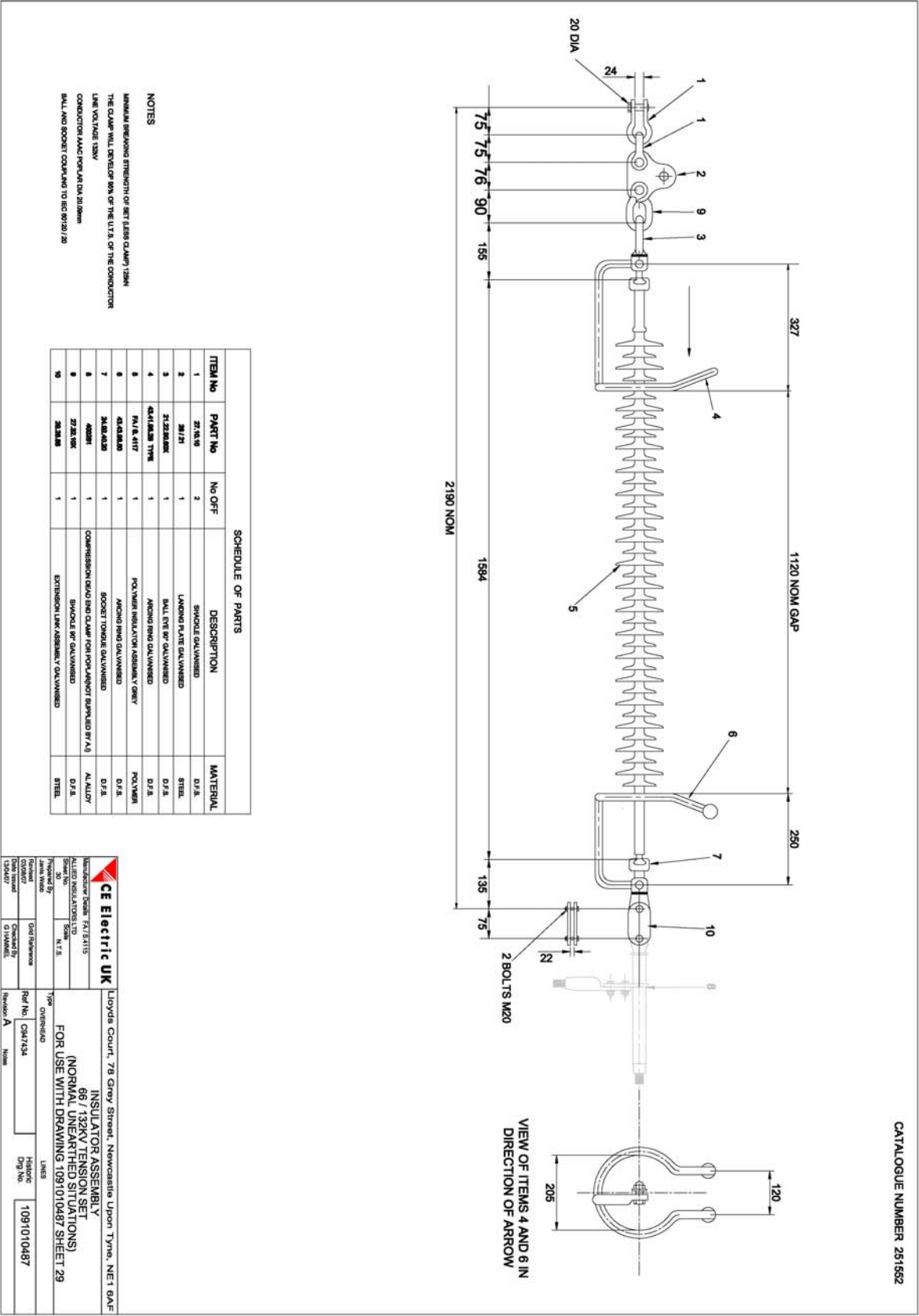
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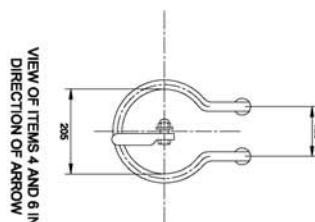
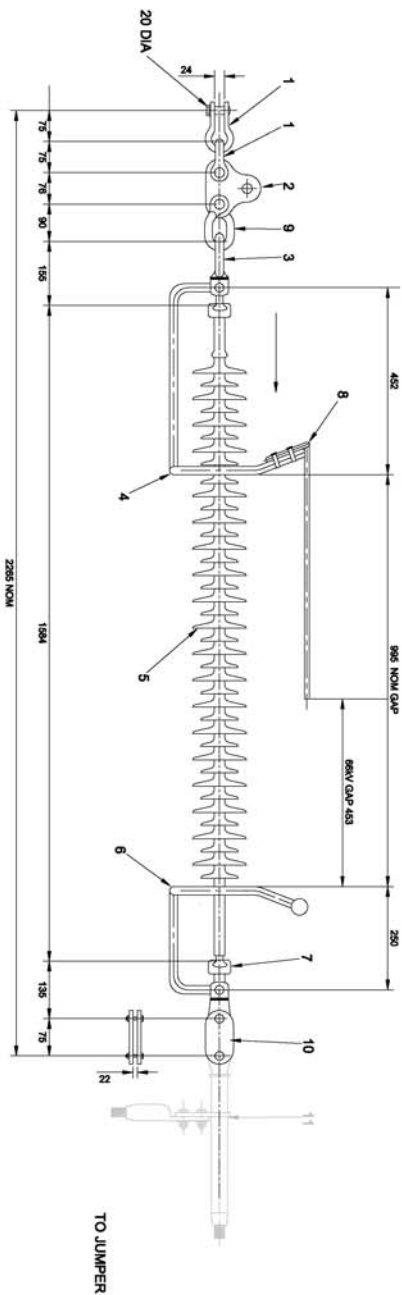
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CATALOGUE No. 251553 132KV OPERATION
251554 66/33KV OPERATION

SCHEDULE OF PARTS				
ITEM NO	PART NO	NO OFF	DESCRIPTION	MATERIAL
1	27-15-10	2	SHOCK DAWNERED	D/S
2	28-17		LANDING PLATE GALVANIZED	STEEL
3	21-22-20-20-20	1	BALL JOINT GALVANIZED	D/S
4	42-11-20-20-20	1	ANCHOR END GALVANIZED	D/S
5	PA-1-4117	1	POLYMER INSULATION ASSEMBLY ONLY	POLYMER
6	42-11-20-20	1	ANCHOR END GALVANIZED	D/S
7	34-11-20-20	1	ROCKET THROAT GALVANIZED	D/S
8	PA-1-4117	1	POWER SOURCE FOR NEW DAW X-400	12 GA STEEL INDO
9	27-15-10	1	SHOCK DAWNERED	D/S
10	28-17	1	EXTENSION LINK ASSEMBLY GALVANIZED	STEEL
11	42-11-20-20	1	COMPRESSION LEAD END FOR POWER (NOT SUPPLIED BY AL)	AL ALLOY

NOTES

THE M 10 POWER DEVICE USED WHEN NEW GAPS IS REQUIRED
MINIMAL BREAKING STRENGTH OF AIR (LESS CLAMP) 150N
THE CLAMP WILL DEVELOP 80% OF THE U.T.S. OF THE CONDUCTOR
LINE VOLTAGE 120V
CONDUCTOR AAC PCHV DIA 20.0mm
BALL AND SOCKET COUPLING TO EC 80/20 / 20

	
Manufactured under Licence No. 11/17/17/18	
Drawing No. 109/1010487	
Designed by	Mr. S. N. S.
Reviewed by	
Drawn by	
Checked by	
Approved by	
Drawn by	
Checked by	
Approved by	

Type Orientation
 Insulator Assembly
 66 / 132KV TENSION SET
 (APPROX. 16 LEATH STATIONS)
 FOR USE WITH DRAWING NO. 109/1010487 SHEET 29

Ref No.	109/1010487
Revision	Initials

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CAT. NO. 251547

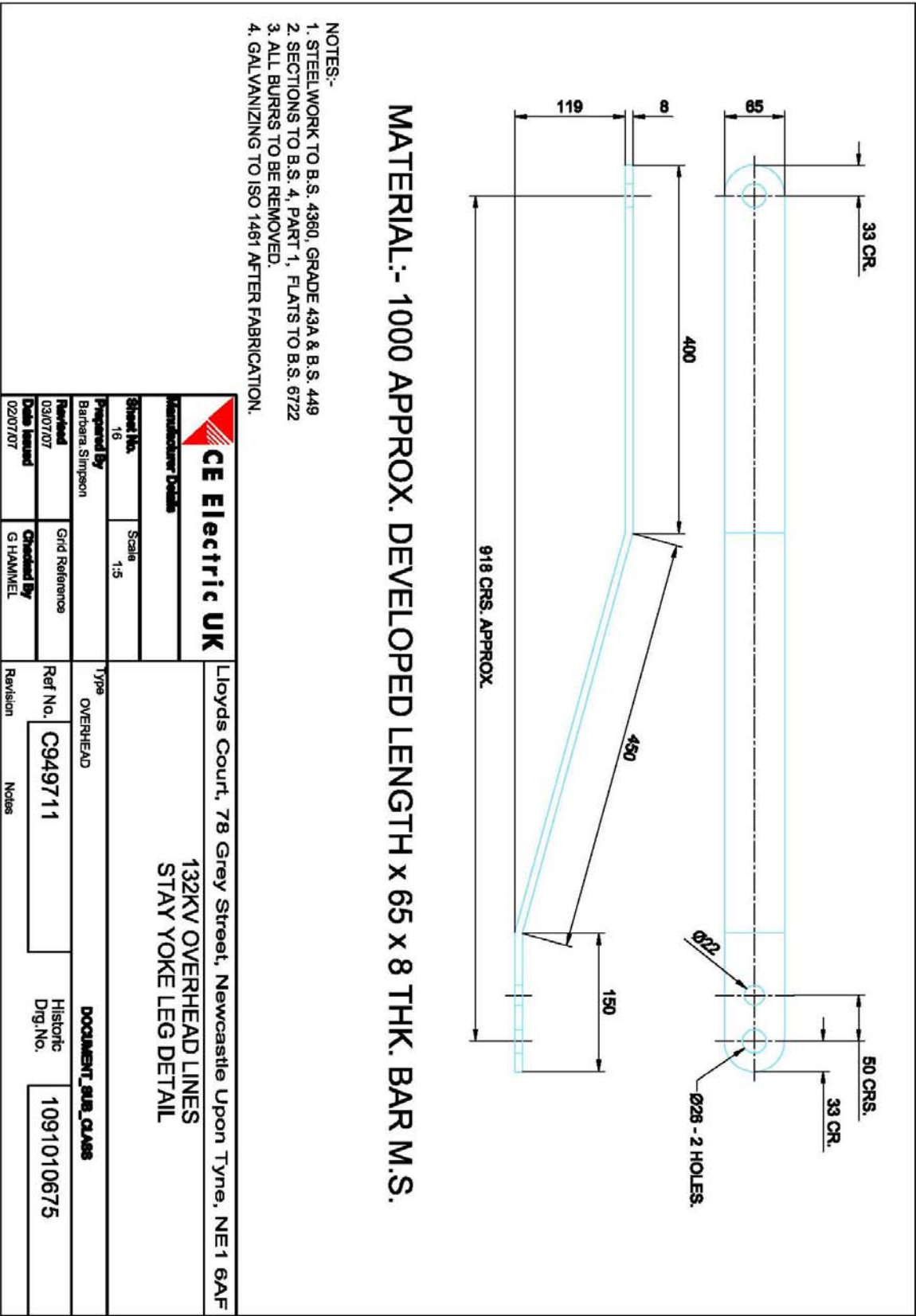
1120mm

NOTES

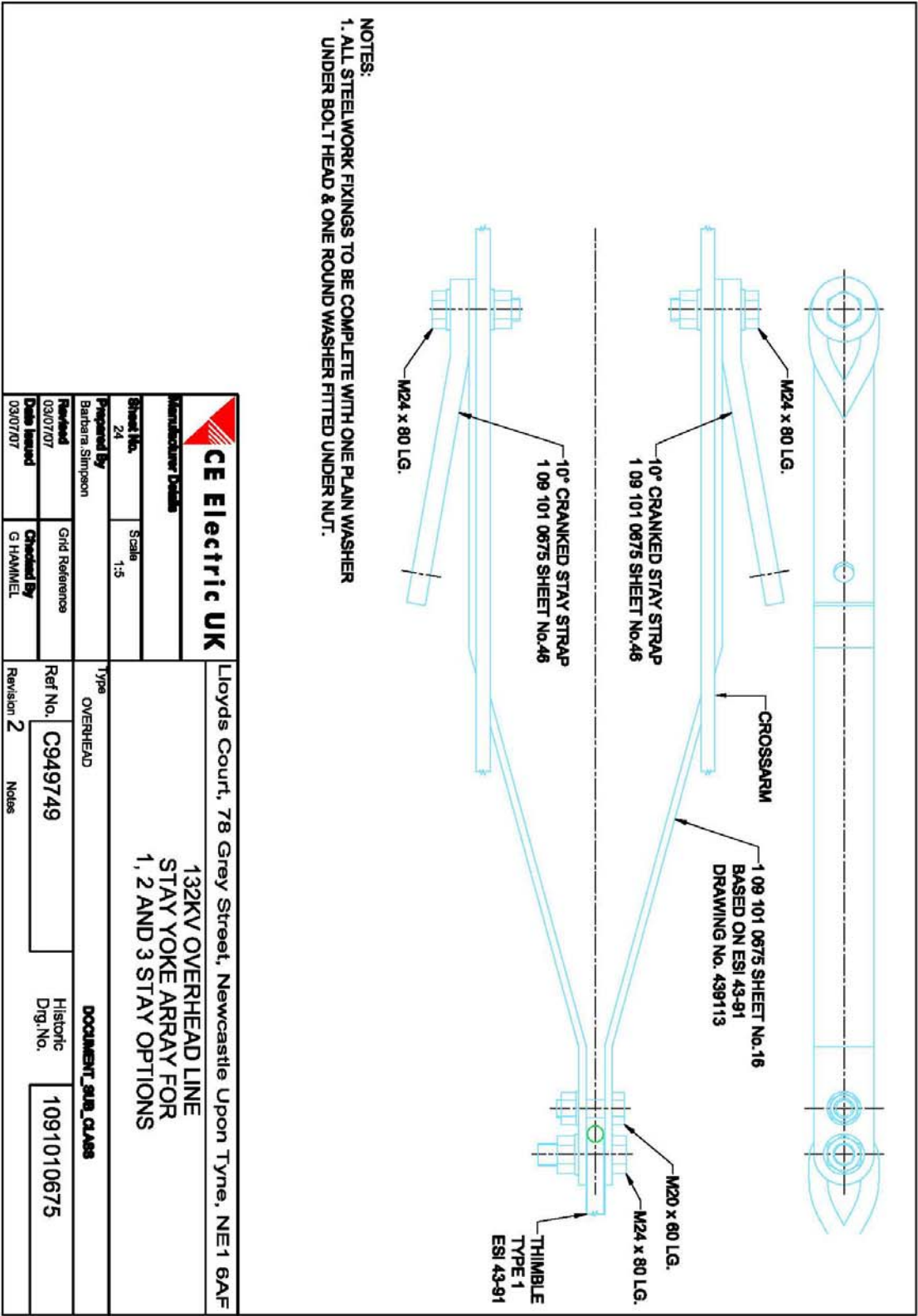
BALL AND SOCKET COUPLING TO IEC 60120 / 20

CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details		INSULATOR STAY ASSEMBLY (UNEARTHED ONLY)	
Sheet No. 2	Scale N.T.S.	(120kV) 66kV AND 132kV COMPOSITE	
Designed By J.M.M.C.B.		Type OVERHEAD	
Reviewed 10/08/07	Grid Reference	Ref No. C318321	Historic Dwg No. 1091010372
Date Issued 10/09/89	Checked By H.M.M.O.D.Y.	Revision D	Notes

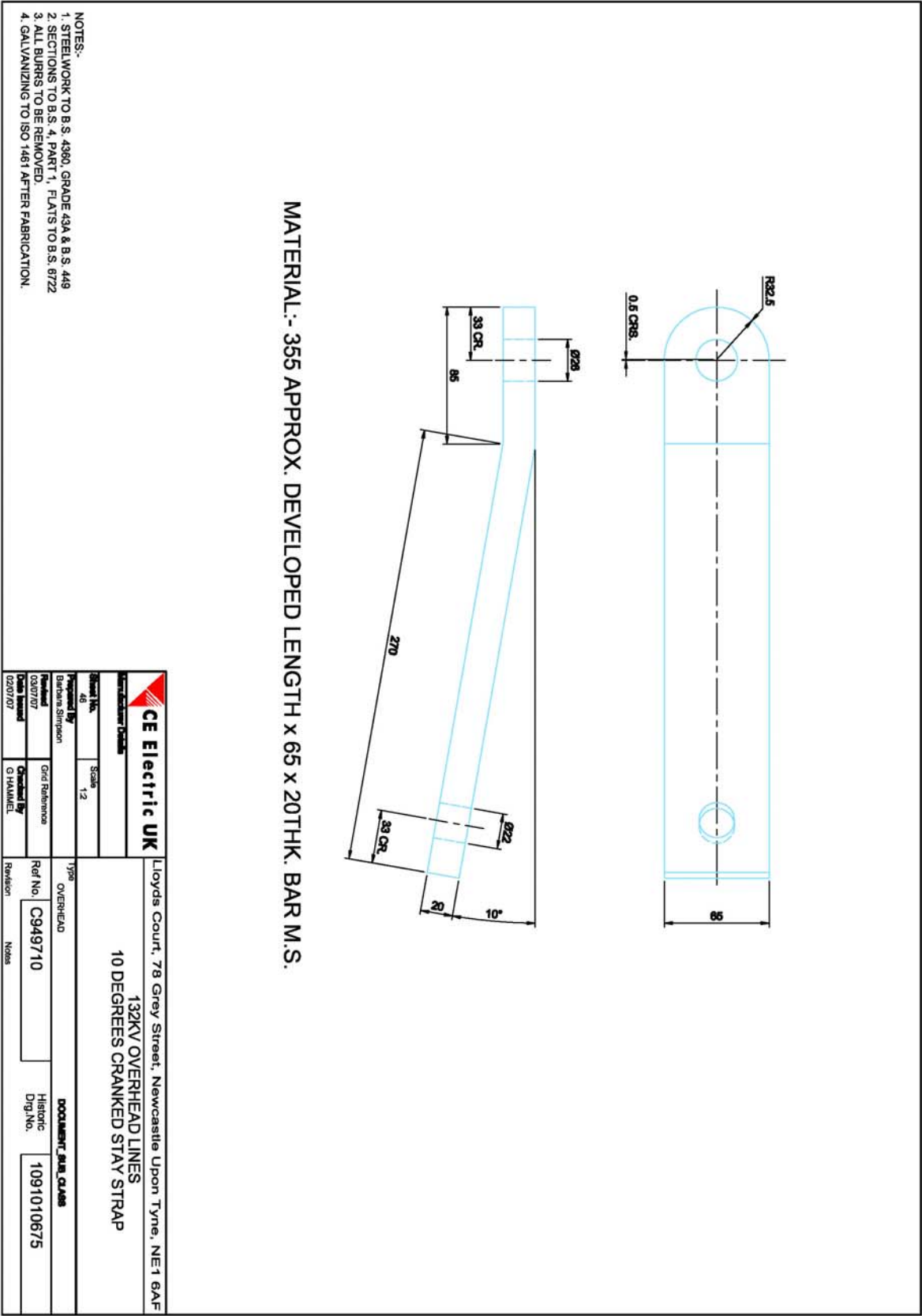
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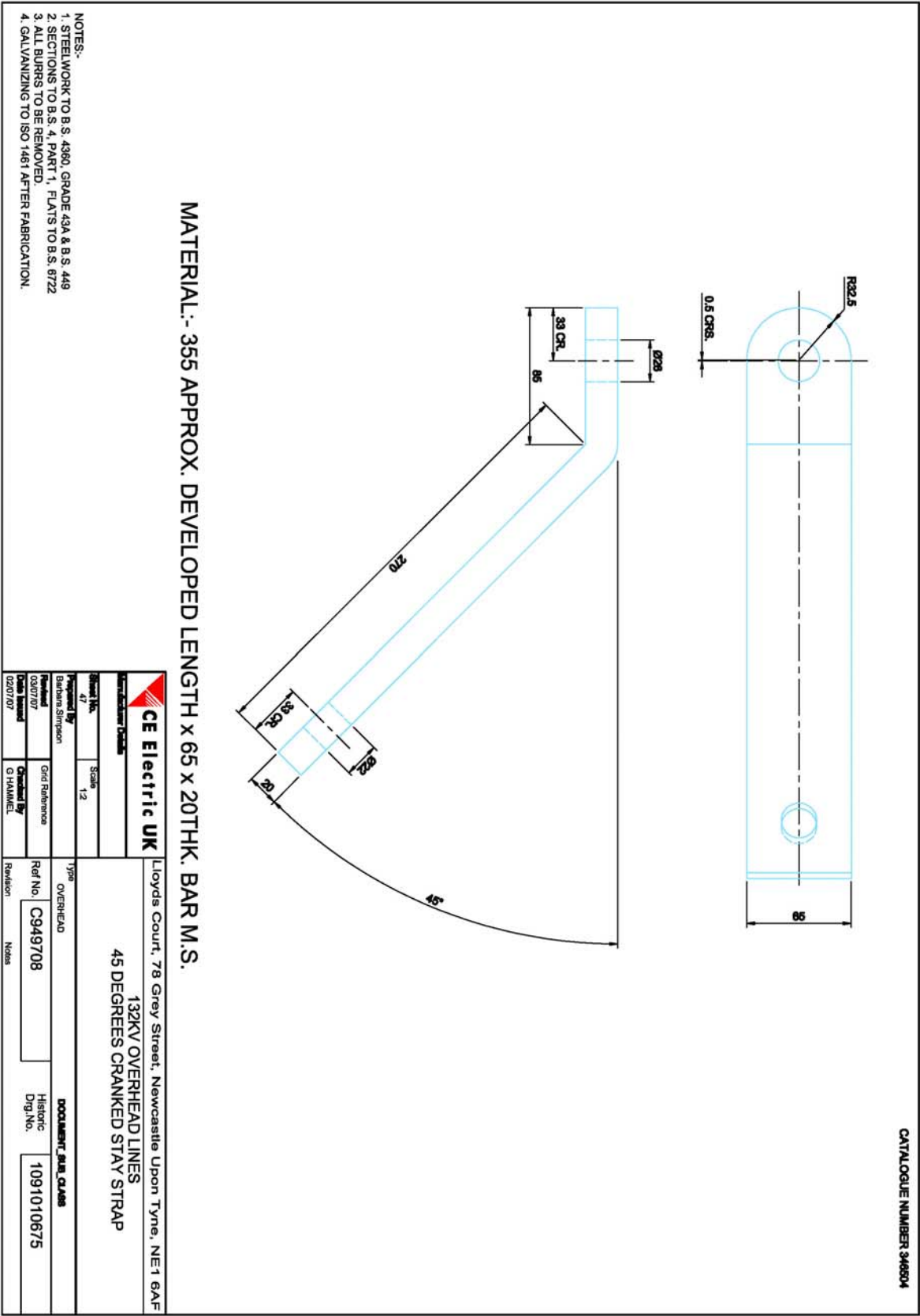
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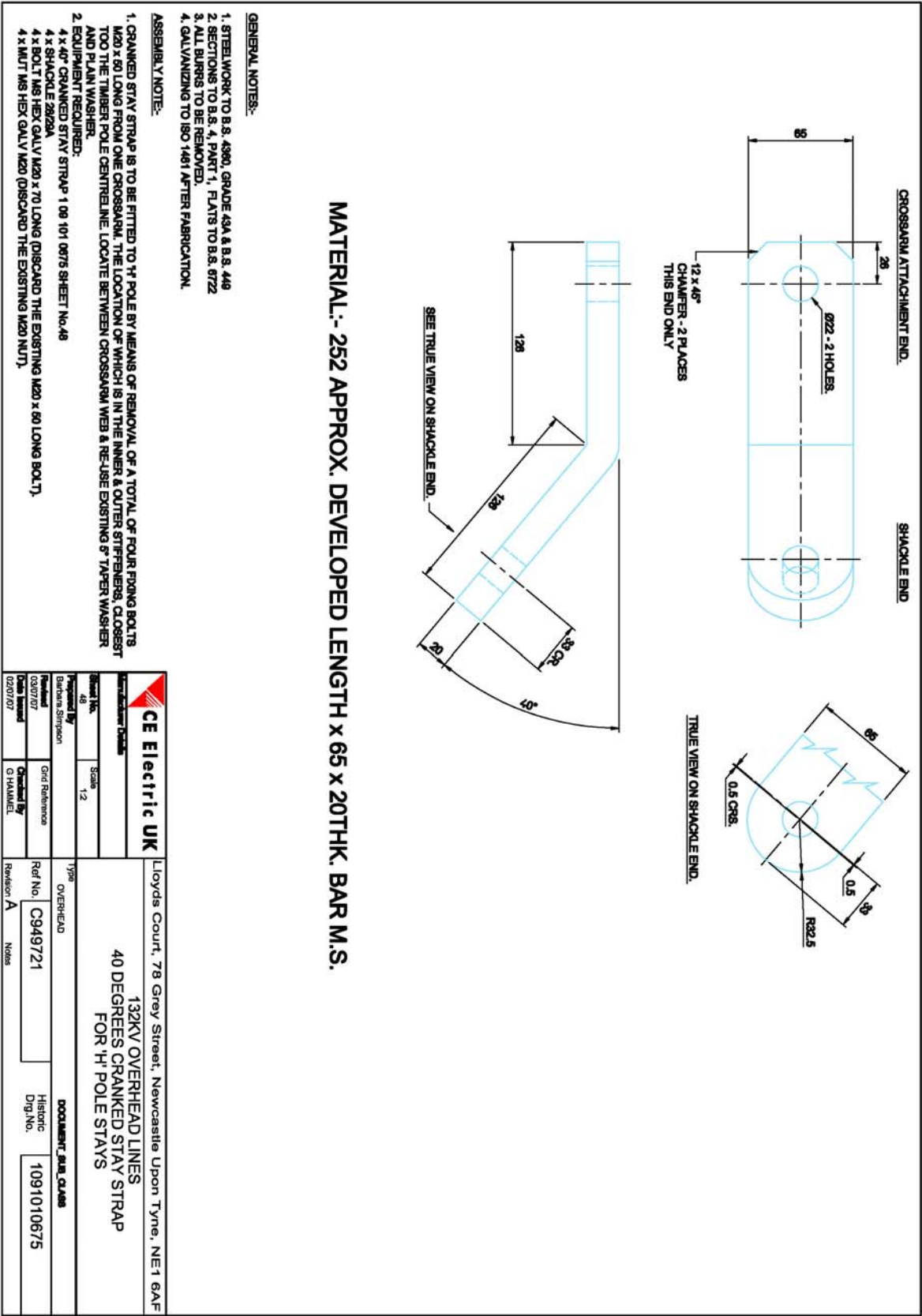
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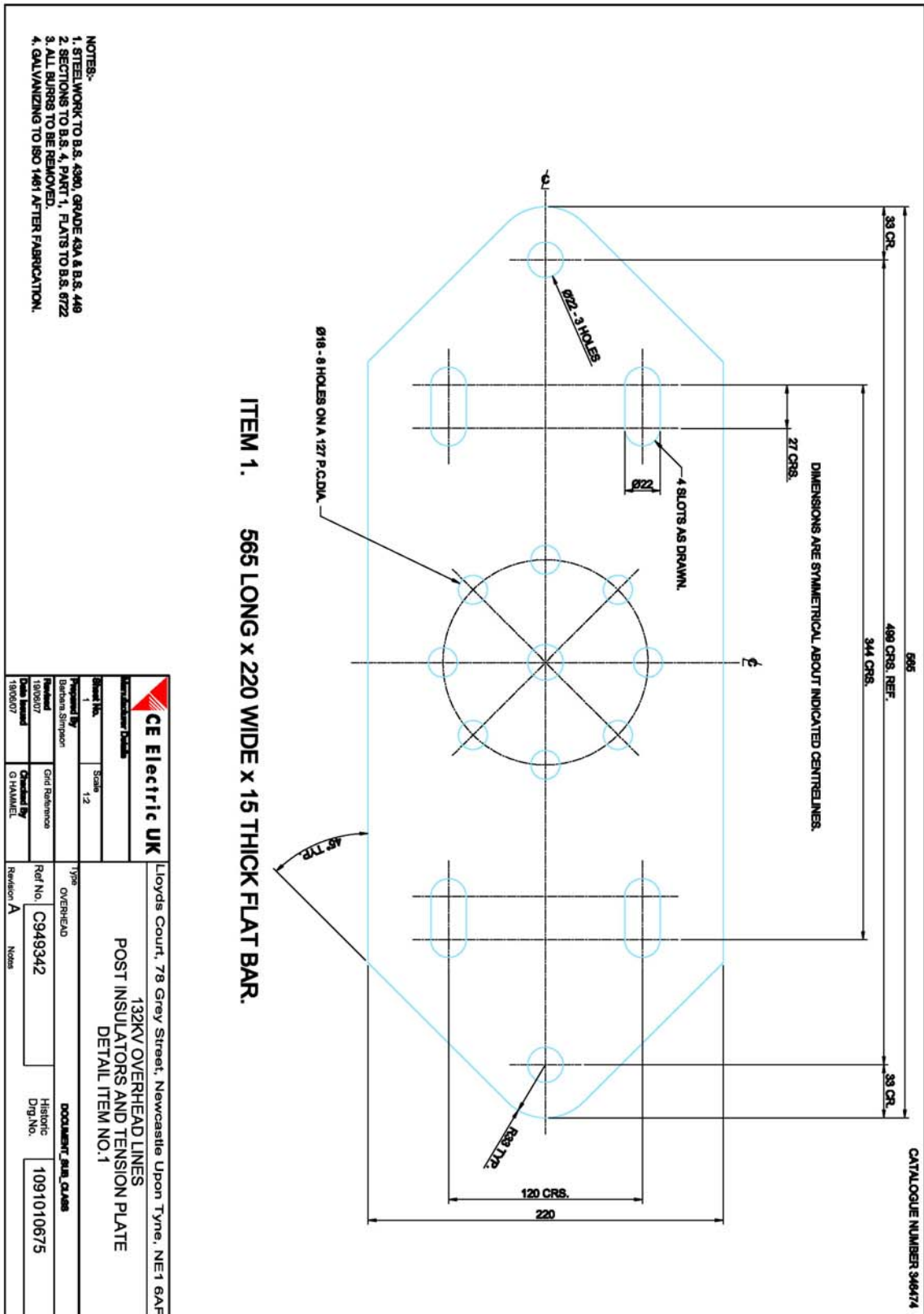
Document reference		NSP/004/045			
Version:-	2.0	Date of Issue:-	Sept 2009	Page	87 of 127



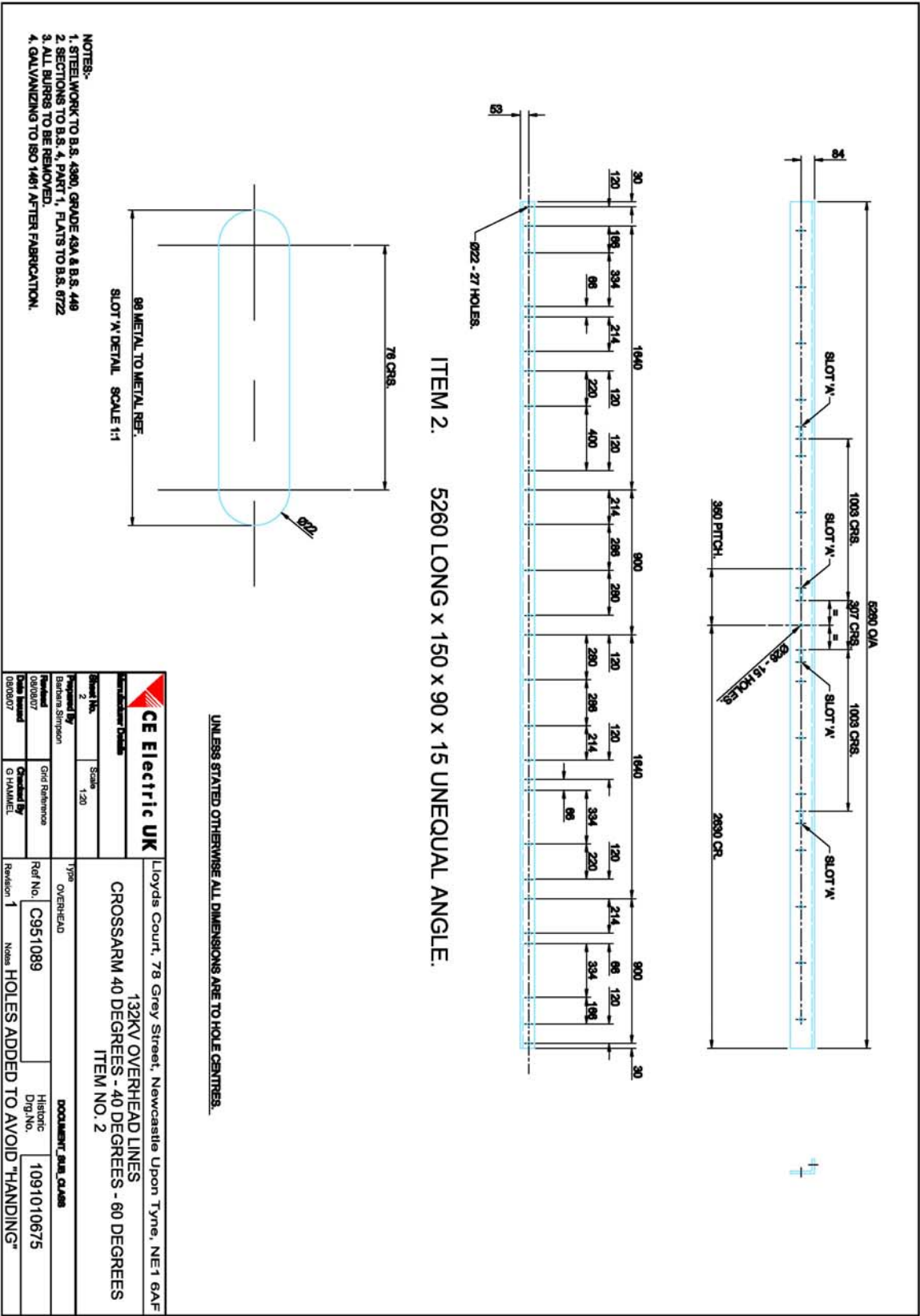
Document reference	NSP/004/045			
Version:-	2.0	Date of Issue:-	Sept 2009	Page 88 of 127



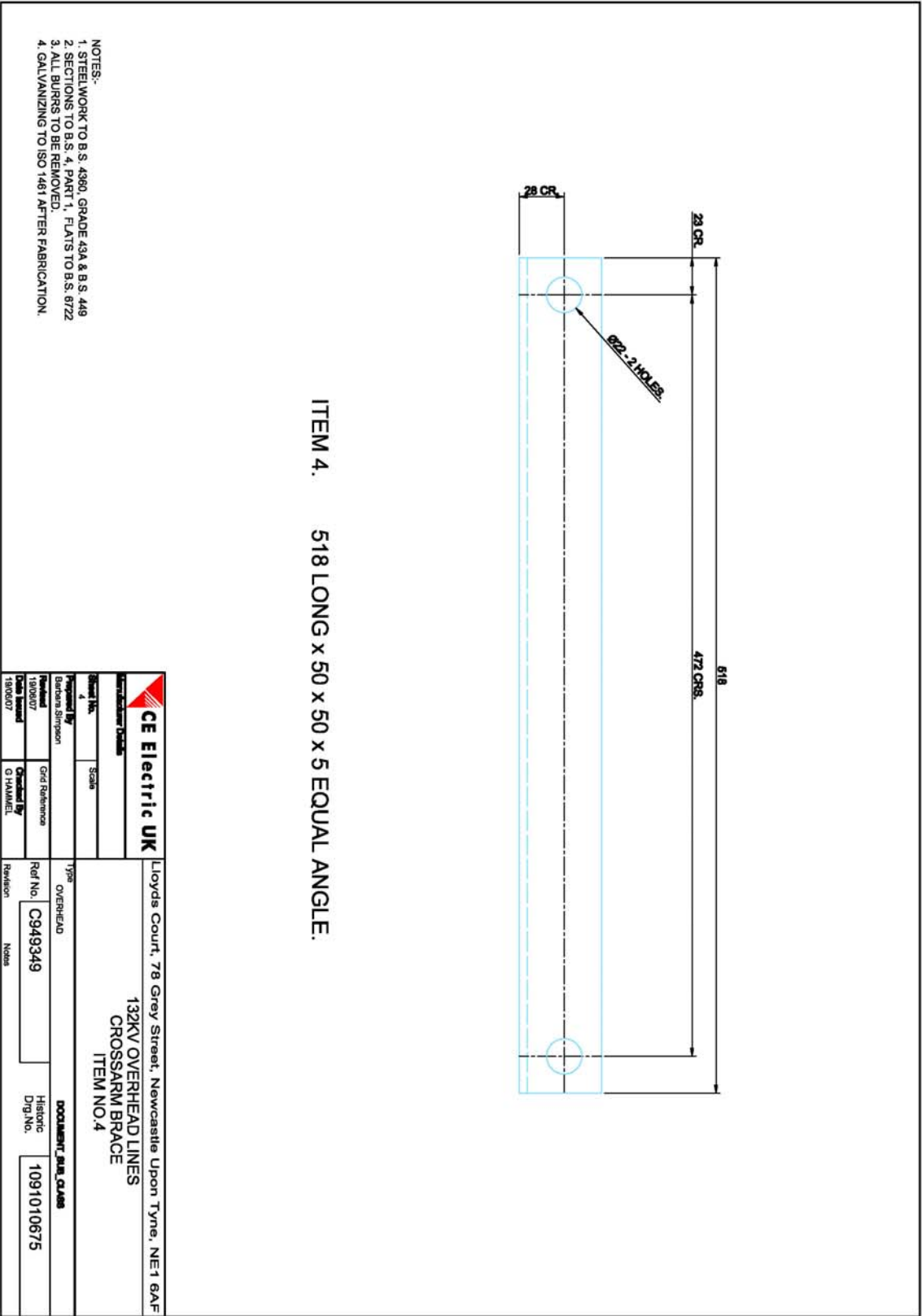
Document reference	NSP/004/045			
Version:-	2.0	Date of Issue:-	Sept 2009	Page 89 of 127



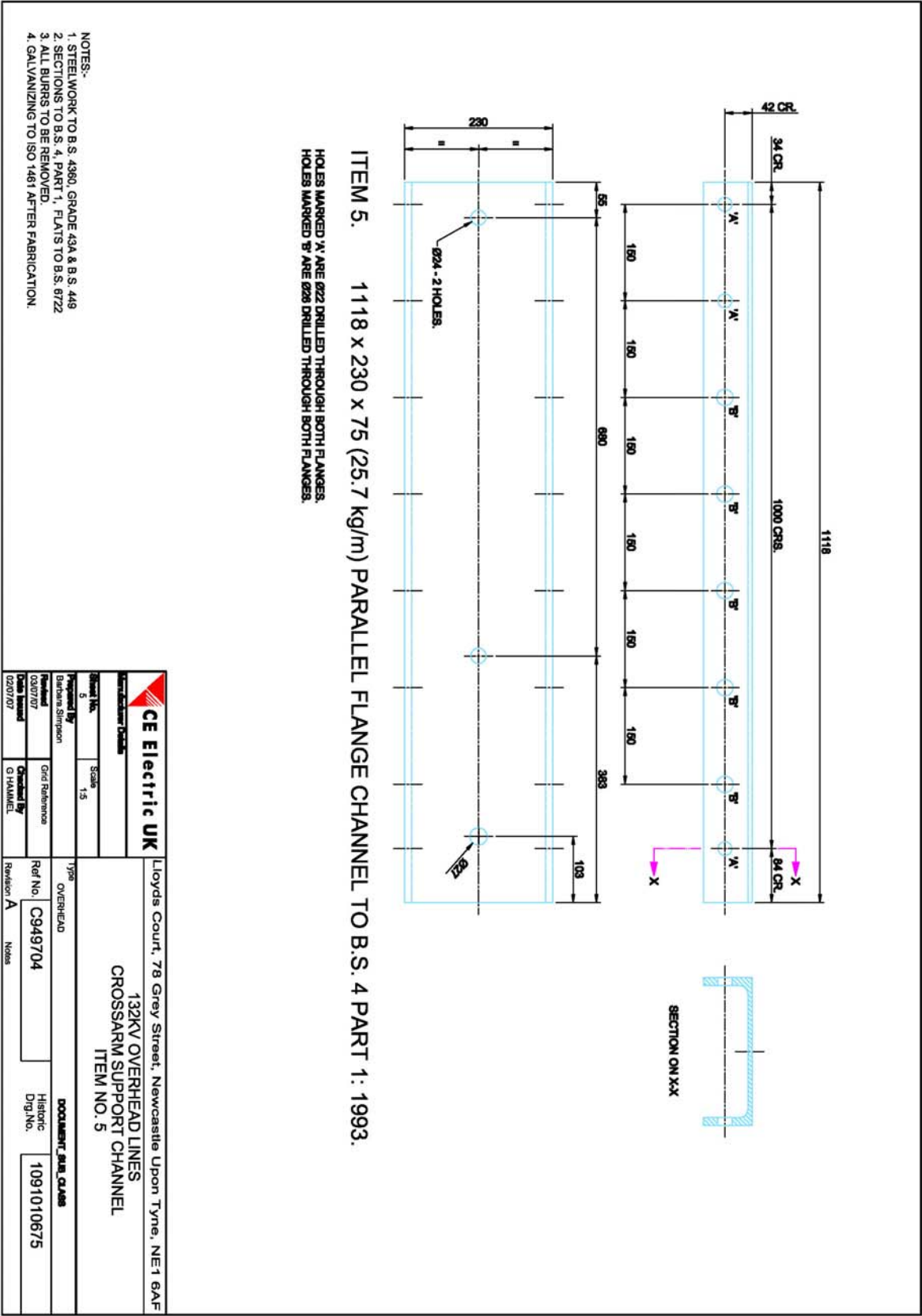
Document reference	NSP/004/045			
Version:-	2.0	Date of Issue:-	Sept 2009	Page 90 of 127



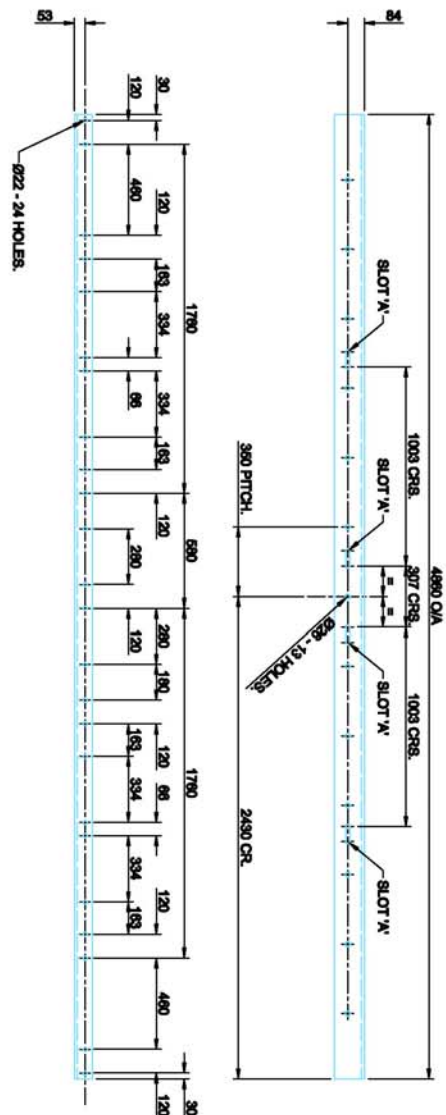
Document reference		NSP/004/045			
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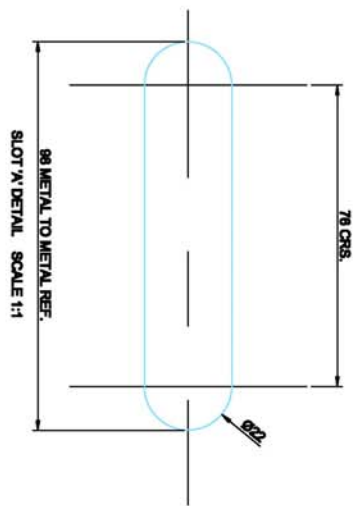
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Document reference		NSP/004/045					
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ITEM 6. 4860 LONG x 150 x 90 x 15 UNEQUAL ANGLE.



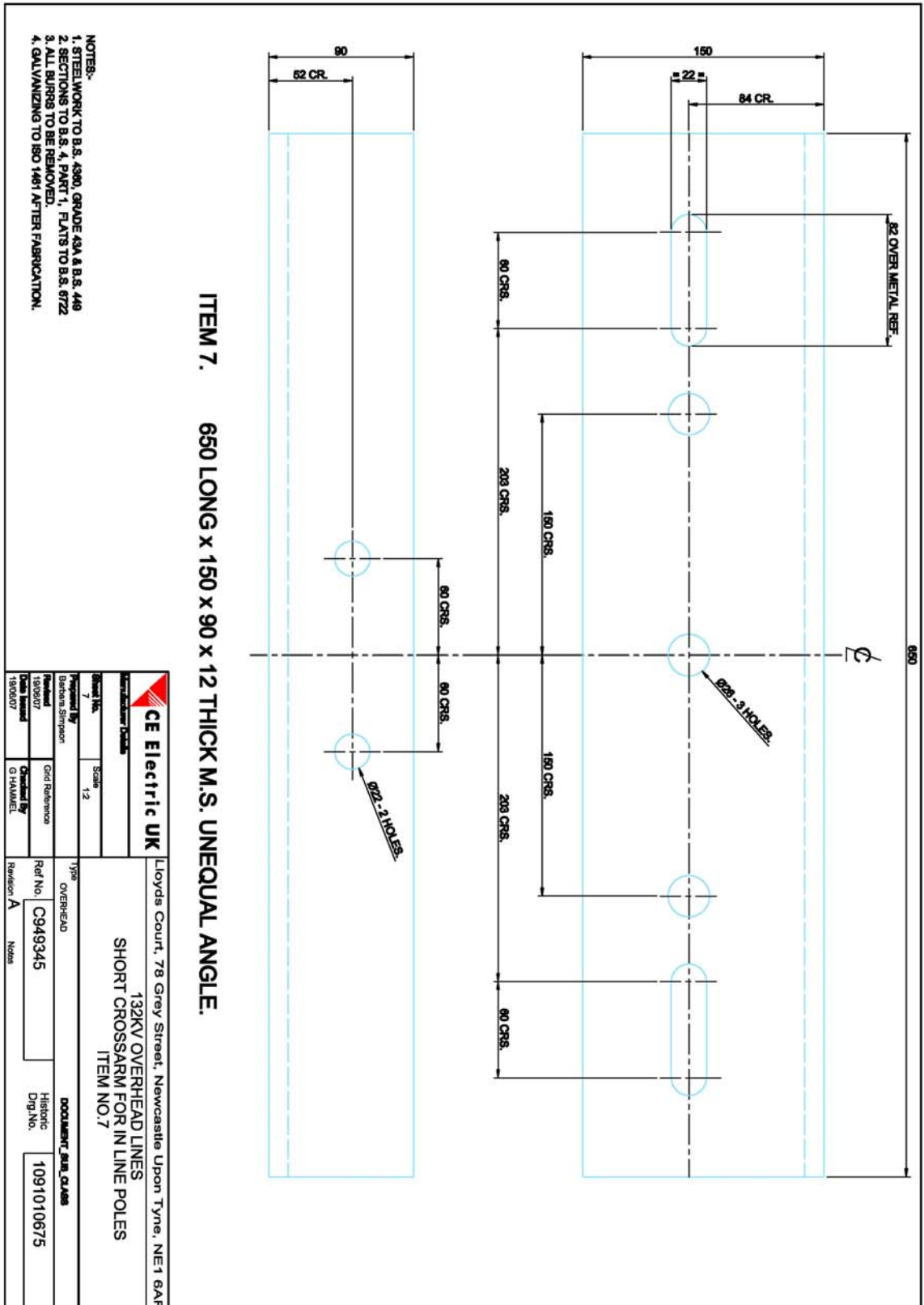
UNLESS STATED OTHERWISE ALL DIMENSIONS ARE TO HOLE CENTRES.

 CE Electric UK 132KV OVERHEAD LINES CROSSARM 0 DEGREES - 40 DEGREES ITEM NO.6		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AA	
Product Details Electrical No. 6 Projected by Barbara Simpson Forecast 03/07/07 Date Forecast 03/07/07		Scale 1:20 Grid Reference G 944067 Checked by G HAMMILL	
Type OVERHEAD		Documentation Date Class	
Ref No. C949705		Historic Dwg No. 1091010875	
Revision A		Notes	

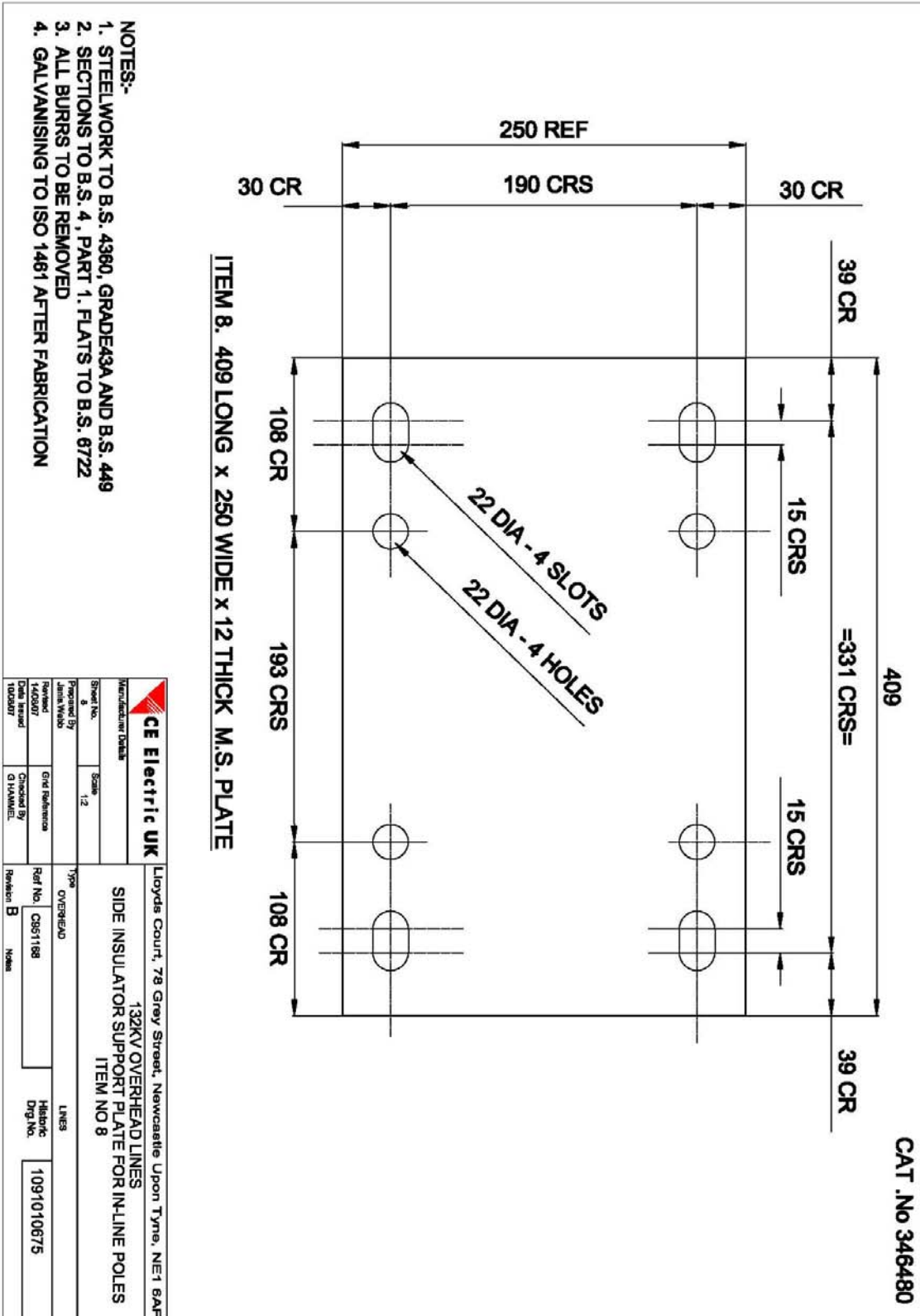
NOTES:-

1. STEELWORK TO B.S. 4360, GRADE 43A & B.S. 449
2. SECTIONS TO B.S. 4, PART 1, FLATS TO B.S. 6722
3. ALL BURRS TO BE REMOVED.
4. GALVANIZING TO ISO 1461 AFTER FABRICATION.

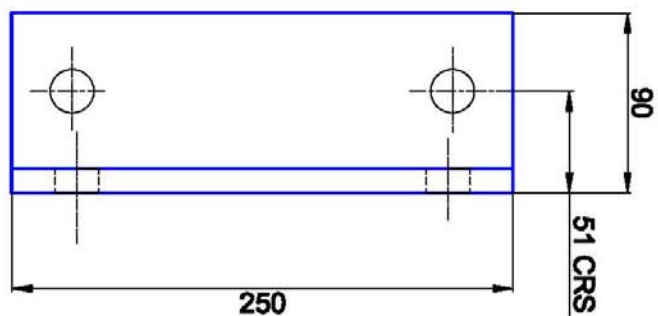
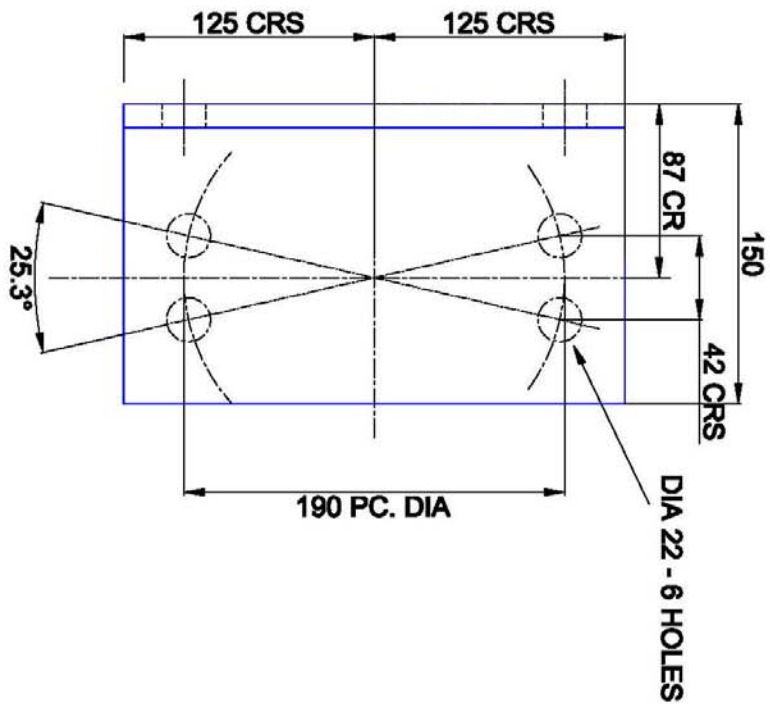
Document reference		NSP/004/045					
Version:-	2.0	Date of Issue:-	Sept 2009	Page	94	of	127



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Version:-	2.0	Date of Issue:-	Sept 2009	Page 95 of 127



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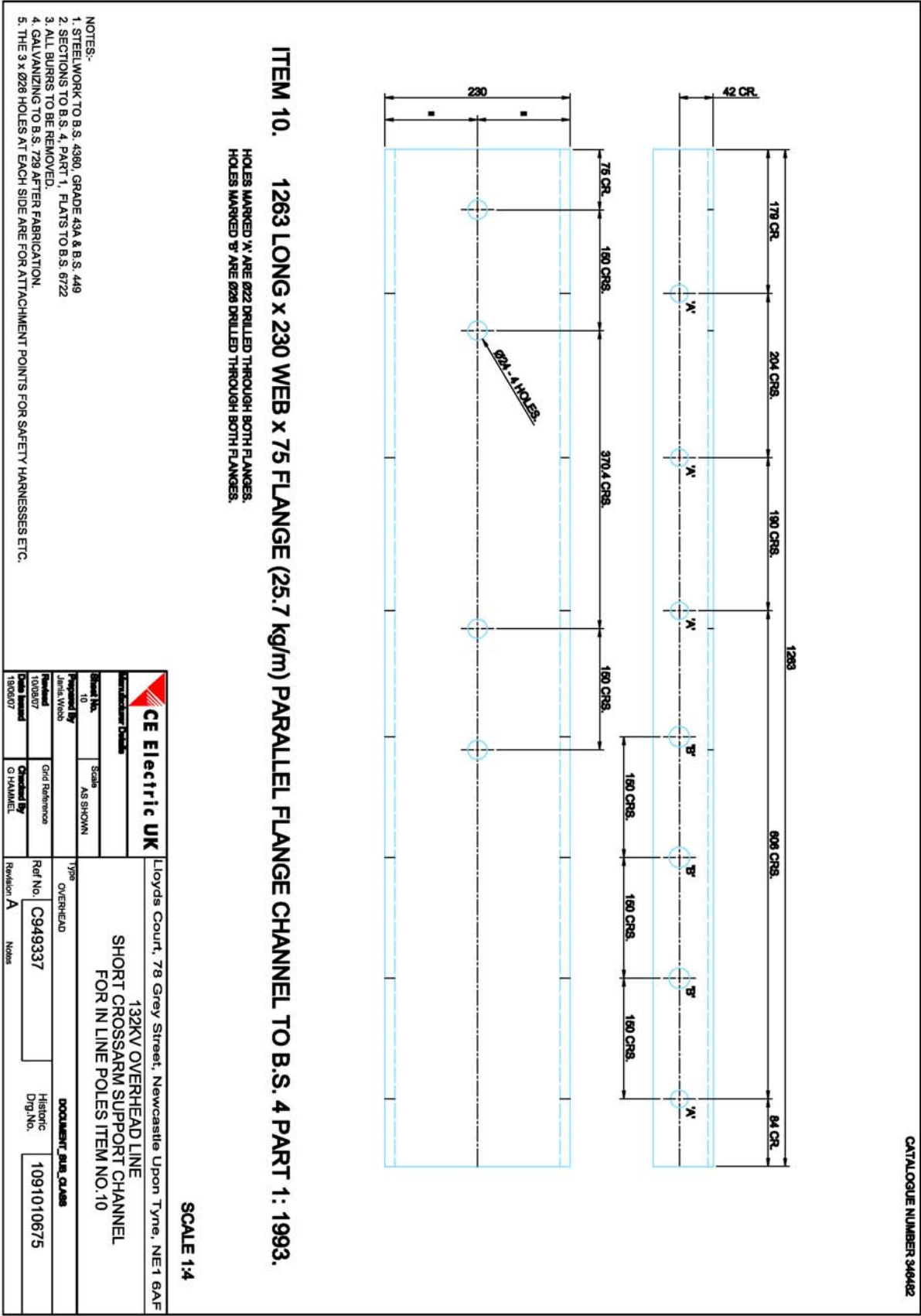
CATALOGUE No 345481

ITEM 9 250 LONG x 150 x 90 x 12 THICK M.S. UNEQUAL ANGLE B.S. 4848: PART 4 : 1972

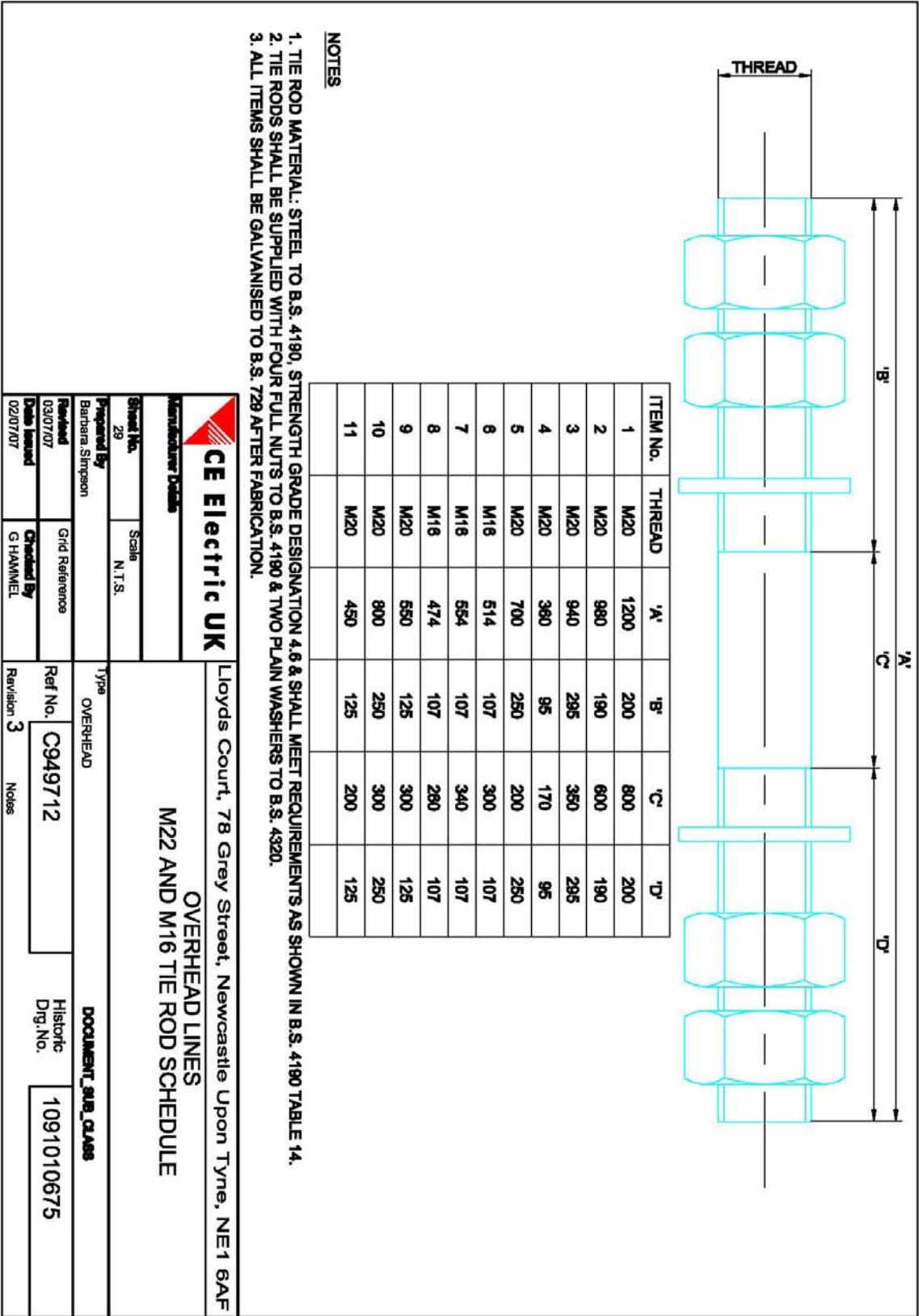
- NOTES:-
1. STEELWORK TO B.S. 4380, GRADE 43A AND B.S.449
 2. SECTIONS TO B.S.4 PART 1, FLATS TO B.S. 6722
 3. ALL BURRS TO BE REMOVED
 4. GALVANISING TO ISO 1461 AFTER FABRICATION

CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details	Scale	1:2	
Sheet No.	9	Type	
Prepared By	James Webb	OVERHEAD	
Reviewed	14/06/07	Ref No.	
Date Issued	14/06/07	C961228	
Checked By	G HAMMILL	Revision	
		B Notes	
		132KV OVERHEAD LINES	
		INSULATOR SUPPORT PLATE BRACKET	
		FOR IN - LINE POLES ITEM NO 9	
		POLES	
		Hole No.	
		1091010675	

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The diagram illustrates the dimensions of a timber block. The top view shows a rectangle with length 'A' and width 'C'. The side view shows a rectangle with height 'B' and width 'C'. The 'D' CENTRES dimension is indicated as the distance between the centers of the two circular holes in the side view. The holes are labeled with a diameter of Ø24.

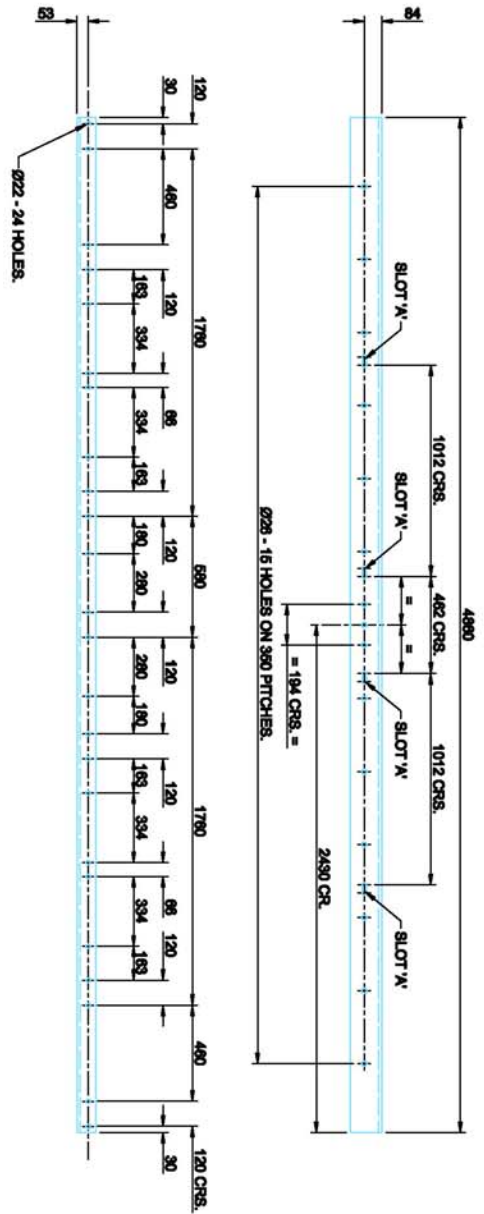
ITEM NO.	LENGTH 'A'	HEIGHT 'B'	THICKNESS 'C'	'D' CENTRES	MATERIAL
1	175	300	100	150	ANGELIM PEDRA
2	175	250	100	86	ANGELIM PEDRA
3	1300	250	125	86	PINUS SYLVESTRIS
4	2500	250	125	86	PINUS SYLVESTRIS
5	1300	250	125	150	PINUS SYLVESTRIS
6	2500	250	125	150	PINUS SYLVESTRIS
7	240	300	130	150	ANGELIM PEDRA
8	210	300	130	150	ANGELIM PEDRA
9	2500	250	125	150	PINUS SYLVESTRIS
10	1300	250	125	150	PINUS SYLVESTRIS
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.

NOTES

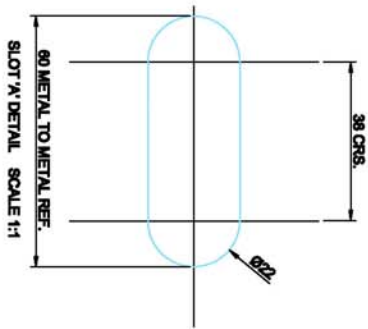
1. SEE MATERIAL COLUMN IN TABLE A SUITABLE ALTERNATIVE MAY BE USED PROVIDED IT IS OF SUITABLE STRENGTH & DURABILITY.
2. BLOCKS TO BE CUT FROM SAWN TIMBER.
3. TIMBER GRAIN TO BE LONGITUDINALLY ALONG LENGTH 'A'.
4. LONGITUDINALLY CUT SURFACES TO BE STRAIGHT & CLEAN & ANNUAL GROWTH RINGS SHALL BE APPROX. PARALLEL.

		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
132KV OVERHEAD LINES RUTTER TYPE POLE SHEAR BLOCK SCHEDULE		132KV OVERHEAD	
Prepared by 30 BURNHAM STATION	Scale N.T.S.	Ref No. C949240	Historic Dwg No. 1091010675
Revised 1306/07 G HAMMEL	Checked by G HAMMEL	Revision 4	Notes ITEMS 7,8,9 AND 10 ADDED

Document reference		NSP/004/045					
Version:-	2.0	Date of Issue:-	Sept 2009	Page	100	of	127



ITEM 31. 4860 LONG x 150 x 90 x 15 UNEQUAL ANGLE.

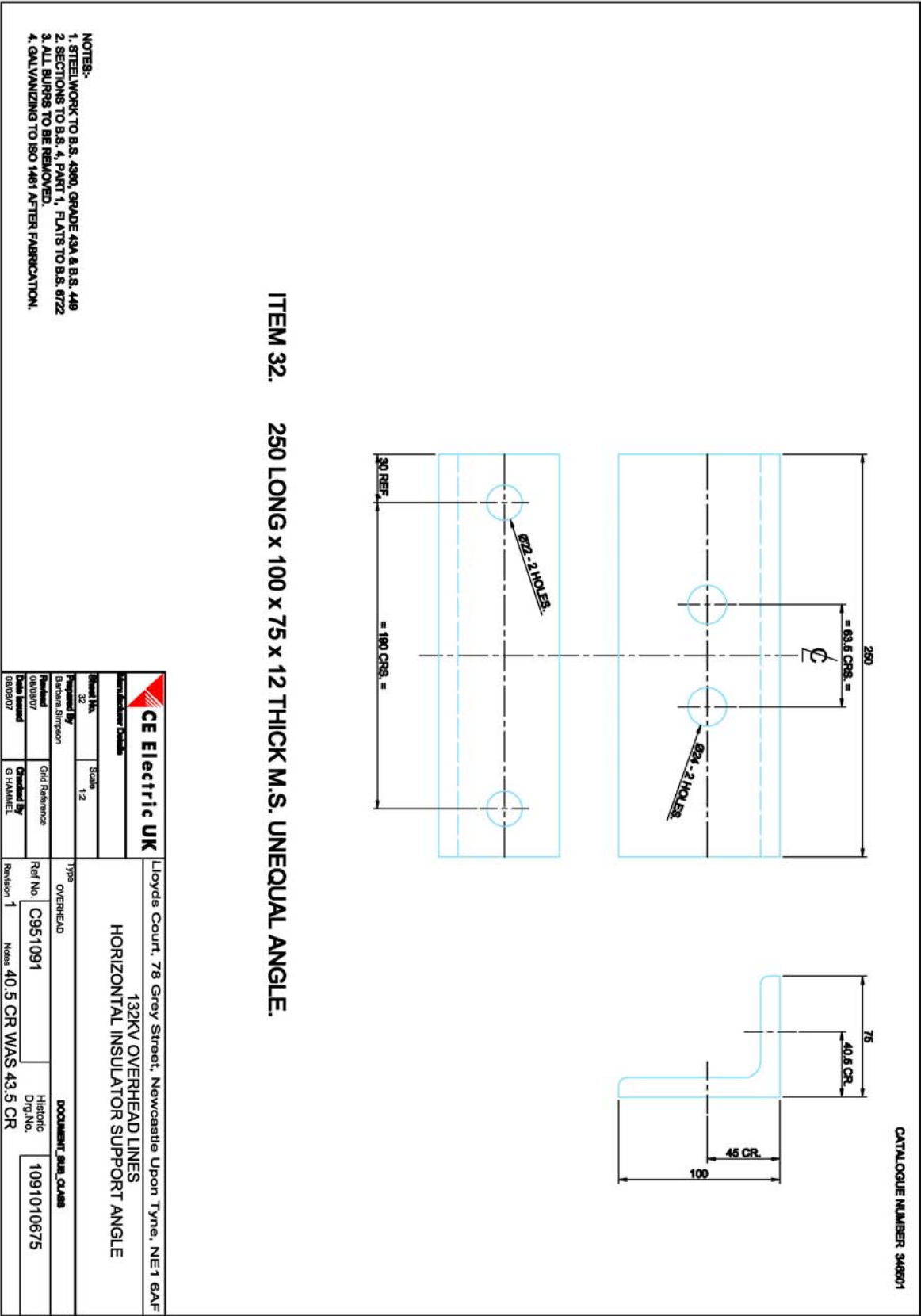


UNLESS STATED OTHERWISE ALL DIMENSIONS ARE TO HOLE CENTRES.

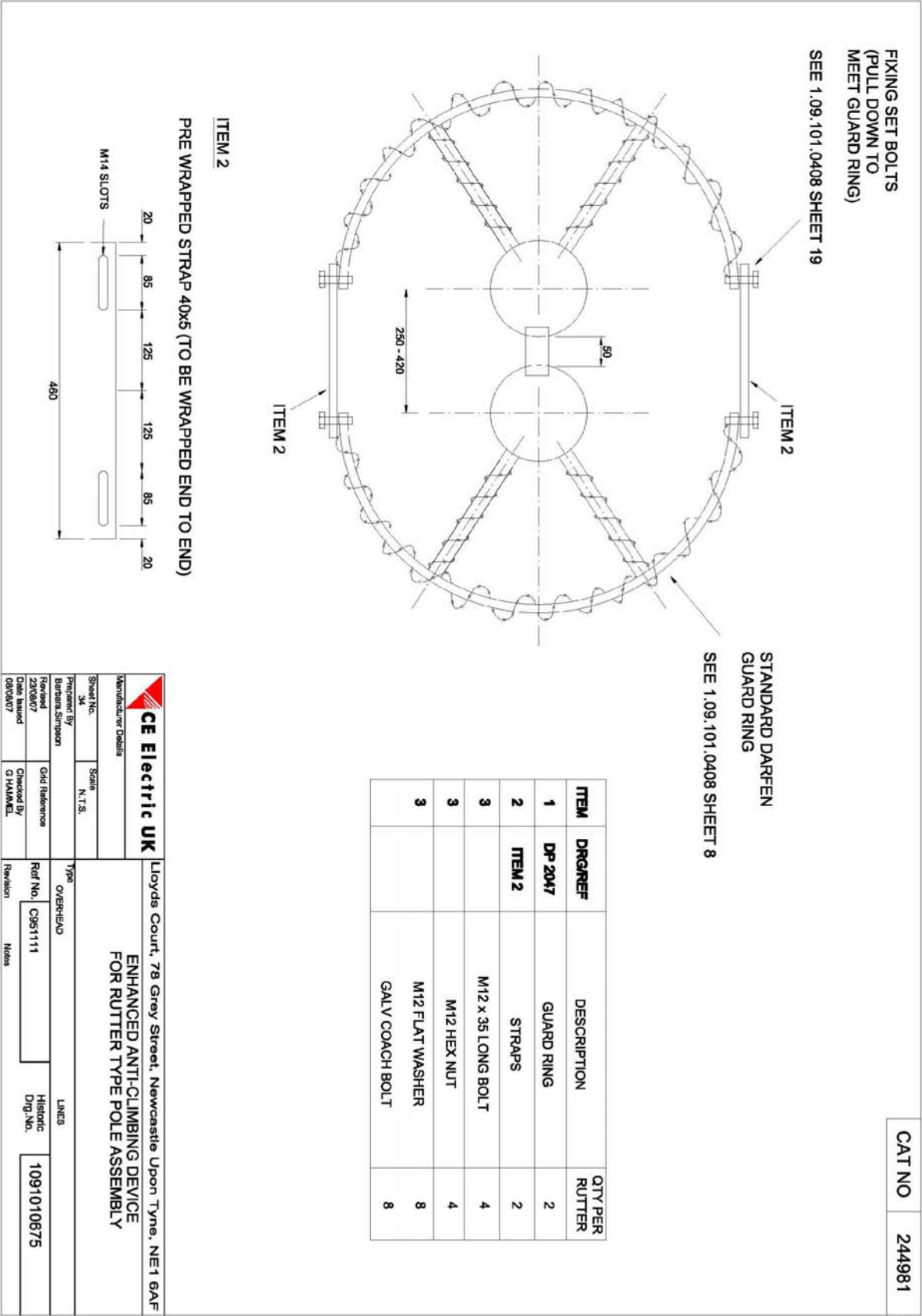
 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
132KV OVERHEAD LINES RUTTER OILE CROSSARM 0 DEGREES - 2 DEGREES ITEM NO. 31			
Production Details		TYPE OVERHEAD	
Order No. 31	Scale 1:20	document path	
Prepared By Barbara Simpson		History	
Reviewed 08/08/07	Grid Reference	Ref No. C9A9348	Hist No. 1091010675
Date Issued 01/06/07	Checked By G HAMMILL	Revision B	Notes

CAT. No. 346500

Document reference	NSP/004/045				
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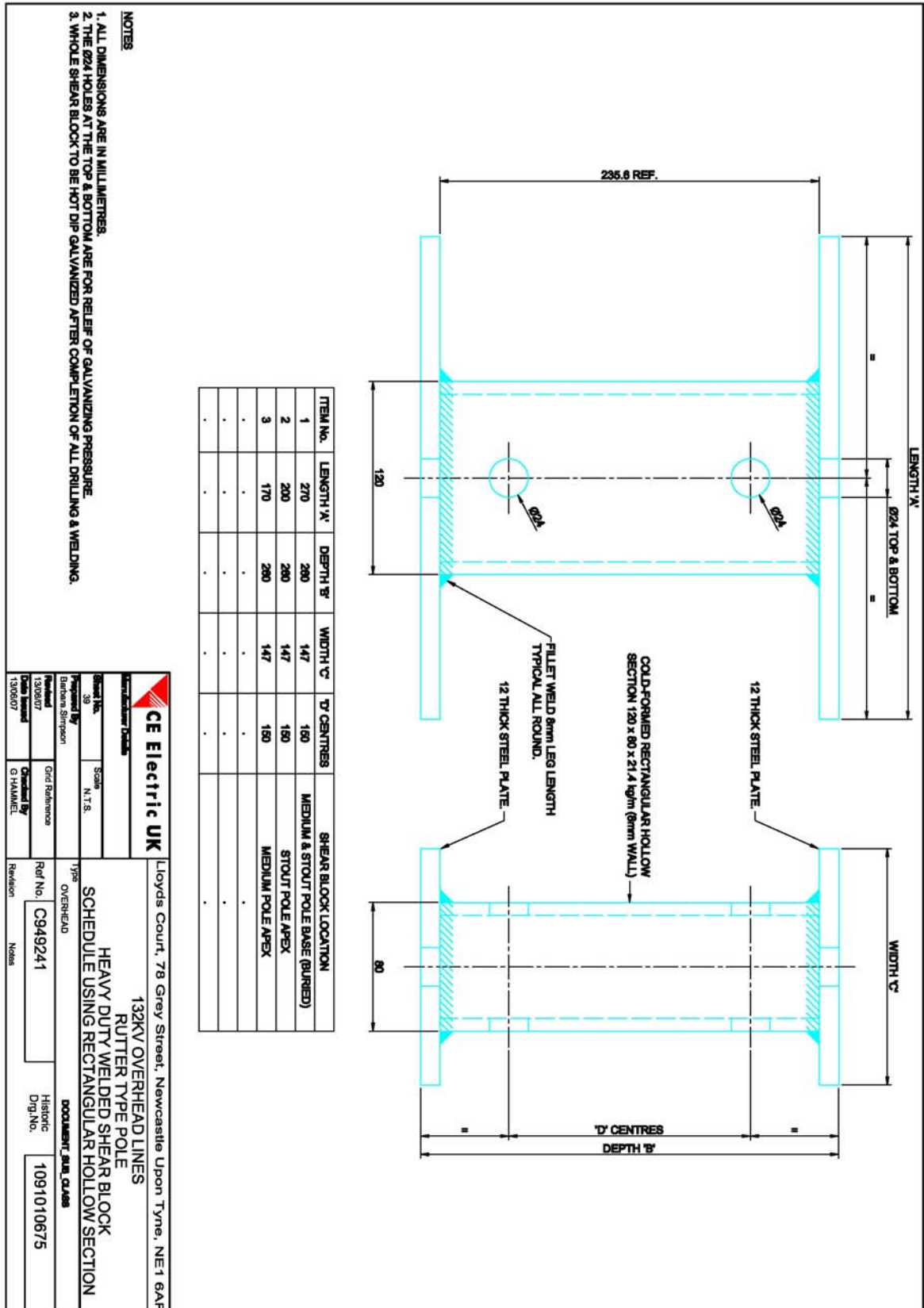


Document reference	NSP/004/045			
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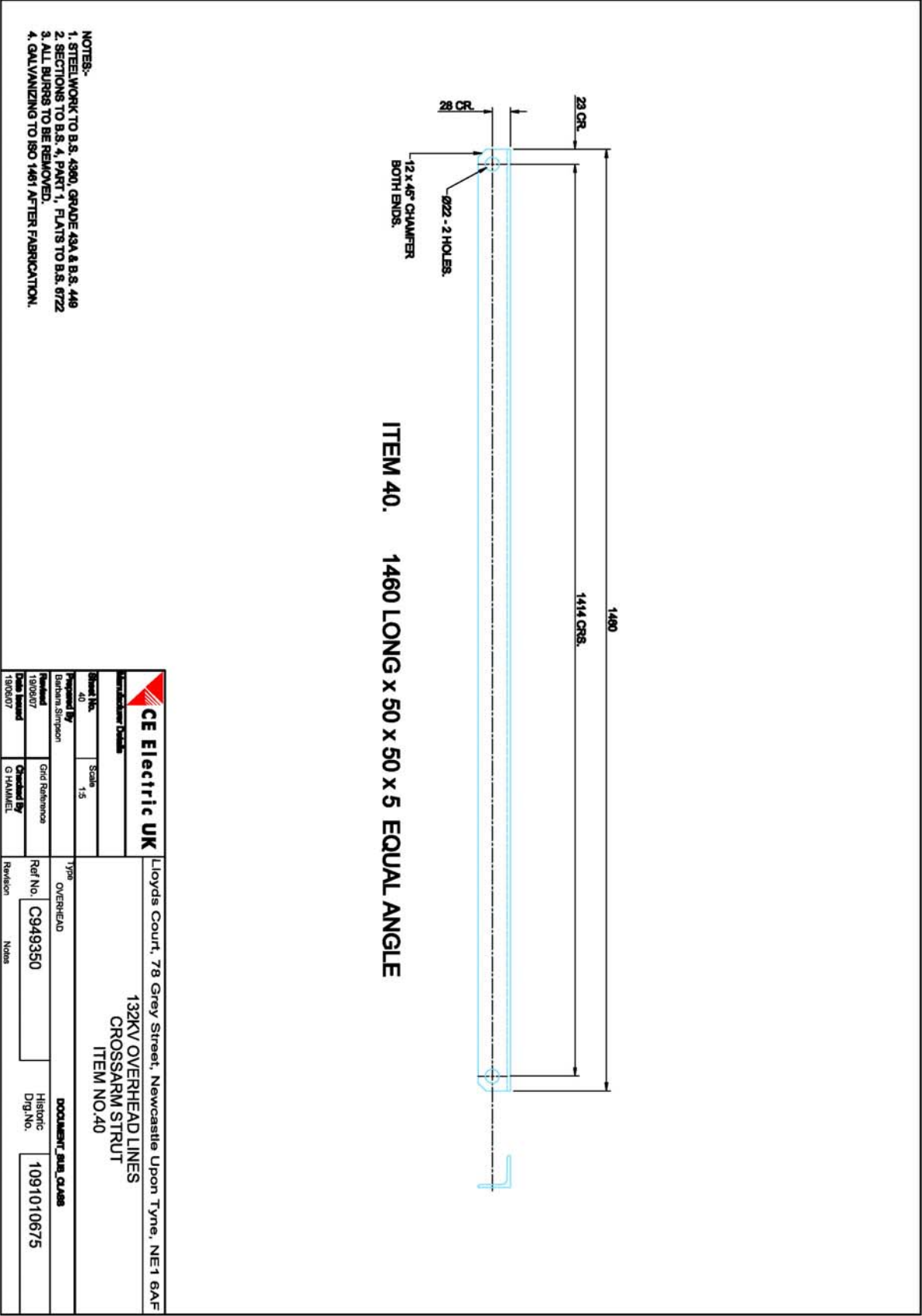


CAT NO 244981

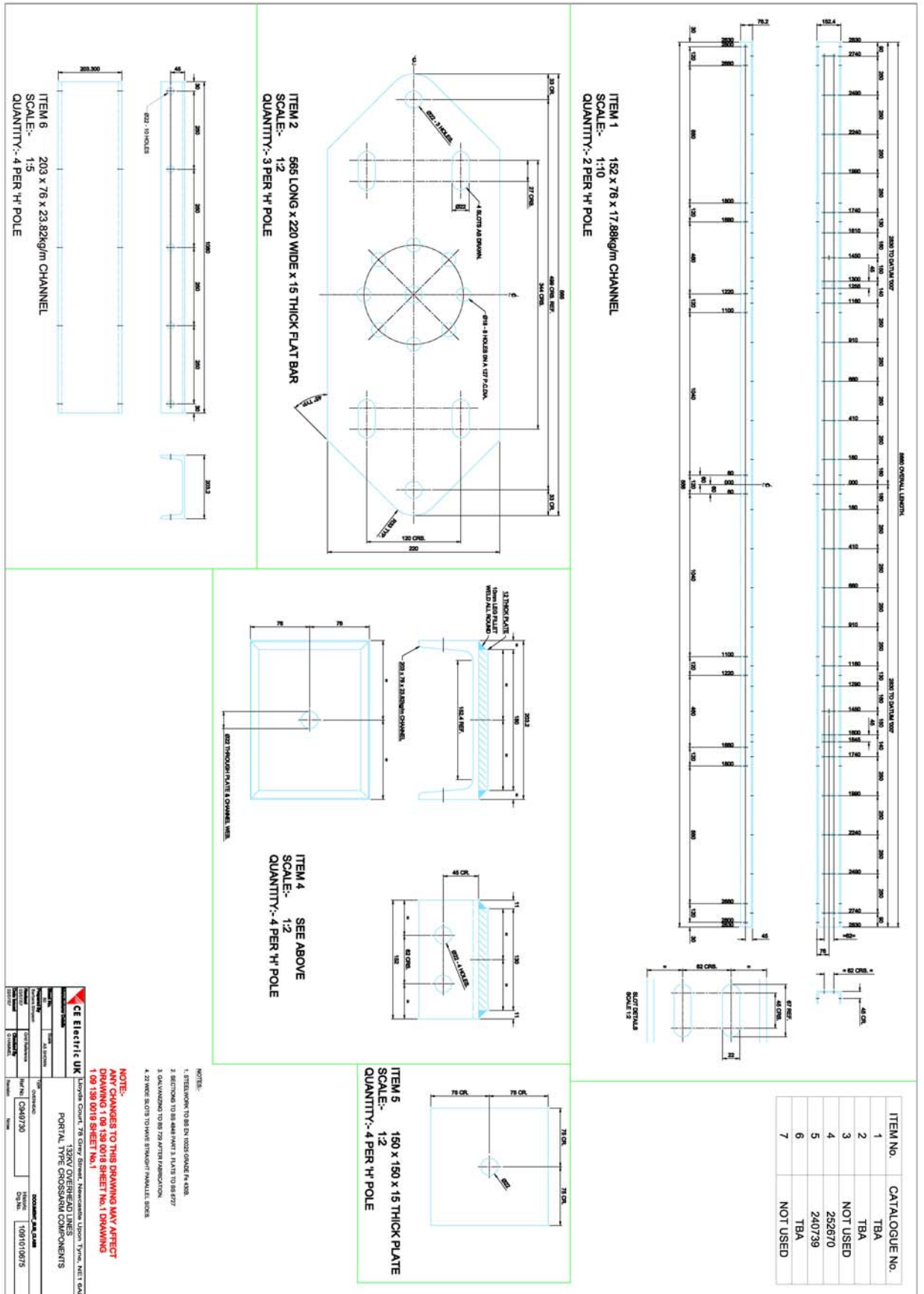
Document reference	NSP/004/045				
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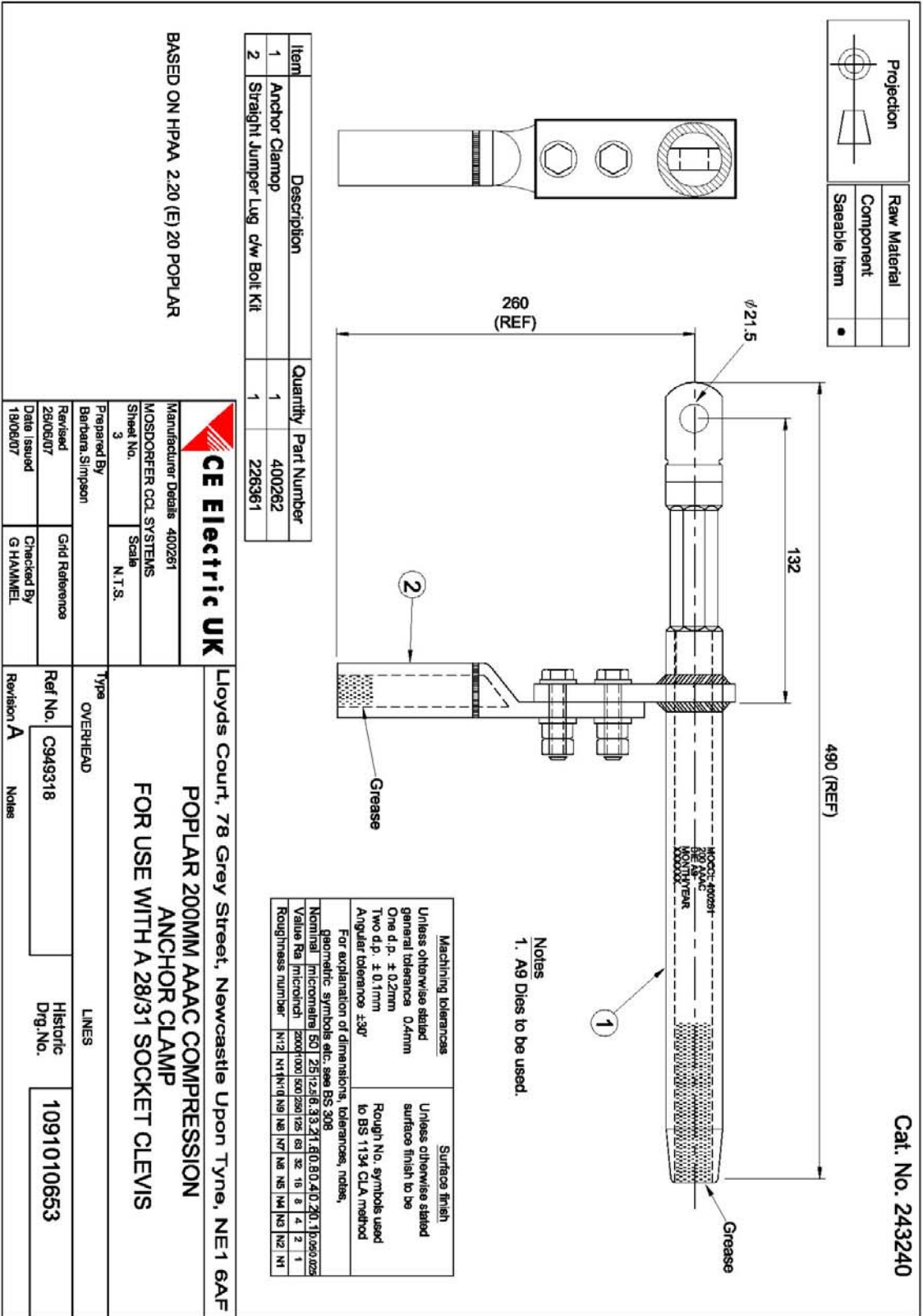
Document reference		NSP/004/045			
Version:-	2.0	Date of Issue:-	Sept 2009	Page	104 of 127



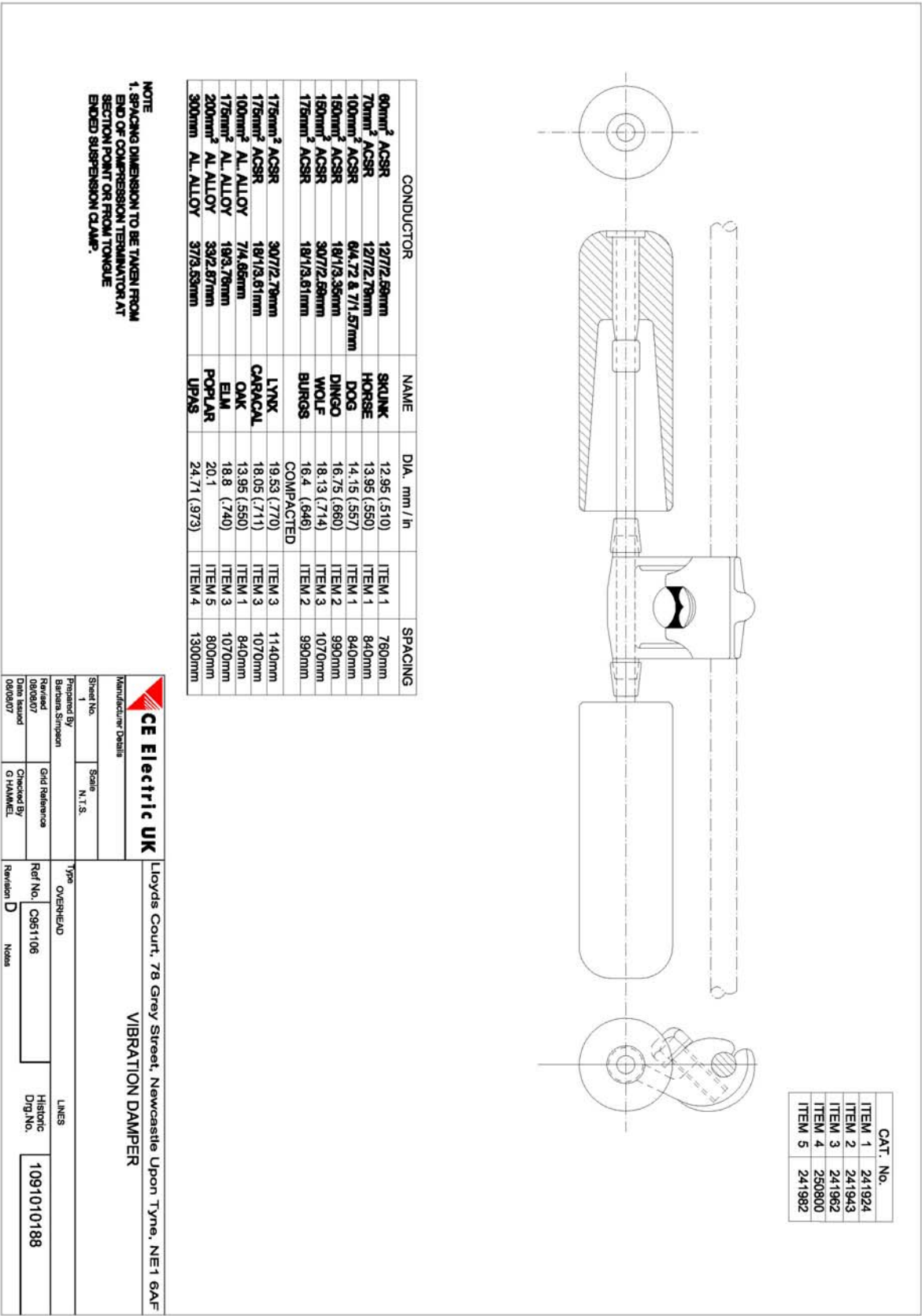
Document reference		NSP/004/045					
Version:-	2.0	Date of Issue:-	Sept 2009	Page	105	of	127



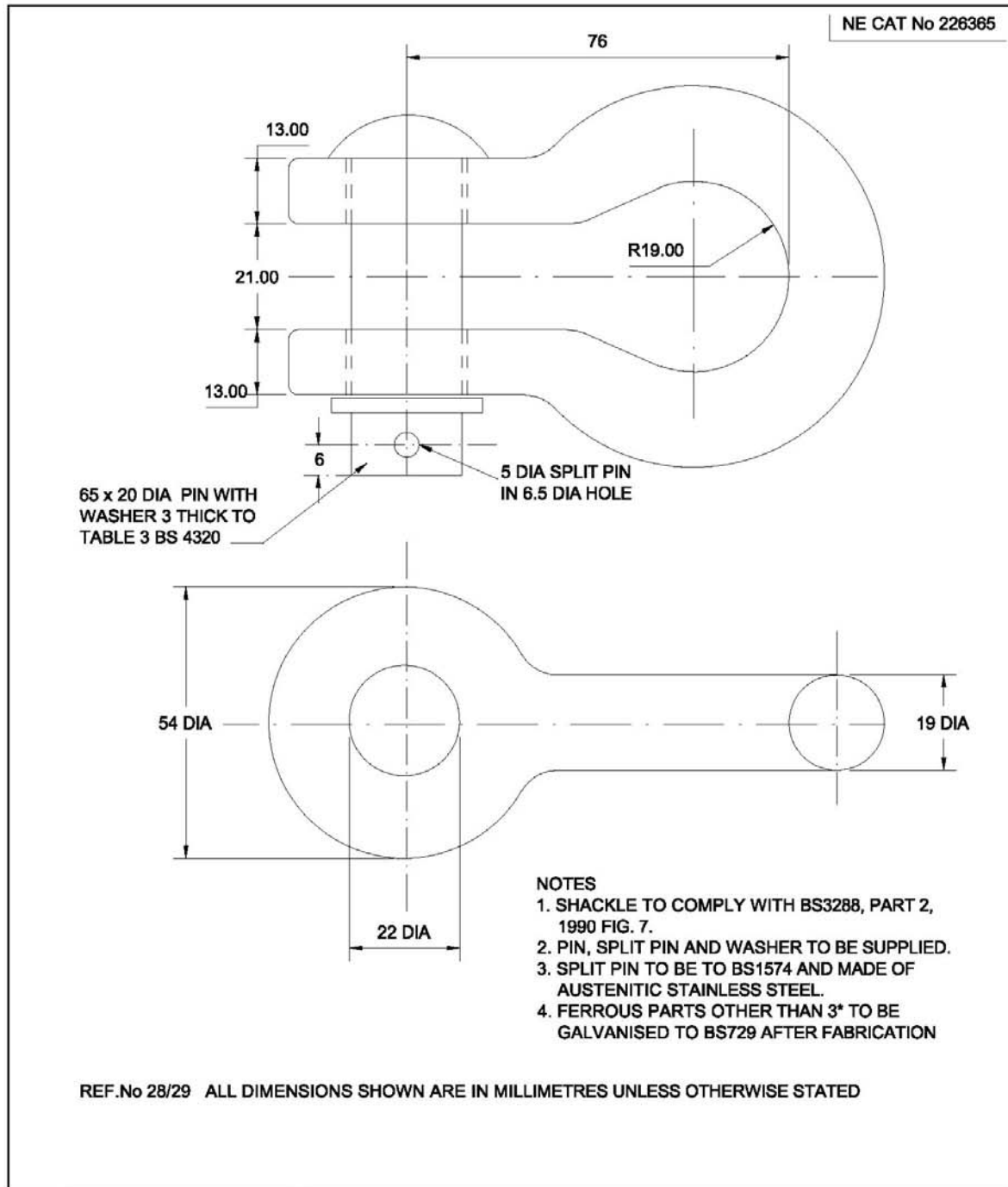
Document reference	NSP/004/045			
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 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		SHACKLE MINIMUM FAILING LOAD 125KN			
Sheet No. 1	Scale 1:1				
Prepared By Barbara.Simpson		Type OVERHEAD	LINES		
Revised 08/08/07	Grid Reference	Ref No.	C951103	Historic Drg.No.	1091010461
Date Issued 08/08/07	Checked By G HAMMEL	Revision	D	Notes	

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Appendix B - User Guide Woodhouse Mast Replacement .XLS

POLE LOADING CALCULATION SPREADSHEET

The spreadsheet is designed to calculate loading on poles based on deterministic loading to . BSEN 50341-3-9. The loading calculation is carried out on the "LOAD CALC" sheet. Analysis of a Rutter pole under the calculated loading is undertaken on sheet "RUTTER POLE". whilst analysis of guyed pole structure is undertaken on sheet "GUYED POLE".

IF "Rutter Pole" is selected in cell B3 on LOAD CALC, then the utilisation and deflection for a Rutter Pole design will be quoted on LOAD CALC. If "Guyed Pole" is selected then the utilisation for a guyed pole will be given.

The calculations are based on data given in databases on sheets "CONDR PROP", "POLE & INSUL DATA" and "RUTTER PROP"

Pole type (Medium, Stout, Extra Stout, etc), pole length, insulator type and embedment depth should be selected in cells B4-B8 on LOAD CALC.

If non standard poles are adopted for guyed poles, then the standard diameters can be overwritten by input into cells D19-D20 in GUYED POLES.

The stay angles between stay and vertical, and in plan, can be specified in cells D11-D14 in GUYED POLES. The angles of the stays in plan are measured from the longitudinal axis so that a stay in the transverse direction will have orientation 90 degrees. Note that some stay orientations may result in loads in stays calculated as compressive. (A warning is given if this is the case.)

The notation adopted for cells is as follows:-

- 1) Where **black** font is adopted, the contents of the cells should not be changed.
- 2) Where **blue** font is adopted within **blue** boxes, the user is expected to input data. (List boxes are provided where only a specific number of acceptable values are possible.)
- 3) Where **green** font is adopted, values may be changed by the user but this will generally be unnecessary and is not recommended without consultation.
- 4) **Red** font is adopted for the most important output values and these should not be altered by the user.

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Appendix C - Self supporting steel masts

C1. Introduction

This Appendix covers the design and constructional requirements for new 3-phase overhead lines on self supporting steel masts to replace the Woodhouse Mast structures and to operate at voltages up to and including 132 kV in accordance with the requirements of BS EN 50341-1:2001 Overhead Electrical Lines Exceeding AC 45 kV. The structures have been designed utilising 200 mm² bare aluminium alloy (AAAC) conductors and comply with the "Empirical" design approach as detailed in BS EN 50423-3-9:2005 and ENA TS 43-97 Steel Supports for Single Circuit 132 kV Overhead Lines with the exception of the partial strength parameters which shall be as detailed below.

C2. Design Criteria

The steel masts detailed in this Appendix are designed to be self supporting structures without guys and designed to resist the loads due to the climatic, accidental-security conditions, construction and maintenance conditions as given in the main specification except as detailed below.

The following clauses differ from those given in the main body of the Woodhouse Mast Replacement Specification. Otherwise the clauses given in the Woodhouse Mast Replacement Specification shall apply.

The supports detailed in this Appendix may be used for different conductors, spans and loading conditions subject to satisfactory design analysis.

The designs may be in accordance with ANSI/NEMA TT1 – "Tapered Tubular Steel Structures" published by the American National Standards Institute or alternative approved design method.

C2.2.1 Type tests

Where manufactures can not provide suitable evidence to demonstrate that they have experience of designing and fabricating similar structures to those detailed in ENA TS 43-97, they will be required to confirm the adequacy of the design by submitting the structures for type tests. The results of the type tests shall be applied to any supports not tested using the same design methods as for the tested pole support. For this purpose the interpretation of the test results, the actual geometry and strength properties of the tested pole support shall be taken into account. One method of achieving this is to be found in BS 5649 part 8.

The tests on complete supports shall be in accordance ENA TS 43-97 Issue 2, Part 2 Test Requirements.

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C2.2 Security Loads

C2.2.1 Broken Wire

The self supporting steel masts detailed here have been designed to withstand longitudinal loads resulting from a “broken wire” condition defined as the static tension of one broken phase conductor.

C2.3 Partial Factors

C2.3.1 Partial Strength Factors

The following partial strength factors shall be used for steel poles and crossarms

Component	Clause	Material Property	γ_m
Steel Members	5.5	Resistance of cross sections and buckling of sections (based on yield strength)	1.15
Mild steel		Nominal yield (275 N/mm ²) 240 N/mm ²	
High yield steel		Nominal yield (355N/mm ²) 310 N/mm ²	
Bolts		Grade 8.8 according to BS 5950	1.15
Welds		Electrodes to E43/E51 according to BS639	1.15

C2.3.2 Deflection

The deflection at the top of the pole shall not exceed 1.5% of its overall length when subjected to its unfactored working loads excluding the effect of foundation rotation.

C3. Construction and Dimensions

The pole shafts shall be octagonal and of folded plate construction as detailed. The steel crossarms shall be tapered and either of hollow section or folded plate. Fabrication, welding and tolerances shall be in accordance with ENA TS 43-97 - Steel Supports for Single Circuit 132 kV Overhead Lines unless detailed otherwise within this

Each pole support shall be permanently marked on the base adjacent to the concreting hatch with the pole type and length, year of manufacture, manufacturers name and the Contract Reference.

C3.1 Pole and Crossarm Assemblies

Poles and Steel crossarms shall be designed in accordance with BS 5950 Structural use of Steelwork in Building, Part 1, Code of Practice for Design – Rolled and Welded Sections, except that partial factors for loads shall be in accordance with Clause 2.5.1 Partial Factors for Actions as given in this specification.

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C 3.1.0 Materials

C3.1.1 Steel

All steel shall comply with the requirements of BS EN 10025, BS EN 10137 or BS EN 10210 as appropriate and shall be suitable for all the usual fabrication processes, including hot and cold working within the specified ranges.

The quality of finished steel shall be in accordance with BS EN 10163 or BS EN 10210 as appropriate. All steel shall be free from blisters, scale, laminations, segregations and other defects. There shall be no rolling laps at the toes of rolled sections or rolled-in mill scale.

Unless stated to the contrary the following grades of steel shall be applicable:

- i) Mild Steel shall be grade S275JR (Fe430B)
- ii) High Tensile steel shall be grade S355JR (Fe510B) for sections less than 20mm thick and grade S355JO (Fe510C) for sections greater or equal to 20mm, except for plates which shall be greater or equal to 40mm thick. The minimum steel thickness shall be 5mm.
- iii) Steel plates for folded, pressed braked or stretch bending pole supports shall be of a suitable grade to meet the recommendations of clause 7.4.3 of part 1
- iv) Mild steel and high tensile hollow sections shall be grade S275J2H and S377J2H respectively.

Steel section profiles shall be in accordance with the requirements of BS4 part 1, BS EN 10056, BS EN 10210 and BS EN 10279 as appropriate.

Hot rolled steel plates greater or equal to 3mm thick shall be in accordance with BS EN 10029 or BS EN 10137 as appropriate.

C3.1.2 Bolts Nuts and washers

Unless stated otherwise in the specification the following grades of steel shall be acceptable:

- i) Steel for bolts smaller or equal to 12mm diameter shall be grade S275JR
- ii) Steel for bolts greater or equal to 16mm diameter shall be to grade S355JR, unless otherwise agreed.
- iii) Steel for non structural step bolts shall be grade S275JR
- iv) Nuts shall be grade S275JR
- v) Holding down bolts shall be strength grade 8.8/10 to BS 4190

Unless stated to the contrary bolts and nuts shall be ISO Metric Black Hexagon to BS 4190, and shall be threaded ISO Metric Course pitch to BS3643 part 2, tolerance class 7H/8g

Unless stated to the contrary holding down bolts shall be supplied complete with two full nuts and one locknut.

Countersunk and other bolts without hexagon heads shall have slotted heads.

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All flat washers shall comply with the requirements of BS4320, Form E, Grade 4.6. Unless specified to the contrary they shall be 3mm thick. Pack washers shall have an external diameter of twice the nominal bolt diameter plus 15mm, a hole with a diameter of the nominal bolt diameter plus 2mm and a thickness as specified on the appropriate fabrication drawing.

C3.1.3 Welding

All welding shall be carried out in accordance with the requirements of BS 5135 and BS EN 1011-1.

C3.1.4 Erection Marks

Before leaving the fabricators works, all members shall be stamped with distinguishing numbers and or letters corresponding to those on the general arrangement or fabrication drawings. Additionally, the agreed fabricators identification symbol must appear adjacent to the erection mark.

Erection marks shall be stamped on before galvanising, using characters at least 10mm high and shall be clearly legible after galvanising.

Erection marks shall be located as follows:

- a) All members – stamped within 600mm of the end but clear of the holes.
- b) Plates or flat bars – stamped at the most suitable position between holes
- c) Horizontal members – stamped at either end of the bar
- d) Steel pole sections – 100mm either side of the joint.

Steel poles shall be marked on the base adjacent to the earthing lug with the year of manufacture, the manufacturers name and the contract reference.

C3.1.6 Slip Joints

Slip joints shall develop the full required design strength. They should be detailed for a minimum lap of 1.5 times the larger diameter with an allowance of 75mm for fabrication tolerance.

C3.1.5 Corrosion Protection

Unless stated to the contrary, after completion of all fabrication processes (including all drilling, punching, stamping, cutting, bending and welding) support steelwork, poles and pole steelwork including nuts, bolts and washers shall be hot dip galvanised and tested in accordance with the requirements of BS EN ISO 1461. Electro-galvanising is not an acceptable alternative.

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Excessively thick or brittle coatings due to high levels of silicon or phosphorous in the steel, which may result in an increased risk of coating damage and/or other features that make the final product “not fit for purpose” shall be cause for rejection.

All materials prior to galvanising shall be free from oil, grease or any other substance which may adversely effect the quality or finish.

All bolts and screwed rods, including the threaded portions, shall be galvanised. The threads shall be cleaned of all surplus spelter by spinning or brushing. Dies shall not be used for the cleaning threads other than nuts. Nuts shall be galvanised and tapped 0.4mm oversize and threads shall be oiled.

C3.2 Cat Ladder

The cat ladder step bolts shall be M16 x 200mm long grade 8.8 bolts and shall commence at 2.0m above ground level and be continuous to within 1m from the pole top. Step bolt spacing shall typically be 300mm.

C3.3 Type and Range of Structures

The range of the steel pole supports have been selected such that the existing Woodhouse Mast lines can be replaced and uprated without the need to insert additional support structures. The new steel pole designs are based on overall pole lengths of 12 m, 14m and 16m. This range of pole lengths will allow the existing poles to be replaced without exceeding the existing structure heights. However, subject to the approval of the Engineer, if any ground clearance infringements are encountered then these can be overcome by using the next height of pole provided the height increase is 10% or less.

The structures detailed provide for the following angles of deviation.

Structure Type	Deviation
Intermediate	0°
Section D10	0° – 10°
Section D30	10° - 30°
Section D45	30° – 45°

C3.4 Foundations

Structure foundations shall be designed for three soil conditions in accordance with Clause 4.10 of the main body of this Specification. Steel mast foundations shall be of the mass concrete monoblock type, installed at the minimum depths shown on the drawings. The detail drawings are based on 628kN/m² or “good” soil on the premise that soil additives will be used when required.

Concrete shall be designated mix FND3 in accordance with BS8500: Part 2 and shall be cast in one pour and fully compacted. The inside of the pole shall be filled with compacted concrete to ground level via the concreting hatch provided.

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C3.4.1 Protection of buried steelwork

Any part of the pole support steelwork buried in the ground is to be given additional anti-corrosion protection. This should consist of a heavy duty bituminous paint with a density of 1kg/litre to be applied at a rate of 1.7 to 2.1sq.m/litre over the whole of the exterior and interior surfaces of the complete base section and to extend to 300mm above known ground level.

C3.5 Earthing

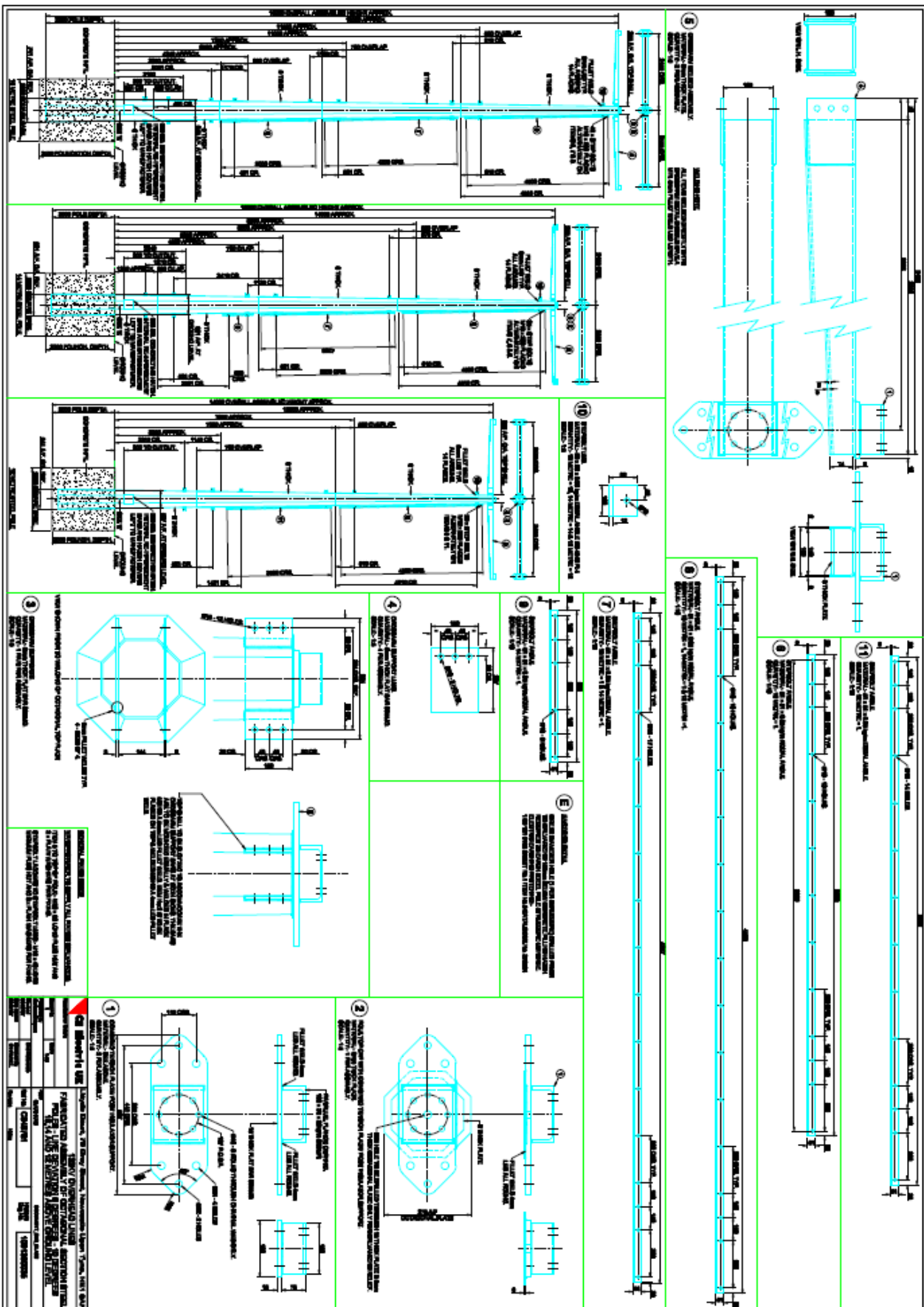
All steel structures shall be earthed with the earth cable connected to the pole and earth rods as required to obtain a maximum earth resistance of not more than 10 ohms. Earth rods and their connectors shall comply with the requirements of ENA TS 43-94.

C3.6 Self Supporting Steel Mast Drawing Arrangements

The following pages detail a selection of self supporting steel mast structures to be used in conjunction with the main body of this document where specific design conditions are encountered

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[illegible]

[illegible]

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Appendix D – Specialist Pole Top Equipment

D1 Crane / Platform

**CLY 620 0500:
132kV Crane Platform
Data Sheet**

The diagram illustrates the CLY 620 0500 132kV Crane Platform in two configurations: Centre Phase Configuration and Outer Phase Configuration. The Centre Phase Configuration shows a crane boom with a sash line roller and a crane upright. The Outer Phase Configuration shows a crane boom with a sash line roller and a crane upright. The crane is equipped with three EN 795 Class A1 fall arrest points for added safety. The first is placed half way up the crane upright to allow attachment to the crane whilst still safely attached to the pole. The second two are placed above the workers shoulders in the crane boom gusset, one either side to allow two men to be attached.

Part Number	CLY 620 0500
SWL	550kg
Platform Half	11kg
Crane Base	25kg
Crane Upright	22kg
Crane Boom	23kg
Total	117kg
Assembled Dims	W1560 x L440 x H2360mm

The Clydesdale 132kV Crane Platform has been developed in conjunction with CE Electric UK to facilitate construction and maintenance of 132kV lines on wood poles.

The device consists of two platform halves which tighten to the steelwork, two cranesbases which are then fastened in place with ratchet straps, the crane upright is pinned in either centre phase or outer phase position, and finally the crane boom is pinned in place atop the crane upright.

The conductor can then be lifted using the ratchet lever hoist connected to the conductor spreader bar. Whilst the crane maintains the conductor in the air, the post insulator can be replaced with the assistance of a sash line through the sash line roller.

The crane is equipped with three EN 795 Class A1 fall arrest points for added safety. The first is placed half way up the crane upright to allow attachment to the crane whilst still safely attached to the pole. The second two are placed above the workers shoulders in the crane boom gusset, one either side to allow two men to be attached.

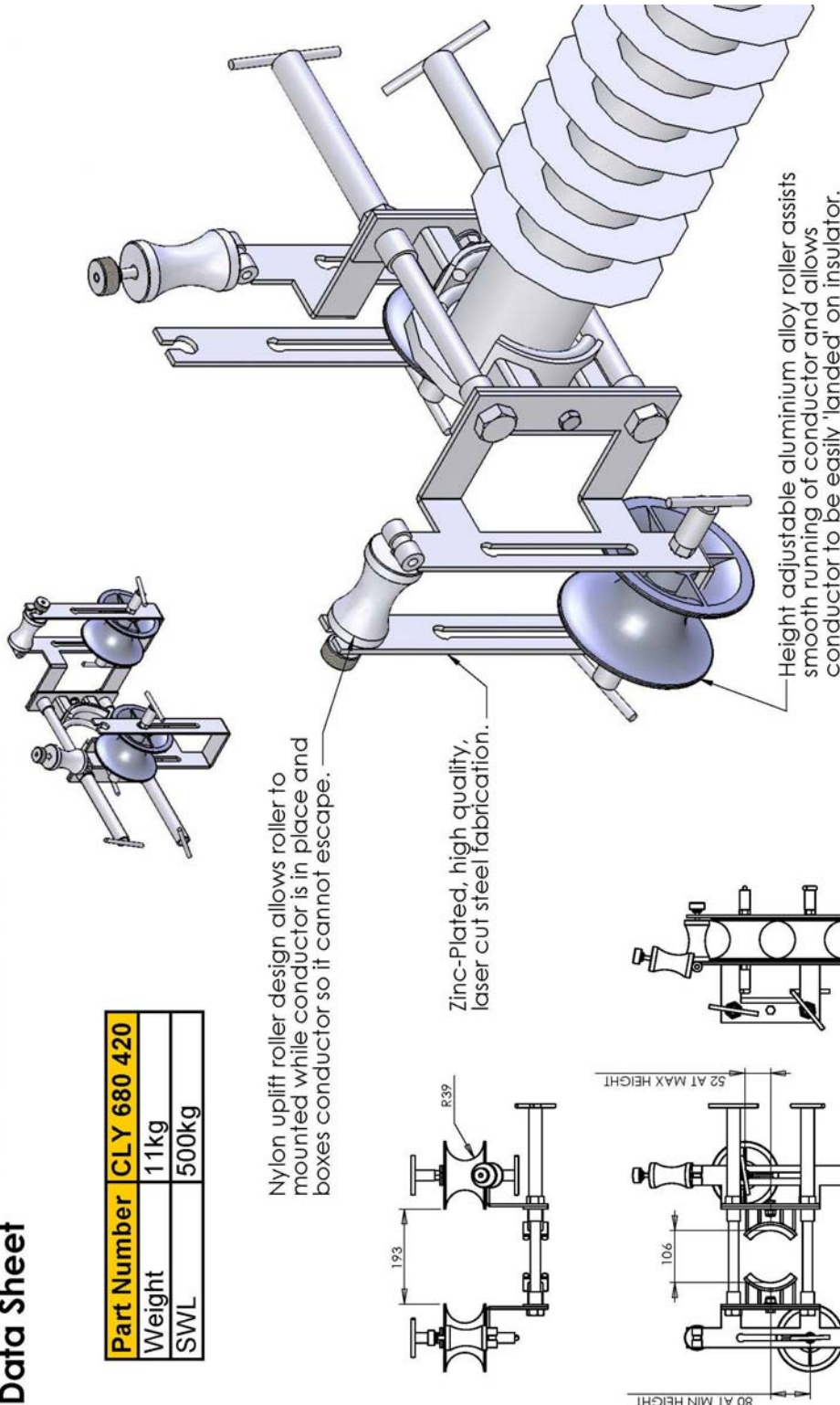
DS CLY 620 0500 17/09/07 REV0 CB

Document reference	NSP/004/045			
Version:-	2.0	Date of Issue:-	Sept 2009	Page 121 of 127

D2 Horizontal Insulator Mounted Roller

CLY 680 420: Horizontal Post Insulator Mounted Roller Data Sheet

Part Number	CLY 680 420
Weight	11kg
SWL	500kg



DS CLY 680 420 14/09/07 REV0 CB

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D3 Vertical Insulator Mounted Roller

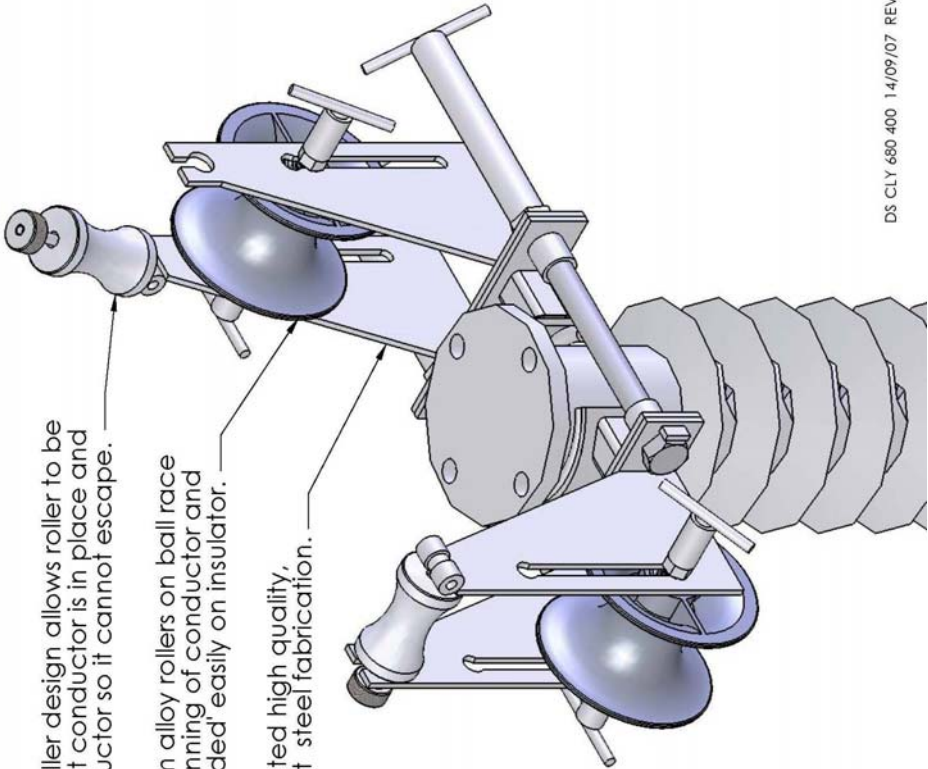
CLY 680 400:
Vertical Post Insulator Mounted Roller
Data Sheet

Part Number	CLY 680 400
Weight	9.5kg
SWL	500kg

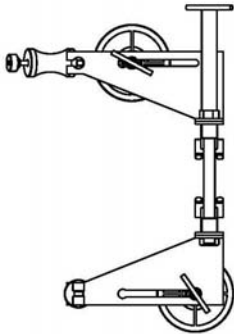
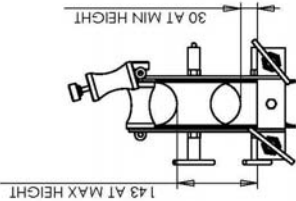
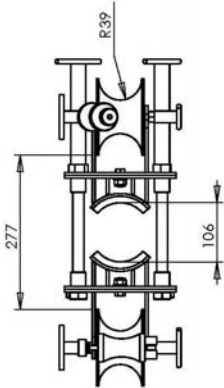
Nylon Uplift Roller design allows roller to be mounted whilst conductor is in place and boxes in conductor so it cannot escape.

Height adjustable aluminium alloy rollers on ball race suspension assists smooth running of conductor and allows conductor to be 'landed' easily on insulator.

Zinc-plated high quality, laser cut steel fabrication.

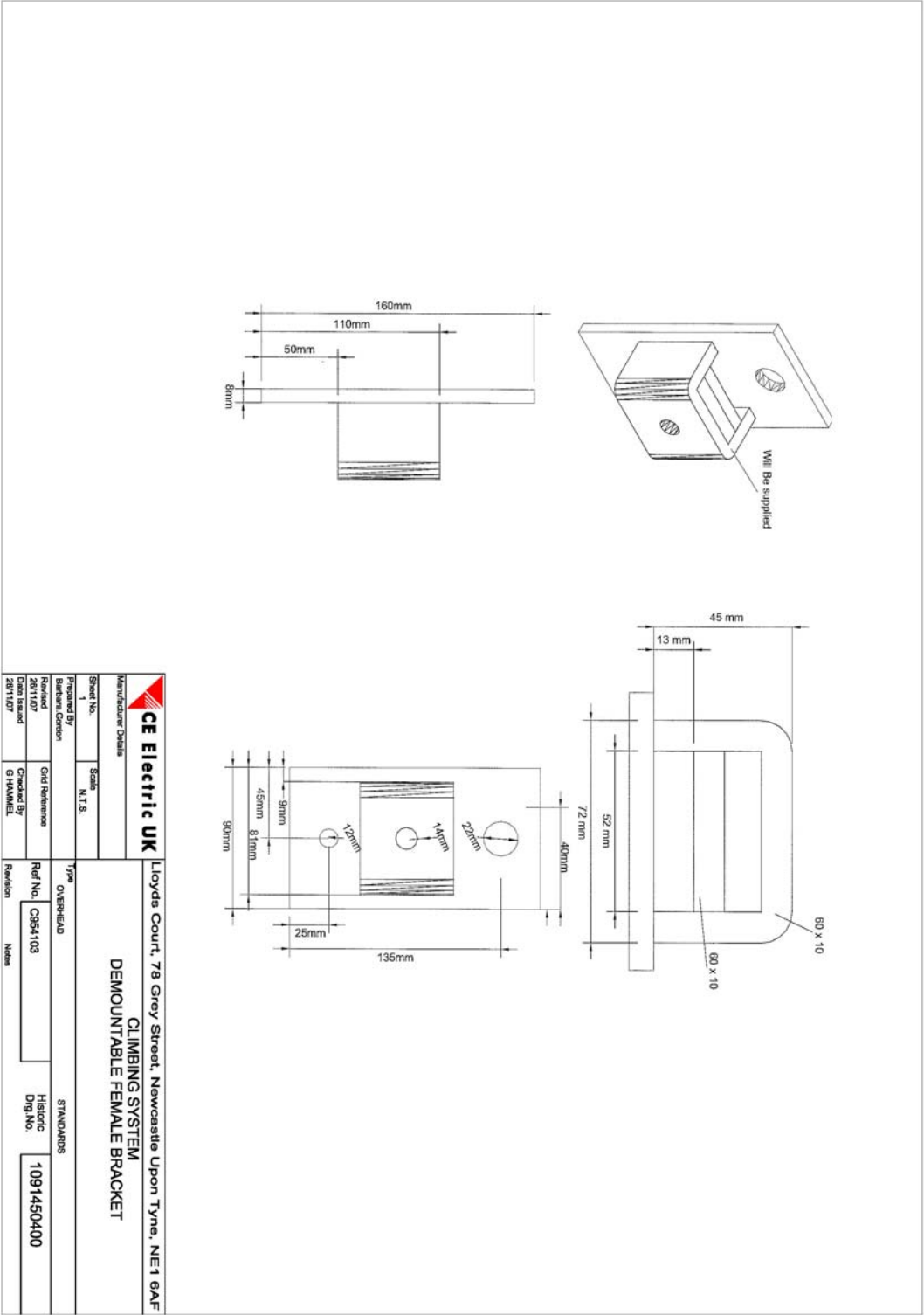


DS CLY 680 400 14/09/07 REV0 CB



Document reference	NSP/004/045				
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D4 Fixed Bracket for Demountable Ladders



Document reference	NSP/004/045			
Version:-	2.0	Date of Issue:-	Sept 2009	Page 124 of 127

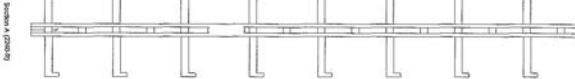
D5 Demountable Ladder System

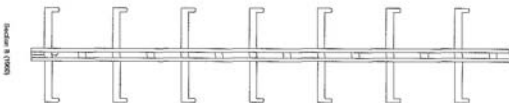
TYPE	DESCRIPTION	CAT NUMBER	No RUNGS	WEIGHT in Kgs
A	Common base ladder section 2240mm in length	313000	8	9.2
B	Intermediate ladder section 1800mm in length	313001	7	8.5
C	Intermediate ladder section 1400mm in length	313002	6	7.4
D	Intermediate ladder section 1400mm in length	313003	5	6.5
E	Short Entry Top section 1120mm in length	313004	1	4.5
-	Permanent Female Mounting Brackets	313005	-	-
-	Oldlock Threader	313006	-	-

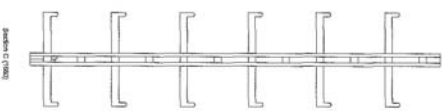
LADDER SELECTION TABLE

STRUCTURE TYPE	(RUTTER POLE)	SINGLE SECTION POLES & H SECTION / TERMINAL POLES
12m	(10A)-(10B)-(10C)	(10A)-(10B)
15m	(10A)-(10C)-(10D)-(10E)	(10A)-(10B)-(10C)-(10D)
16m	(10A)-(10C)-(10E)	(10A)-(10C)-(10D)
16m	(10A)-(10C)-(10E)	(10A)-(10D)
16m	(10A)-(10C)-(10E)	(10A)-(10C)-(10D)
17m	(10A)-(10D)-(10E)	(10A)-(10C)-(10D)


Diagram of a ladder section with a single pole and a terminal pole.


Diagram of a ladder section with a single pole and a terminal pole.


Diagram of a ladder section with a single pole and a terminal pole.


Diagram of a ladder section with a single pole and a terminal pole.

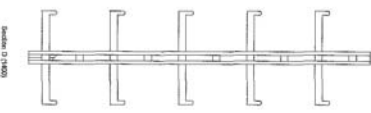

Diagram of a ladder section with a single pole and a terminal pole.

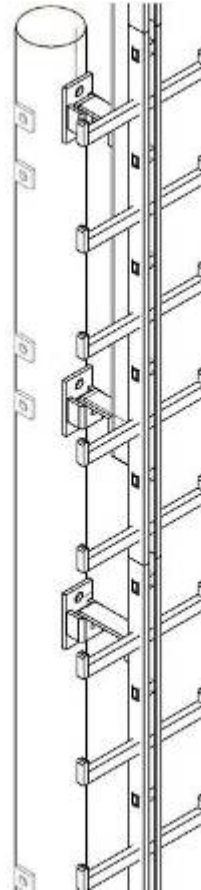

Diagram of a ladder section with a single pole and a terminal pole.

		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
CE Electric UK		CLIMBING SYSTEM	
Manufacture Details		DEMOUNTABLE LADDER TYPES	
Stock No.	Scale	STANDARDS	
1	N.T.S.		
Prepared By	Barbara Gordon	OVERHEAD	
Revised	27/11/07	Grid Reference	
Date Issued	27/11/07	Ref No. C954131	
Checked By	G HAMMILL	Hilroyc Dig No.	
Revision		Notes	

1091450402

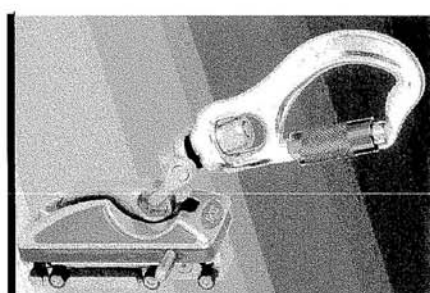
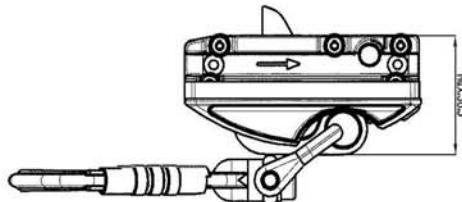
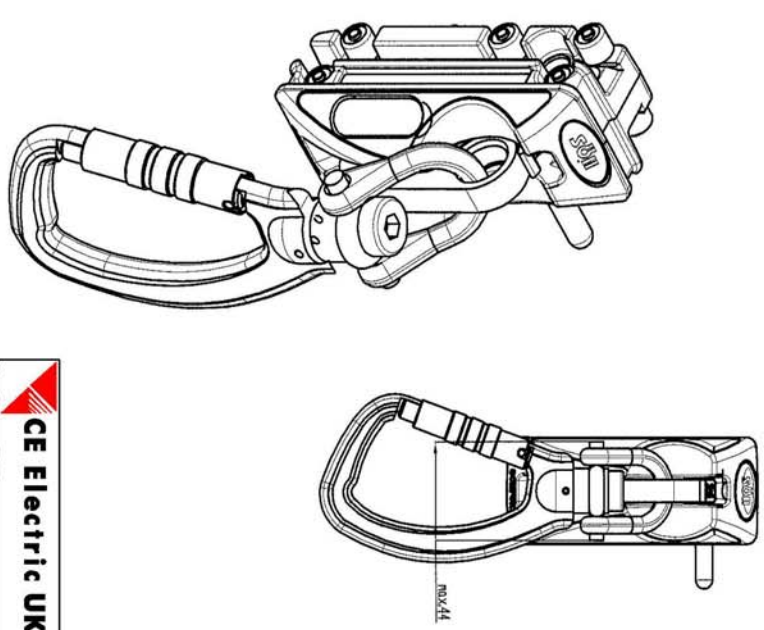
Document reference		NSP/004/045			
Version:-	2.0	Date of Issue:-	Sept 2009	Page	125 of 127

D5.1 Demountable Ladder System



Document reference		NSP/004/045			
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D6 Traveller Attachment Device



 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details		CLIMBING SYSTEM TRAVELLER ATTACHMENT DEVICE	
Sheet No.	1	Scale	N.T.S.
Prepared By	Barbara Gordon	Gtd Reference	
Revised	27/1/07	Checked By	G HAMMEL
Date Issued	27/1/07	Revision	Notes
Type		STANDARDS	
OVERHEAD		Historic Drg.No.	
Ref No. C954128		1091450401	

CE ELECTRIC UK

Document reference		NSP/004/045					
Version:-	2.0	Date of Issue:-	Sept 2009	Page	127	of	127
